Characterization and Convergence of Markov Processes via Martingale Problems

A blueprint for formalization

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Abstract

This manuscript collects, in a self-contained and formalization-oriented form, four results on martingale problems: (i) existence of a càdlàg modification of a solution (Ethier–Kurtz [4], Theorem 4.3.6); (ii) the fact that uniqueness of the one-dimensional distributions already implies uniqueness of the finite-dimensional distributions and the (strong) Markov property ([4], Theorem 4.4.2); (iii) duality as a tool for proving uniqueness ([4], Lemma 4.4.10, Theorem 4.4.11 and its corollaries); (iv) existence: five routes, of which two produce a solution out of nothing — an explicit construction of jump processes, and the correspondence with stochastic differential equations — one constructs it from a *dual* process (Depperschmidt–Greven–Pfaffelhuber [3]), and one obtains it as a weak limit of solutions of approximating martingale problems ([4], Lemma 4.5.1 and Remark 4.5.2), together with the strictly more general convergence theorem of Criens–Pfaffelhuber–Schmidt [2]. Each of (i)–(iii) is first proved in the abstract setting of [2], without any operator, and [4]'s statement is then recovered as a corollary; the abstract versions are new, since [2] prove no characterization results. A Markov structure is introduced only where a result needs one: uniqueness of the finite-dimensional distributions does not, and is proved for an arbitrary family of solution measures, with the Markov property falling out afterwards from a shift hypothesis. Two non-Markovian examples of [2] are used as test objects. Throughout, the time index is an abstract ordered set carrying a *clock*, the state space is graded from "measurable space" to "Polish", and local martingale problems are treated wherever the statement permits it. Every statement is tagged with the weakest hypotheses on the index and on the state space that it uses, and Section 2.9 tabulates the result: the partially ordered index survives in (ii) for the Markov property but provably not for uniqueness, in (iii) for an atomic clock, and in the duality route to (iv) throughout; a merely measurable state space suffices for (ii) and (iii) throughout. Section 1 explains, and justifies, the choice of the underlying setting; Section 8 records the design decisions that are relevant for the Lean formalization.

Contents

1 Scope, and the choice of the setting

1.1 The two candidate settings

There are two natural frameworks in which the results of this manuscript can be phrased.

(EK) The Ethier–Kurtz setting [4], Chapter 4. A martingale problem is attached to an *operator* $A \subset M(E) \times M(E)$ on a metric space E: a measurable E-valued process X solves the martingale problem for A if

$$f(X(t)) - \int_0^t g(X(s))\,\mathrm{d}s, \qquad t \geq 0, \tag{1}$$

is a martingale for every $(f, g) \in A$. This is the setting in which *characterization* results are traditionally stated: uniqueness, the Markov property, duality, and the correspondence with semigroups and generators.

(A) The abstract setting. A martingale problem is attached to a *family* $\mathbb{X}$ *of real-valued test processes* on a filtered space carrying, as yet, no measure: a probability measure P solves the martingale problem $\mathrm{MP}(\mathbb{X})$ if every $Y \in \mathbb{X}$ is a P-martingale, and the local martingale problem if every $Y \in \mathbb{X}$ is a P-local martingale. The state space is Polish and the path space is an arbitrary Polish subset of $E^{\mathbb{T}}$; no operator and no Markov structure are assumed.

It is worth being accurate about where this second setting comes from, since this manuscript leans on it heavily. It is not new with [2]. In essentially the form used here it is [5], Definition III.1.3, which attaches a martingale problem to a family of optional processes and does so with *local* martingales as the default rather than as a variant, and with the initial condition given on an initial σ-field rather than by an initial distribution. [2] rediscover it in the shape that suits weak convergence — canonical test processes, determining sets, P-continuity — and it is their formulation that Section 3 follows, but Remark 3.6 records what [5] already had and what is worth importing from them. [7], Chapter 32 works in the same spirit for diffusions, again with local martingales throughout.

What each setting is good for. The division of labour is not the one the first draft of this manuscript assumed, and the correction is the main structural finding here. Setting (A) is where *all four* target results are naturally proved — including the three that [2] do not state. Setting (EK) is where they are naturally *applied*: in practice one is handed an operator, not a family of test processes, and the hypotheses of Section 5.6 are the ones that can be checked on A directly. Section 1.2 sets out the resulting three-layer architecture.

1.2 Recommendation

We use a *hybrid*, organised in three layers. The reader should be warned that the balance between them shifted while this manuscript was written, and that the outcome is not the one the first draft anticipated.

(1) **Definitional layer: (A).** "Solution of a martingale problem" is defined as in [2], Definition 3.2, relative to a family $\mathbb{X}$ of test processes, with a *local* variant available from the outset (Definition 3.2). This costs one line and is what makes local martingale problems and non-càdlàg path spaces expressible at all.

(2) **Abstract working layer: also (A).** Each of the three characterization results is stated and proved in this setting, with no operator anywhere: Theorem 4.3 for the càdlàg modification, Proposition 5.9 and Theorems 5.16, 5.19 for uniqueness and the (strong) Markov property, and Lemma 6.1 with Theorem 6.5 for duality. This layer is itself graded: a *Markov structure* is introduced only where it is needed, namely in Section 5.2, and Remark 5.42 tabulates what lies on either side of that line. Uniqueness, in particular, lies on the non-Markovian side.

(3) **Markovian layer: (EK).** The martingale problem for an operator A is the special case $\mathbb{X} = \mathbb{X}_A(X)$ of (1) (Definition 3.5), and the four target results are recovered on it: Theorem 4.5, Theorem 5.43, Theorem 6.10 and Lemma 7.23. Each is a corollary of layer (2), and each proof is a verification of hypotheses rather than an argument. Section 7 sits partly outside this scheme: the existence results there are not characterizations and have no abstract counterpart to be corollaries of, except Theorem 7.27, which is itself the abstract one.

Layer (2) is the substance of this manuscript, and it is the point on which the first draft was wrong. It assumed that uniqueness, the Markov property and duality are *intrinsically* statements about an operator A and its domain, and that (A) could therefore serve only as packaging. That is not so. Each of the three proofs, read carefully, uses of the operator only that it produces test processes of a particular shape, and each hypothesis it really needs can be stated directly:

- the càdlàg theorem needs a class Φ for which $f(X)$ splits into a martingale plus a time-regular remainder — Definition 4.2;
- uniqueness and the Markov property need the test family to be stable under the shift — Definition 5.10 — and nothing else, the whole proof reducing to one lemma applied four times (Lemma 5.13);
- duality needs only two increment representations of a function of two time variables (Lemma 6.1), and no probability at all.

Remarks 4.4, 5.21 and 6.8 record what each abstraction buys. Briefly: path-dependent and atomic compensators, which cannot be written with an operator $A \subset C_b(E) \times B(E)$ at all; the disappearance of the auxiliary filtration (4) from everything except Proposition 3.8; and a precise account of which time indices and which clocks each result admits (Section 2.9).

Two caveats keep this from being an argument for (A) alone.

- *[2] contains none of the three.* That paper is concerned exclusively with identifying weak limits; it proves no uniqueness statement, no Markov property and no path regularity statement. For the càdlàg theorem the omission is structural rather than accidental: their Setting 3.1 *presupposes* a Polish path space F and right continuous test processes, which is exactly what a regularization theorem has to produce. Layer (2) is therefore new here, not imported.
- *Layer (3) is not decoration.* The abstract hypotheses are the ones a proof needs; the operator is the one a user has. In applications one is handed A, and the value of Theorem 5.43 over Theorem 5.16 is that its hypotheses are checkable on A without first reconstructing $\mathbb{X}_A$, θ_r and a determining set.

For the fourth target result — existence by convergence, [4], Lemma 4.5.1 — the [2] formulation is a strict generalization (Theorem 7.30): it drops boundedness of f and g, replaces exact solutions of approximating martingale problems by *approximate* martingales, and replaces uniform boundedness by uniform integrability. Here [2] is used directly, and Theorem 7.27 isolates the abstract statement behind it, which needs neither càdlàg paths nor the Skorokhod topology and is proved here in full.

1.3 Generality: time index, Polish state space, local martingale problems

Four deliberate strengthenings relative to [4] are made.

Abstract time index. [4] and [2] index time by $[0, \infty)$. Here the index is an abstract set $\mathbb{T}$, and every statement is tagged with the weakest of the bundles (T0)–(T4) of Definition 2.1 that it needs; the compensator becomes a separate datum, a *clock* in the sense of Definition 2.3. Section 2.9 tabulates the result. There are two reasons for the effort. The first is that the classical formulation conflates a topological hypothesis on $\mathbb{T}$ with an algebraic one: Section 4 needs the former, Sections 5 and 6 the latter, and $[0, \infty)$ happens to supply both. The second is that Mathlib indexes `Filtration`, `Martingale` and `IsStoppingTime` by an arbitrary `[Preorder ι]`, so fixing $\mathbb{T} = \mathbb{R}_+$ at the outset would mean working against the library rather than with it.

The gain is not merely cosmetic. Discrete time ($\mathbb{T} = \mathbb{N}_0$, q counting measure) and processes with fixed times of discontinuity (q with atoms) become special cases rather than analogues, and the partially ordered case survives in Section 3, in the Markov half of Section 5, in the abstract convergence theorem and in the duality route to existence (Section 7.2). It fails in exactly two places, and for reasons worth knowing: the uniqueness induction needs the times to form a chain (Remark 5.18), and the rectification argument behind Theorem 6.5 needs the down-sets to be nested (Remark 6.8) — a limitation of the proof only, since for an atomic clock duality survives on a partially ordered index (Remark 6.7).

State space: measurable, not Polish, where possible. [4] and [2] take E Polish throughout. Here E is graded exactly as the time index is: (E0) a measurable space, (E1) standard Borel, (E2) separable metrizable, (E3) Polish (Definition 2.7). The grading is not decoration. Sections 3 and 6 and almost all of Section 5 run under (E0); regular conditional distributions and the Borel structure, and hence (E1), are needed at five places listed in Remark 2.8; the topology of E is used only in Sections 4 and 7. The payoff is concrete: martingale problems for distribution-valued processes, where $E = \mathcal{S}'(\mathbb{R}^d)$ is standard Borel but not metrizable, are covered by the first three sections with no new proof. See Remarks 2.8 and 2.9.

Where a topology is used, E is *not* assumed locally compact. On a general Polish space there is no space $C_c(E)$ and no one-point compactification argument, and the rôle of local compactness is taken over throughout by the *compact containment condition* (29) resp. (15). This is the essential mechanism, and it is worth isolating it as a definition (Definition 3.20) in the formalization.

Local martingale problems. Whether "martingale" may be weakened to "local martingale" differs from result to result, and it is important not to overclaim. The situation is:

- *Theorem 4.5 (càdlàg modification):* must be stated for genuine martingales. Localization requires stopping times, stopping times require a right-continuous filtration and a progressively measurable process, and path regularity is precisely what is being proved. The argument is circular otherwise. See Remark 4.7.
- *Theorem 5.43 (uniqueness and Markov property):* the proof conditions on events of positive probability and uses the martingale property under the conditioned measures; boundedness of f and g enters through the integrability needed for (21). A

local version is available, but only after localization along stopping times that are *common* to all solutions; see Remark 5.45.

- *Theorem 6.10 (duality):* is already stated for unbounded $f, g, h \in M(E_1 \times E_2)$, controlled by the integrability hypotheses (57)–(58). Nothing has to be changed.

- *Section 7.3 (jump processes):* the local version is not an afterthought but the general case — with unbounded rates the process explodes, and the jump times are a localizing sequence by construction; see Remark 7.15.

- *Section 7.10 (path-dependent rates):* here the local version is not merely the general case but the *primary* one. With a path-dependent rate the compensator need not be integrable even when the process does not explode, so "is a martingale" has no content before stopping; see Remark 7.53. This is the cleanest instance in the manuscript of why Definition 3.2 carries a local variant from the outset.

- *Theorem 7.30 (convergence):* concludes that the limit solves the *global* martingale problem. A local martingale problem is obtained in the limit by applying the theorem to a localized family of test processes; see Remark 7.31.

Section 5.3 carries out the local theory in the abstract setting, following [7], Theorems 32.10 and 32.11. Three hypotheses are needed and are isolated there as (L1)–(L3): a localizing sequence that is a *strict* stopping time and common to all P; covariance of that sequence under the shift; and integrability of the increments after a restart. The point of isolating them is that the local martingale property is *not* linear in P, so that the two workhorse lemmas of Section 5.1 — mixture stability and the restart lemma — do not survive localization for free.

2 Preliminaries

Following the project plan, all material of [4], Chapters 1–3, may be cited. This section makes that interface explicit: it collects, in the form of numbered *Facts*, precisely those definitions and results of [4], Chapters 1–3 (together with a few items from the Appendixes of [4]) that are used later on. Nothing here is proved; the rôle of the section is to fix statements and to serve as the list of prerequisites for the formalization. Section 2.8 records where each item is used.

2.1 Notation

Let $(E, \mathcal{E})$ be a measurable space; the topological assumptions on E are made bundle by bundle in Definition 2.7, and $\mathcal{B}(E)$ is written for $\mathcal{E}$ whenever E carries a topology. We write

- $M(E)$ for the real-valued Borel measurable functions on E;
- $B(E) \subset M(E)$ for the bounded ones, with $\|f\| = \sup_{x \in E} |f(x)|$;
- $C(E)$ for the continuous functions, $C_b(E) = C(E) \cap B(E)$, and, for E locally compact, $\hat{C}(E)$ for the continuous functions vanishing at infinity;
- $\mathcal{P}(E)$ for the Borel probability measures on E and $V(E)$ for the finite signed Borel measures on E;
- $D_E[0, \infty)$ for the space of càdlàg functions $[0, \infty) \to E$ and $C_E[0, \infty)$ for the continuous ones.

L always denotes a real Banach space with norm $\|\cdot\|$.

An *operator* is a subset $A \subset M(E) \times M(E)$; it need not be single-valued nor linear. Its domain is $\mathcal{D}(A) = \{f : (f, g) \in A \text{ for some } g\}$, its range is $\mathcal{R}(A) = \{g : (f, g) \in A \text{ for some } f\}$, and A_S denotes its linear span inside $M(E) \times M(E)$. For $\lambda \in \mathbb{R}$ we write $\lambda - A = \{(f, \lambda f - g) : (f, g) \in A\}$.

Throughout, references of the form "[4], Theorem 3.9.1" mean Chapter 3, Section 9, item 1 of [4].

2.2 The time index and the clock

[4] and [2] both index time by $\mathbb{R}_+$. Almost nothing in this manuscript needs that, and the parts that do need it need it for visibly different reasons: Section 4 needs a *topological* property of the index, Sections 5 and 6 an *algebraic* one. We therefore fix an abstract index $\mathbb{T}$ and record, for each statement, the weakest bundle of assumptions it uses. Two motives: it separates the topological from the algebraic hypotheses, which the classical formulation conflates; and it matches Mathlib, which indexes `Filtration`, `Martingale` and `IsStoppingTime` by an arbitrary `[Preorder ι]`, so that committing to $\mathbb{R}_+$ at the outset would mean working against the library.

Structure on the index

Definition 2.1. We use the following bundles of assumptions on $\mathbb{T}$.

- **(T0)** $\mathbb{T}$ is a preordered set with a least element 0.
- **(T1)** (T0), and $\mathbb{T}$ is a directed lattice: $s \wedge t$ and $s \vee t$ exist.
- **(T1′)** (T1) in the concrete form of [4], Section 2.8: a metric lattice whose order intervals are separable from above.
- **(T2a)** (T0) and $\mathbb{T}$ is linearly ordered.
- **(T2b)** (T2a), $\mathbb{T}$ carries the order topology and admits a countable dense subset D, and every non-maximal $t \in \mathbb{T}$ is a limit from the right: $D \cap (t, u) \neq \emptyset$ whenever $t < u$. A maximal element is allowed — $\mathbb{T} = [0, T]$ satisfies (T2b) — and is the one point at which a right limit is not available.
- **(T3)** $\mathbb{T} = [0, \infty)$ or $\mathbb{T} = [0, T]$.
- **(T4)** (T0) and $\mathbb{T}$ is a cancellative ordered commutative monoid: $r + \cdot$ is order preserving, and $t - s$ exists for $s \leq t$.

Three remarks on the design of Definition 2.1.

(T4) is independent of (T2a)–(T3). The monoid structure is what the shift $X(r + \cdot)$ needs; it presupposes no linear order. Keeping it separate makes visible that Sections 5 and 6 rest on an algebraic property of $\mathbb{T}$ whereas Sections 4 and 7 rest on a topological one.

Why (T2a) and (T2b) are separate. The uniqueness half of Theorem 5.16 and the whole of Section 6 need the order to be linear but use no topology at all; only Section 4 and the concrete convergence results need the topology. Bundling the two, as one is tempted to do, hides that.

(T1′) is not used in this manuscript. It is listed because it is the only place where [4] themselves treat a general index set, and because it is easy to misremember what it is for; Remark 2.6 sets that out. But every statement here that involves a stopping time — the strong Markov property of Theorem 5.19, and the localized statements of Remarks 5.45, 4.7 and 7.31 — already assumes (T2b), hence a linear order, for which the lattice structure is automatic. (T1′) would become relevant only for a strong Markov property on a genuinely

directed index, and Remark 5.18 shows that such a statement is blocked earlier, on the uniqueness side.

Lemma 2.2 (Shifts are order embeddings; (T0) + (T4))**.** *Assume (T4). Then for every $r \in \mathbb{T}$ the map $r + \cdot$ is an order embedding: $a \le b \iff r + a \le r + b$. If moreover (T2a) holds, then $r + (a \wedge b) = (r + a) \wedge (r + b)$.*

Proof. "$\Rightarrow$" is order preservation. For "$\Leftarrow$" let $r + a \le r + b$ and put $c = (r + b) - (r + a)$, which exists by (T4). Then $(r + a) + c = r + b$, and cancelling r gives $a + c = b$. Since $0 \le c$ and $a + \cdot$ is order preserving, $a = a + 0 \le a + c = b$. The second assertion is immediate: under (T2a) the meet is the minimum, and an injective order preserving map between linear orders carries minima to minima. $\square$

The clock

The index set alone does not carry a martingale problem. The test processes of Definition 3.5 have the form $Y_t = f(X_t) - C_t^{f,g}$, and what distinguishes the compensator of a *generator* from an arbitrary adapted process is that it is additive in time and generated pointwise from g. That is what carries Proposition 3.8, and without a finite-dimensional characterization the phrase "martingale problem for the operator A" has no content, since every process solves the martingale problem for *some* family of test processes, namely the family of its own martingales.

Definition 2.3 (Clock)**.** A *clock* on $\mathbb{T}$ is a σ-field $\mathcal{T}$ on $\mathbb{T}$ for which the down-sets

$$\mathbb{T}_{\le t} = \{u \in \mathbb{T} : u \le t\}, \qquad \mathbb{T}_{<t} = \mathbb{T}_{\le t} \setminus \{u \in \mathbb{T} : t \le u\}$$

are measurable for every t, together with a measure q on $(\mathbb{T}, \mathcal{T})$ such that $q(\mathbb{T}_{\le t}) < \infty$ for every t. For $s \le t$ we write

$$(s,t] \;\coloneqq\; \mathbb{T}_{\le t} \setminus \mathbb{T}_{\le s}, \qquad [s,t) \;\coloneqq\; \mathbb{T}_{<t} \setminus \mathbb{T}_{<s}, \tag{2}$$

and call these the *optional* and the *predictable* interval. Both $t \mapsto \mathbb{T}_{\le t}$ and $t \mapsto \mathbb{T}_{<t}$ are monotone, so both intervals are additive:

$$(s,u] = (s,t] \uplus (t,u], \qquad [s,u) = [s,t) \uplus [t,u), \qquad (s \le t \le u). \tag{3}$$

We write $\iota \in \{\mathrm{o}, \mathrm{p}\}$ for the choice of convention and $\langle s,t \rangle_\iota$ for $(s,t]$ resp. $[s,t)$.

The clock is *shift invariant* if $\mathbb{T}$ satisfies (T4) and $q(r + B) = q(B)$ for all $r \in \mathbb{T}$ and $B \in \mathcal{T}$; it is *reflective* if in addition $\mathbb{T}$ satisfies (T2a); and it is *atomless* if $q(\{u : t \le u \le t\}) = 0$ for every t, which is exactly the condition under which the two conventions coincide.

The additivity in (2) is not an assumption: it follows from transitivity of $\le$, which is why a preorder suffices. This is the point at which a measure is genuinely needed rather than an additive interval function $q_{s,u} = q_{s,t} + q_{t,u}$: the proof of Proposition 3.8 uses Fubini in the form

$$E\Big[\int_{(s,t]} g(X_u)\, q(\mathrm{d}u) \cdot Z\Big] \;=\; \int_{(s,t]} E\big[g(X_u) Z\big]\, q(\mathrm{d}u),$$

and additivity along chains does not give that. On $\mathbb{T} = \mathbb{R}_+^d$, for instance, chain additivity is strictly weaker than d-monotonicity, and only the latter produces a measure on boxes.

Example 2.4. (i) $\mathbb{T} = \mathbb{R}_+$, q Lebesgue measure: $\mathbb{T}_{\le t} = [0,t]$ and $C_t = \int_0^t g(X_u)\,\mathrm{d}u$. This is [4] and [2] throughout.

(ii) $\mathbb{T} = \mathbb{N}_0$, q counting measure: $C_n = \sum_{0<k\le n} g(X_k)$, discrete time.
(iii) $\mathbb{T} = \mathbb{R}_+$, q with atoms: processes with fixed times of discontinuity, the subject of [2], Section 5.3.
(iv) $\mathbb{T} = \mathbb{R}_+^d$, q Lebesgue measure: $\mathbb{T}_{\le t}$ is the box $[0,t]$ and $C_t = \int_{[0,t]} g(X_u)\,\mathrm{d}u$, the multiparameter martingale problem.

All four are clocks in the sense of Definition 2.3; (i), (ii) and (iv) are shift invariant, and only (i) and (ii) are reflective.

Remark 2.5 (Optional or predictable: both). The compensator of Definition 3.5 is $C_t = \int_{\langle 0,t\rangle_\iota} g(X_u)\,q(\mathrm{d}u)$, and the two choices of ι genuinely differ. For $\iota = \mathrm{o}$ the compensator is right continuous; for $\iota = \mathrm{p}$ it is predictable and left continuous. They agree exactly when the clock is atomless, in particular in Example 2.4(i) — which is why the choice is invisible in [4] and [2], and why neither source has to make it.

We carry both. The reason is that neither serves the whole manuscript: Theorem 4.3 holds only for $\iota = \mathrm{o}$, because an atom of q at t must lie inside the compensator up to t for (R3) to hold, and it is that limit which makes the càdlàg process a *modification*; Proposition 6.3 is stated for $\iota = \mathrm{p}$, because with $\iota = \mathrm{o}$ the chain identity telescopes to $\gamma(t,0) - \gamma(0,t)$ instead of to 0; but that quantity vanishes as well (Remark 6.7), so the *conclusion* of Section 6 is convention-agnostic, as are Sections 3 and 5 and Lemma 6.1 itself, all of which use only the additivity (3). Section 2.9 records which ι each result is most naturally stated in.

Two special cases are worth keeping in view. For $\mathbb{T} = \mathbb{N}_0$ with counting measure, $\iota = \mathrm{p}$ gives $C_n = \sum_{k<n} g(X_k)$, the Doob decomposition, with $A = P - I$; $\iota = \mathrm{o}$ gives $C_n = \sum_{k\le n} g(X_k)$ and the condition $Pf - f = Pg$. For $\mathbb{T} = \mathbb{R}_+$ with Lebesgue measure the distinction is empty.

Remark 2.6 (Comparison with [4], Section 2.8). [4], Section 2.8 is the only part of [4] that works with a general index set, and it is worth recording exactly what it does, because three things about it are easy to get wrong.

(i) What [4] assume. Their index $\mathscr{I}$ is a *partially* ordered set — their (8.1)–(8.3) include antisymmetry — which is moreover a *metric lattice*: a metric space in which $u \wedge v$ and $u \vee v$ exist, are unique, and depend *continuously* on (u,v). Separability from above of the intervals is a further hypothesis, imposed where it is needed (their Propositions 8.4, 8.5(c) and Theorem 8.7), not globally. Our (T0) is weaker on one point: we do not assume antisymmetry, because nothing below uses it and Mathlib's `Preorder` does not have it.

(ii) The lattice is needed for $\vee$, not for $\wedge$. It is tempting to say that stopping times require $\tau_1 \wedge \tau_2$. The opposite is true. [4], Proposition 8.1(a) shows that $\max_{k\le n} \tau_k$ is a stopping time — $\{\max_k \tau_k \le u\} = \bigcap_k \{\tau_k \le u\}$ — whereas their Remark 8.3 warns that $\tau \wedge a$ *need not be* one. The reason is that in a lattice $x \wedge y \le u$ is strictly weaker than "$x \le u$ or $y \le u$": in $\mathbb{R}_+^2$ one has $(1,0)\wedge(0,1) = (0,0) \le (0,0)$ with neither factor below $(0,0)$. So $\{\tau \wedge a \le u\}$ is genuinely larger than $\{\tau \le u\} \cup \{a \le u\}$ and there is no reason for it to lie in $\mathcal{F}_u$. [4] therefore replace $\tau \wedge a$ by the truncation $\tau^a = \tau$ on $\{\tau \le a\}$ and $\tau^a = a$ otherwise, their (8.10). Nor does $\mathcal{F}_\tau$ require infima: their (8.6) is the usual $\{A : A \cap \{\tau \le u\} \in \mathcal{F}_u\}$, verbatim.

(iii) What it is for. [4] build Section 2.8 for their Chapter 6, and it is worth being precise about how it is used there. In their Theorem 6.3.4 the multiparameter index carries the *filtration* $\{\mathcal{F}_u : u \in \mathbb{R}_+^k\}$ and a nondecreasing family of $\mathbb{R}_+^k$-valued stopping times $\tau(t)$; the processes themselves live in $\prod_k D_{E_k}[0,\infty)$, and so does the time-changed process $Z(t) = Y(\tau(t))$. That is, [4] never index a *process* by $\mathbb{R}_+^d$: the multiparameter index is a device inside a one-parameter theory, and what Section 2.8 delivers for it is optional sampling and nothing else. This is consistent with — indeed is indirect evidence

for — Remark 5.18: there is no multiparameter uniqueness or Markov theorem in [4] to measure ours against, because there is none to be had.

(iv) A conflation we avoid. [4] declare "we assume throughout this section that $\mathscr{I}$ is a metric lattice" before defining filtration, stopping time, $\mathcal{F}_\tau$, adaptedness and, in their (8.15), the martingale property itself — none of which uses the lattice, and all of which are correct on a bare preorder. The lattice is first used in their Proposition 8.1. This is exactly the conflation that Definition 2.1 is meant to remove, and it is reassuring rather than surprising: the entire martingale layer of this manuscript rests on [4] (8.15) alone, which is stated for a partially ordered index.

Notation for processes

For an E-valued measurable process X on $(\Omega, \mathcal{F}, P)$, indexed by $\mathbb{T}$, we write $\mathcal{F}_t^X = \sigma(X(s) : s \le t)$ and, following [4], (4.3.2),

$$ {}^*\mathcal{F}_t^X \;=\; \mathcal{F}_t^X \vee \sigma\Big(\int_{\langle 0,s\rangle_\iota} h(X(u))\, q(\,\mathrm{d}u) \;:\; s \le t,\; h \in B(E)\Big). \tag{4} $$

If X is progressively measurable — in particular if $\mathbb{T}$ satisfies (T2b) and X is right continuous — then ${}^*\mathcal{F}_t^X = \mathcal{F}_t^X$. Only (T0) and a clock are needed for (4) to make sense.

2.3 The state space

[4] and [2] take the state space E to be Polish throughout. As with the time index, almost nothing needs that, and the parts that do need it need it for visibly different reasons. We therefore separate them.

Definition 2.7. We use the following bundles of assumptions on the state space.

(E0) $(E, \mathcal{E})$ is a measurable space.

(E1) (E0), and $(E, \mathcal{E})$ is *standard Borel*: $\mathcal{E}$ is the Borel σ-field of some Polish topology on E, equivalently $(E, \mathcal{E})$ is Borel isomorphic to a Borel subset of a Polish space.

(E2) (E1), and E carries a separable metrizable topology with $\mathcal{E} = \mathcal{B}(E)$.

(E3) E is Polish.

Thus (E3) $\Rightarrow$ (E2) $\Rightarrow$ (E1) $\Rightarrow$ (E0), and both gaps are inhabited: a Borel non-G_δ subset of $\mathbb{R}$ satisfies (E2) but not (E3), and the space $\mathcal{S}'(\mathbb{R}^d)$ of tempered distributions satisfies (E1) but not (E2), being a Lusin space and hence standard Borel while not metrizable.

Remark 2.8 (What each bundle is for). The four bundles correspond to four different uses, and Section 2.9 records which result makes which.

(i) *(E0): nothing but measurability.* Sections 3 and 6, and the whole of Section 5.1 apart from Lemma 5.3, mention E only through $B(E)$ and $\mathcal{E}$. [2] already state their duality theorem for measurable E_1, E_2; the observation here is that uniqueness and the Markov property are of the same kind.

(ii) *(E1): regular conditional distributions, and the Borel structure.* Five items need it: Lemma 5.3 and its local counterpart Lemma 5.24(c); Fact 2.46; the identification of a law on F by its finite-dimensional distributions; the measurability of the event $\{a_T\beta = a_T\alpha\}$ in Definition 5.35, which needs $\mathcal{E}$ countably generated and separating; and the Kolmogorov extension of Fact 2.45, used in Theorem 7.5. Each is a statement about the Borel structure, not about the topology, and standard Borel is what makes all five available.

(iii) *(E2): separating classes and compact sets.* Section 4 needs a topology: the compact containment condition (29) refers to compact subsets of E, and the two hypotheses on $\mathcal{D}(A)$ refer to $C_b(E)$ separating points and separating measures.

(iv) *(E3): weak convergence.* Only Section 7, and there only because $D_E[0,\infty)$ has to be Polish for Prohorov's theorem and the Skorokhod representation.

Remark 2.9 (The point of (E1)). The gap between (E1) and (E2) is not academic. Martingale problems for distribution-valued processes — the martingale problem formulation of an SPDE, with $E = \mathcal{S}'(\mathbb{R}^d)$ or $E = \mathcal{D}'(\mathbb{R}^d)$ — live on a state space that is standard Borel but not metrizable. Under (E0) and (E1), Sections 3, 5 and 6 apply to them verbatim, with no new proof. Sections 4 and 7 do not, and the repair is a genuine piece of work rather than a change of hypotheses: one needs a substitute for $D_E[0,\infty)$ and for tightness in it, which for nuclear duals is Mitoma's theorem. That is out of scope here, and is flagged rather than attempted.

2.4 From Chapter 1: operator semigroups

Definition 2.10 ([4], Section 1.1)**.** A one-parameter family $\{T(t) : t \geq 0\}$ of bounded linear operators on L is a *semigroup* if $T(0) = I$ and $T(s+t) = T(s)T(t)$ for all $s, t \geq 0$. It is *strongly continuous* if $\lim_{t\to 0} T(t)f = f$ for every $f \in L$, and a *contraction semigroup* if $\|T(t)\| \leq 1$ for all $t \geq 0$.

A (possibly unbounded) *linear operator* A on L is a linear map whose domain $\mathcal{D}(A)$ is a subspace of L and whose range $\mathcal{R}(A)$ lies in L. The *(infinitesimal) generator* of a semigroup $\{T(t)\}$ on L is the linear operator A defined by

$$Af \;=\; \lim_{t\to 0} \tfrac{1}{t}\big(T(t)f - f\big), \tag{5}$$

with $\mathcal{D}(A)$ the set of $f \in L$ for which the limit exists.

Definition 2.11 ([4], Section 1.2)**.** A linear operator A on L is *dissipative* if $\|\lambda f - Af\| \geq \lambda\|f\|$ for every $f \in \mathcal{D}(A)$ and $\lambda > 0$. For $A \subset L \times L$ linear, A is dissipative if $\|\lambda f - g\| \geq \lambda\|f\|$ for all $(f, g) \in A$ and $\lambda > 0$.

Definition 2.12 ([4], Section 1.5)**.** Let L be a function space of the type considered in [4], Section 1.5 (for instance $L = B(E)$ with the sup norm). A semigroup $\{T(t)\}$ on L is *measurable* if $t \mapsto T(t)f$ is measurable for each $f \in L$. The *full generator* of a measurable contraction semigroup $\{T(t)\}$ on L is the multivalued operator

$$\hat{A} \;=\; \Big\{(f, g) \in L \times L \;:\; T(t)f - f = \int_0^t T(s)g\,\mathrm{d}s \text{ for all } t \geq 0\Big\}. \tag{6}$$

Fact 2.13 ([4], Proposition 1.5.1). The full generator $\hat{A}$ of a measurable contraction semigroup $\{T(t)\}$ on L is linear and dissipative, and $(\lambda - \hat{A})^{-1}h = \int_0^\infty e^{-\lambda t}T(t)h\,\mathrm{d}t$ for all $h \in \mathcal{R}(\lambda - \hat{A})$ and $\lambda > 0$.

Remark 2.14. Nothing further from [4], Chapter 1 is used below. In particular the Hille–Yosida theorem, cores, multivalued dissipative operators and the exponential formula ([4], Theorems 1.2.6 and 1.4.3, Propositions 1.3.1 and 1.3.3, Corollary 1.6.8) are *not* needed. They enter only through the semigroup-based uniqueness criteria [4], Theorem 4.4.1 and Corollary 4.4.4, which are deliberately not part of this manuscript: uniqueness is obtained here from Theorem 5.43 together with duality (Corollary 6.18), which requires no semigroup theory at all. Should those criteria be wanted later, this subsection is the place to extend.

2.5 From Chapter 2: stochastic processes and martingales

Definition 2.15 ([4], Section 2.1; (T0), except where noted)**.** Let $\mathbb{T}$ carry a clock (Definition 2.3), with σ-field $\mathcal{T}$. An E-valued process X on $(\Omega, \mathcal{F}, P)$ is *measurable* if $X : \mathbb{T}\times\Omega \to E$ is $\mathcal{T}\otimes\mathcal{F}$-measurable. A *filtration* is an increasing family $(\mathcal{F}_t)_{t\in\mathbb{T}}$ of sub-σ-fields of $\mathcal{F}$. X is $(\mathcal{F}_t)$*-adapted* if $X(t)$ is $\mathcal{F}_t$-measurable for each t, and $(\mathcal{F}_t)$*-progressive* if for each $t \in \mathbb{T}$ the restriction of X to $\mathbb{T}_{\le t} \times \Omega$ is $(\mathcal{T} \cap \mathbb{T}_{\le t}) \otimes \mathcal{F}_t$-measurable.

Y is a *version* of X if the two have the same finite-dimensional distributions; a *modification* of X if $P\{X(t) = Y(t)\} = 1$ for each $t \in \mathbb{T}$; and *indistinguishable* from X if $P\{X(t) = Y(t) \text{ for all } t \in \mathbb{T}\} = 1$.

A $\mathbb{T}\cup\{\infty\}$-valued random variable τ is an $(\mathcal{F}_t)$*-stopping time* if $\{\tau \le t\} \in \mathcal{F}_t$ for every $t \in \mathbb{T}$, and then $\mathcal{F}_\tau = \{F \in \mathcal{F} : F \cap \{\tau \le t\} \in \mathcal{F}_t\ \forall t \in \mathbb{T}\}$.

Under (T2b) one says in addition that $(\mathcal{F}_t)$ is *right continuous* if $\mathcal{F}_t = \mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$ for all t, and that X is *right continuous* if its paths are; every right continuous adapted process is then progressive. All of the above except this last paragraph needs only (T0), and is exactly [4] (8.1)–(8.6) of their Section 2.8; see Remark 2.6.

Fact 2.16 ([4], Propositions 2.1.2 and 2.1.4; (T2b)). Assume (T2b). If $\tau, \tau_1, \tau_2, \ldots$ are $(\mathcal{F}_t)$-stopping times and $c \in \mathbb{T}$, then $\tau_1 + c$, $\tau_1 \wedge c$, $\sup_n \tau_n$ and $\min_{k\le n} \tau_k$ are $(\mathcal{F}_t)$-stopping times; if $(\mathcal{F}_t)$ is right continuous, so are $\inf_n \tau_n$, $\underline{\lim}_n \tau_n$ and $\overline{\lim}_n \tau_n$. If X is $(\mathcal{F}_t)$-progressive and τ is an $(\mathcal{F}_t)$-stopping time, then $X(\tau)$ is $\mathcal{F}_\tau$-measurable on $\{\tau < \infty\}$ and $X(\cdot \wedge \tau)$ is $(\mathcal{F}_t)$-progressive. The linear order is used throughout: on a merely directed index $\tau_1 \wedge \tau_2$ need not be a stopping time at all (Remark 2.6(ii)).

Definition 2.17 ([4], Section 2.2 and (8.15); (T0))**.** A real-valued $(\mathcal{F}_t)$-adapted process X with $E[|X(t)|] < \infty$ for all t is an $(\mathcal{F}_t)$*-martingale* if $E[X(t) \mid \mathcal{F}_s] = X(s)$ a.s. for all $s \le t$ in $\mathbb{T}$, and an $(\mathcal{F}_t)$*-submartingale* if $E[X(t) \mid \mathcal{F}_s] \ge X(s)$ a.s. This is [4] (8.15) verbatim, and it is the whole of what the martingale layer of this manuscript uses; it is also `MeasureTheory.Martingale` of Mathlib, which is stated for `[Preorder ι]`.

Fact 2.18 (Regularization along a dense set; [4], Lemma 2.2.8; (T2b)). Let (E, r) be a metric space, $F \subset \mathbb{T}$ dense, and $x : F \to E$. If $y(t) := \lim_{s\to t+,\, s\in F} x(s)$ exists for every $t \in \mathbb{T}$, then y is right continuous; if $y^-(t) := \lim_{s\to t-,\, s\in F} x(s)$ exists for every $t > 0$, then y^- is left continuous on $\mathbb{T} \setminus \{0\}$; and if both exist for every $t > 0$, then $y^-(t) = y(t-)$ for all $t > 0$. In particular y is càdlàg, and $y \in D_E[0, \infty)$ when $\mathbb{T}$ satisfies (T3).

Fact 2.19 (Submartingale regularization; [4], Proposition 2.2.9; (T2b)). Let X be a real-valued $(\mathcal{F}_t)$-submartingale and let $F \subset \mathbb{T}$ be countable and dense. Then there is Ω_0 with $P(\Omega_0) = 1$ such that for every $\omega \in \Omega_0$ the limits

$$Y(t,\omega) = \lim_{s\to t+,\, s\in F} X(s,\omega) \quad (t \in \mathbb{T}), \qquad Y^-(t,\omega) = \lim_{s\to t-,\, s\in F} X(s,\omega) \quad (t > 0) \tag{7}$$

exist. Moreover $\Gamma = \{t : P\{Y(t) \ne Y(t-)\} > 0\}$ is countable, $P\{X(t) = Y(t)\} = 1$ for $t \notin \Gamma$, and setting $\tilde{X}(t) = Y(t)$ for $t \notin \Gamma$ and $\tilde{X}(t) = X(t)$ for $t \in \Gamma$ defines a modification of X almost all of whose sample paths have right and left limits at all $t \in \mathbb{T}$ and are right continuous at all $t \notin \Gamma$.

Fact 2.20 (Optional sampling; [4], Theorem 2.2.13, Remark 2.2.14, and Chapter 2, Problem 11; (T2b)). Let X be a right continuous $(\mathcal{F}_t)$-submartingale and let τ_1, τ_2 be $(\mathcal{F}_t)$-stopping times. Then for each $T > 0$,

$$E\big[X(\tau_2 \wedge T) \bigm| \mathcal{F}_{\tau_1}\big] \;\ge\; X(\tau_1 \wedge \tau_2 \wedge T). \tag{8}$$

If in addition τ_2 is a.s. finite, $E[|X(\tau_2)|] < \infty$ and $\lim_{T\to\infty} E[|X(T)|\mathbf{1}_{\{\tau_2>T\}}] = 0$, then $E[X(\tau_2) \mid \mathcal{F}_{\tau_1}] \geq X(\tau_1 \wedge \tau_2)$. Equality holds throughout if X is a martingale. In particular, a right continuous martingale with bounded increments satisfies the optional sampling identity at arbitrary a.s. finite stopping times.

Fact 2.21 (Doob's inequalities; [4], Corollary 2.2.17; (T2b)). Let X be a right continuous martingale. Then for $x > 0$, $T > 0$ and $\alpha > 1$,

$$P\Big\{\sup_{t\leq T} |X(t)| \geq x\Big\} \leq x^{-1} E[|X(T)|], \qquad E\Big[\sup_{t\leq T} |X(t)|^\alpha\Big] \leq \Big(\tfrac{\alpha}{\alpha-1}\Big)^\alpha E[|X(T)|^\alpha]. \tag{9}$$

Definition 2.22 ([4], Section 2.3)**.** A real-valued process X is an $(\mathcal{F}_t)$*-local martingale* if there exist $(\mathcal{F}_t)$-stopping times $\tau_1 \leq \tau_2 \leq \cdots$ with $\tau_n \to \infty$ a.s. such that $X^{\tau_n} = X(\cdot \wedge \tau_n)$ is an $(\mathcal{F}_t)$-martingale for each n. The sequence (τ_n) is a *localizing sequence.* Local submartingales are defined analogously.

Fact 2.23 ([4], Proposition 2.3.1). If X is a right continuous $(\mathcal{F}_t)$-local martingale and τ is an $(\mathcal{F}_t)$-stopping time, then $X^\tau = X(\cdot \wedge \tau)$ is an $(\mathcal{F}_t)$-local martingale.

2.6 From Chapter 3: weak convergence and the Skorokhod space

Everything in this subsection assumes (T3), $\mathbb{T} = [0, \infty)$, and is stated for it without further comment: the Skorokhod topology is defined through time changes $\lambda : [0, \infty) \to [0, \infty)$ (Definition 2.36) and has no counterpart on a general index. This is the one block of prerequisites that does not abstract, and Section 2.9 records which results inherit the restriction.

Throughout this subsection (S, d) is a metric space.

Definition 2.24 ([4], Section 3.3)**.** $P_n \Rightarrow P$ in $\mathcal{P}(S)$ (*weak convergence*) if $\int f \, \mathrm{d}P_n \to \int f \, \mathrm{d}P$ for every $f \in C_b(S)$. S-valued random variables X_n *converge in distribution* to X, written $X_n \Rightarrow X$, if $PX_n^{-1} \Rightarrow PX^{-1}$.

Fact 2.25 (Portmanteau; [4], Theorem 3.3.1). For $\{P_n\} \subset \mathcal{P}(S)$ and $P \in \mathcal{P}(S)$, conditions (b)–(f) below are equivalent and are implied by (a); if S is separable, all six are equivalent.

(a) $\rho(P_n, P) \to 0$, where ρ is the Prohorov metric;
(b) $P_n \Rightarrow P$;
(c) $\int f \, \mathrm{d}P_n \to \int f \, \mathrm{d}P$ for all uniformly continuous $f \in C_b(S)$;
(d) $\overline{\lim}_n P_n(F) \leq P(F)$ for all closed $F \subset S$;
(e) $\underline{\lim}_n P_n(G) \geq P(G)$ for all open $G \subset S$;
(f) $P_n(A) \to P(A)$ for all P-continuity sets $A \subset S$, i.e. all $A \in \mathcal{B}(S)$ with $P(\partial A) = 0$.

Fact 2.26 (Continuous mapping theorem; [4], Corollary 3.1.9 and Corollary 3.3.3). Let (S, d) and (S', d') be separable metric spaces and $h : S \to S'$ Borel measurable. Let C_h be the set of points at which h is continuous. If $P_n \Rightarrow P$ and $P(C_h) = 1$, then $P_n h^{-1} \Rightarrow P h^{-1}$. Moreover, if $X_n \Rightarrow X$ and $d(X_n, Y_n) \to 0$ in probability, then $Y_n \Rightarrow X$.

Fact 2.27 ([4], Theorems 3.1.7 and 3.1.8). If S is separable, then $\mathcal{P}(S)$ is separable; if (S, d) is complete and separable, then so is $\mathcal{P}(S)$ with the Prohorov metric. Moreover (*Skorokhod representation*), if S is separable and $P_n \Rightarrow P$, there are S-valued random variables X_n, X on a common probability space with laws P_n, P and $X_n \to X$ a.s.

Fact 2.28 (Prohorov; [4], Lemma 3.2.1 and Theorem 3.2.2). Call $\mathcal{M} \subset \mathcal{P}(S)$ *tight* if for every $\varepsilon > 0$ there is a compact $K \subset S$ with $\inf_{P\in\mathcal{M}} P(K) \geq 1 - \varepsilon$. If (S, d) is complete and separable, then every $P \in \mathcal{P}(S)$ is tight, and $\mathcal{M} \subset \mathcal{P}(S)$ is tight if and only if it is relatively compact.

Definition 2.29 ([4], Section 3.4 and Appendix 3)**.** A sequence $\{f_n\} \subset B(S)$ *converges boundedly and pointwise* to $f \in B(S)$, written bp-$\lim_n f_n = f$, if $\sup_n \|f_n\| < \infty$ and $f_n(x) \to f(x)$ for every $x \in S$. A set $M \subset B(S)$ is *bp-closed* if it is closed under bp-limits of sequences, and the *bp-closure* of M is the smallest bp-closed set containing it. The same terminology is used for subsets of $B(S) \times B(S)$, applying bp-lim in each coordinate.

Fact 2.30 ([4], Lemma 3.4.1, Proposition 3.4.2, and Appendix 3, Proposition 3.1)*.* The bp-closure of a subspace of $B(S)$ is a subspace.

Remark 2.31. [4] prove more: $C_b(S)$ is bp-dense in $B(S)$; if S is separable there is a sequence of nonnegative functions in $C_b(S)$ whose linear span is bp-dense in $B(S)$; and on bounded sequences bp-convergence coincides with convergence in the weak* topology induced by $V(S)$. None of this is used below. Fact 2.30 is invoked at exactly one place, Corollary 3.13, and there only in the form stated; see Remark 3.14 for why even that is optional.

Definition 2.32 ([4], Section 3.4)**.** $M \subset C_b(S)$ is *separating* if, for $P, Q \in \mathcal{P}(S)$, $\int f \,\mathrm{d}P = \int f \,\mathrm{d}Q$ for all $f \in M$ implies $P = Q$; it is *convergence determining* if, for $\{P_n\} \subset \mathcal{P}(S)$ and $P \in \mathcal{P}(S)$, $\int f \,\mathrm{d}P_n \to \int f \,\mathrm{d}P$ for all $f \in M$ implies $P_n \Rightarrow P$. M *separates points* if for $x \neq y$ there is $f \in M$ with $f(x) \neq f(y)$, and *strongly separates points* if for every $x \in S$ and $\delta > 0$ there are $h_1, \ldots, h_k \in M$ with $\inf_{y: d(y,x) \geq \delta} \max_{i \leq k} |h_i(y) - h_i(x)| > 0$. Every convergence determining set is separating.

Fact 2.33 ([4], Proposition 3.4.4)*.* If (S, d) is separable, then the set of uniformly continuous $f \in C_b(S)$ with bounded support is convergence determining. If S is in addition locally compact, then $C_c(S)$ is convergence determining.

Fact 2.34 (Stone–Weierstrass for separating classes; [4], Theorem 3.4.5)*.* Let (S, d) be complete and separable and let $M \subset C_b(S)$ be an algebra. If M separates points, then M is separating. If M strongly separates points, then M is convergence determining.

Fact 2.35 ([4], Proposition 3.4.6 and Proposition 3.7.1)*.* Let (S_k, d_k), $k \geq 1$, be separable and $S = \prod_{k \geq 1} S_k$. If $M_k \subset C_b(S_k)$ are separating, then

$$M = \Big\{ x \mapsto \textstyle\prod_{k=1}^n f_k(x_k) \ : \ n \geq 1, \ f_k \in M_k \cup \{1\} \Big\} \tag{10}$$

is separating on S; if the (S_k, d_k) are complete and separable and the M_k are convergence determining, so is M. In particular the finite-dimensional distributions of a process determine its law. Moreover, for $\pi_t : D_E[0,\infty) \to E$, $\pi_t(x) = x(t)$, one has $\mathcal{B}(D_E[0,\infty)) = \sigma(\pi_t : t \geq 0) = \sigma(\pi_t : t \in D)$ for any dense $D \subset [0,\infty)$, provided E is separable; hence the finite-dimensional distributions of a process with sample paths in $D_E[0,\infty)$ determine its law on $D_E[0,\infty)$.

Definition 2.36 ([4], Section 3.5)**.** $D_E[0,\infty)$ is the space of right continuous $x : [0,\infty) \to E$ having left limits $x(t-)$ for all $t > 0$ (with $x(0-) = x(0)$). Let $q = r \wedge 1$, let Λ' be the set of strictly increasing continuous $\lambda : [0,\infty) \to [0,\infty)$ onto with $\lambda(0) = 0$, and let $\Lambda \subset \Lambda'$ be those λ which are Lipschitz with

$$\gamma(\lambda) \ = \ \operatorname*{ess\,sup}_{t \geq 0} |\log \lambda'(t)| \ = \ \sup_{s > t \geq 0} \left| \log \tfrac{\lambda(s) - \lambda(t)}{s - t} \right| \ < \ \infty. \tag{11}$$

The *Skorokhod (J_1) metric* on $D_E[0,\infty)$ is

$$d(x, y) \ = \ \inf_{\lambda \in \Lambda} \Big[\gamma(\lambda) \vee \int_0^\infty e^{-u} d(x, y, \lambda, u) \,\mathrm{d}u \Big], \tag{12}$$

where

$$d(x,y,\lambda,u) = \sup_{t\ge 0} q\big(x(t\wedge u), y(\lambda(t)\wedge u)\big). \tag{13}$$

Fact 2.37 ([4], Lemma 3.5.1 and Theorem 3.5.6). Every $x \in D_E[0,\infty)$ has at most countably many discontinuities. If E is separable, then $D_E[0,\infty)$ is separable; if (E,r) is complete, then $(D_E[0,\infty), d)$ is complete. In particular, $D_E[0,\infty)$ is Polish whenever E is Polish.

Fact 2.38 (Compact sets of $D_E[0,\infty)$; [4], Lemma 3.6.2 and Theorem 3.6.3). For $x \in D_E[0,\infty)$, $\delta > 0$ and $T > 0$ define the modulus

$$w'(x,\delta,T) \;=\; \inf_{\{t_i\}} \max_i \sup_{s,t\in[t_{i-1},t_i)} r\big(x(s),x(t)\big), \tag{14}$$

the infimum being over all partitions $0 = t_0 < t_1 < \cdots < t_n$ with $T \le t_n$ and $\min_i(t_i - t_{i-1}) > \delta$. Then $w'(x,\delta,T) \to 0$ as $\delta \to 0$, $w'(\cdot,\delta,T)$ is Borel measurable, and, for (E,r) complete, the closure of $A \subset D_E[0,\infty)$ is compact if and only if

(a) for every rational $t \ge 0$ there is a compact $\Gamma_t \subset E$ with $x(t) \in \Gamma_t$ for all $x \in A$, and
(b) $\lim_{\delta\to 0}\sup_{x\in A} w'(x,\delta,T) = 0$ for each $T > 0$.

Fact 2.39 ([4], Lemma 3.7.7). If X has sample paths in $D_E[0,\infty)$, then $D(X) \coloneqq \{t \ge 0 : P\{X(t) = X(t-)\} = 1\}$ has at most countable complement in $[0,\infty)$. In particular $D(X)$ is dense.

Fact 2.40 ([4], Theorem 3.7.8). Let E be separable and let X_n, X have sample paths in $D_E[0,\infty)$.

(a) If $X_n \Rightarrow X$ in $D_E[0,\infty)$, then $(X_n(t_1),\dots,X_n(t_k)) \Rightarrow (X(t_1),\dots,X(t_k))$ for every finite $\{t_1,\dots,t_k\} \subset D(X)$.
(b) If $\{X_n\}$ is relatively compact and there is a dense $D \subset [0,\infty)$ such that the convergence in (a) holds for every finite subset of D, then $X_n \Rightarrow X$ in $D_E[0,\infty)$.

Fact 2.41 (Relative compactness, I; [4], Theorem 3.9.1). Let (E,r) be complete and separable and let $\{X_n\}$ be a family of processes with sample paths in $D_E[0,\infty)$ satisfying the compact containment condition: for every $\eta, T > 0$ there is a compact $\Gamma_{\eta,T} \subset E$ with

$$\inf_n P\{X_n(t) \in \Gamma_{\eta,T} \text{ for } 0 \le t \le T\} \;\ge\; 1-\eta. \tag{15}$$

Let $H \subset C_b(E)$ be dense in the topology of uniform convergence on compact sets. Then $\{X_n\}$ is relatively compact if and only if $\{f\circ X_n\}$ is relatively compact in $D_{\mathbb{R}}[0,\infty)$ for each $f \in H$.

Fact 2.42 (Relative compactness, II; [4], Theorem 3.9.4). Let (E,r) be arbitrary and let X_n have sample paths in $D_E[0,\infty)$, adapted to filtrations $(\mathcal{F}^n_t)$. Let $\mathcal{L}_n$ be the Banach space of real-valued $(\mathcal{F}^n_t)$-progressive processes with $\|Y\| = \sup_{t\ge 0} E[|Y(t)|] < \infty$ and put

$$\mathcal{A}_n \;=\; \Big\{(Y,Z) \in \mathcal{L}_n\times\mathcal{L}_n : Y(t) - \int_0^t Z(s)\,\mathrm{d}s \text{ is an } (\mathcal{F}^n_t)\text{-martingale}\Big\}. \tag{16}$$

Let $C_a \subset C_b(E)$ be a subalgebra and let D be the collection of $f \in C_b(E)$ such that for every $\varepsilon, T > 0$ there is $(Y_n, Z_n) \in \mathcal{A}_n$ with

$$\sup_n E\Big[\sup_{t\in[0,T]\cap\mathbb{Q}} |Y_n(t) - f(X_n(t))|\Big] < \varepsilon, \qquad \sup_n E\big[\|Z_n\|_{p,T}\big] < \infty \text{ for some } p \in (1,\infty]. \tag{17}$$

If C_a is contained in the sup-norm closure of D, then $\{f\circ X_n\}$ is relatively compact for each $f \in C_a$; more generally $\{(f_1,\dots,f_k)\circ X_n\}$ is relatively compact in $D_{\mathbb{R}^k}[0,\infty)$ for all $f_1,\dots,f_k \in C_a$.

2.7 From the Appendixes of [4]

Fact 2.43 (Uniform integrability; [4], Appendix 2). A family $\{X_i\}$ of real-valued random variables is *uniformly integrable* if $\sup_i E[|X_i|] < \infty$ and for every $\varepsilon > 0$ there is $\delta > 0$ such that $P(A) < \delta$ implies $|E[X_i \mathbf{1}_A]| < \varepsilon$ for all i. Equivalently, $\lim_{N\to\infty} \sup_i E[|X_i| - |X_i| \wedge N] = 0$; equivalently, there is an increasing convex φ on $[0, \infty)$ with $\varphi(x)/x \to \infty$ and $\sup_i E[\varphi(|X_i|)] < \infty$. If $X_n \Rightarrow X$ and $\{X_n\}$ is uniformly integrable, then $E[X_n] \to E[X]$.

Fact 2.44 (Monotone class theorem; [4], Appendix 4). Let H be a linear space of bounded real-valued functions on Ω containing the constants and closed under bounded monotone limits, and let $K \subset H$ be closed under multiplication. Then H contains every bounded $\sigma(K)$-measurable function.

Fact 2.45 (Kolmogorov extension; [4], Theorem 4.1.1; (T0) + (E1)). Assume (E1), let $\nu \in \mathcal{P}(E)$ and let $(\mu_t)_{t\in\mathbb{T}}$ be a family of probability kernels on E with $\mu_0(x, \cdot) = \delta_x$, with $(t, x) \mapsto \mu_t(x, \Gamma)$ jointly measurable for every $\Gamma \in \mathcal{E}$, and with the Chapman–Kolmogorov property $\mu_{s+t}(x, \cdot) = \int \mu_s(x, \mathrm{d}x')\mu_t(x', \cdot)$ (for (T4)); more generally, let the finite-dimensional distributions built from ν and (μ_t) be a projective family. Then there is a probability measure on $E^{\mathbb{T}}$ under which the coordinate process is Markov with initial distribution ν and transition kernels (μ_t).

Standard Borel is what the theorem needs: on a general measurable space the projective limit may be empty.

The Mathlib situation as of `v4.33.1` is worth recording precisely, because it is easy to assume more than is there. Present are

- the predicate `IsProjectiveLimit`, with its uniqueness statement;
- the additive content `projectiveFamilyContent` built from a projective family;
- the cylinders with closed compact bases, on which the classical proof runs;
- the Ionescu–Tulcea theorem `Kernel.traj`, which needs no topological hypothesis but only a *sequential* index $\mathbb{N}$.

Absent is the Kolmogorov extension theorem itself, for an arbitrary index set under topological assumptions — the source file says so in as many words. So Fact 2.45 is a genuine prerequisite to be built, unless $\mathbb{T}$ is countable and can be enumerated, in which case Ionescu–Tulcea suffices.

Fact 2.46 (Conditional determination by separating sets; [4], Chapter 3, Problem 7). Assume (E1) and (E2), let $\mathcal{G} \subset \mathcal{F}$ be a sub-σ-field, let $M \subset C_b(E)$ be separating, and let U, V be E-valued random variables with V $\mathcal{G}$-measurable. If $E[f(U) \mid \mathcal{G}] = f(V)$ a.s. for every $f \in M$, then $P\{U = V\} = 1$.

Remark 2.47 ([4], Chapter 3, Problem 7 — a proof). This is stated as a *problem* in [4] and is used in the last step of Theorem 4.3, so we prove it. The point worth making is that the countability the argument needs comes from the *state space*, not from M: no countable subfamily of M is assumed, and none is available in general.

Step 1: M separating gives conditional equality in law. Let $G \in \mathcal{G}$ with $P(G) > 0$. For $f \in M$, the hypothesis gives $E[\mathbf{1}_G f(U)] = E[\mathbf{1}_G E[f(U) \mid \mathcal{G}]] = E[\mathbf{1}_G f(V)]$. The two probability measures

$$\nu_1(B) = \frac{P(\{U \in B\} \cap G)}{P(G)}, \qquad \nu_2(B) = \frac{P(\{V \in B\} \cap G)}{P(G)}, \qquad B \in \mathcal{E},$$

therefore satisfy $\int f \, \mathrm{d}\nu_1 = \int f \, \mathrm{d}\nu_2$ for every $f \in M$, and since M is separating, $\nu_1 = \nu_2$. Hence

$$P\big(\{U \in B\} \cap G\big) = P\big(\{V \in B\} \cap G\big) \qquad \text{for all } B \in \mathcal{E},\ G \in \mathcal{G}, \tag{18}$$

i.e. $P(U \in B \mid \mathcal{G}) = \mathbf{1}_B(V)$ a.s. for each fixed B.

Step 2: one null set, using (E1). By (E1) the σ-field $\mathcal{E}$ is countably generated; let $\mathcal{E}_0$ be a countable algebra generating it. Again by (E1) there is a regular conditional distribution $\mu(\omega, \cdot)$ of U given $\mathcal{G}$. Applying (18) to the countably many $B \in \mathcal{E}_0$ yields a single set of full measure off which $\mu(\omega, B) = \mathbf{1}_B(V(\omega))$ for all $B \in \mathcal{E}_0$ simultaneously. Two probability measures agreeing on a generating algebra agree, so $\mu(\omega, \cdot) = \delta_{V(\omega)}$ a.s.

Step 3. The diagonal $\{(x, \omega) : x = V(\omega)\}$ is $\mathcal{E}\otimes\mathcal{G}$-measurable, again because $\mathcal{E}$ is countably generated and separates points, so $P\{U = V\} = E\big[\mu(\cdot, \{V(\cdot)\})\big] = E\big[\delta_V(\{V\})\big] = 1$.

Only (E2) is used, in Step 1, to make sense of "$M \subset C_b(E)$ separating"; all the measure theory is (E1).

2.8 Where the prerequisites are used

Prerequisite	Used in
Definition 2.11	Proposition 3.18, Fact 2.13
Fact 2.13	Remark 3.19
Fact 2.16	Theorem 5.43(b), Corollary 6.16
Fact 2.18, 2.19	Theorem 4.3
Fact 2.20	Theorem 5.19
Fact 2.21	Remark 7.25 (via Fact 2.42)
Fact 2.23	Remarks 5.45, 7.31
Fact 2.25, 2.26	Lemma 7.23, Theorem 7.30
Fact 2.27, 2.28	Remark 7.25
Fact 2.30	Corollary 3.13 only
Fact 2.33	Remark 4.8
Fact 2.34	Remark 7.25, Proposition 7.7
Fact 2.45	Theorem 7.5
Fact 2.35	Theorem 5.16, Corollary 5.44, Example 3.4
Fact 2.37	Setting 3.1, Theorem 7.30
Fact 2.38	Fact 2.41
Fact 2.39	Lemma 7.23, Theorem 7.30
Fact 2.40	Remark 7.25
Fact 2.41, 2.42	Remark 7.25
Fact 2.43	Theorem 7.30
Fact 2.44	Proposition 3.8, Example 3.4
Fact 2.46	Theorem 4.3

2.9 Which result needs which bundle

The counterpart of Section 2.8 for Definitions 2.1, 2.3 and 2.7. "Clock" means Definition 2.3 with no further property; "reflective" is as defined there, and ι is the convention of Definition 2.3, shown only where it matters. Bold entries mark the places where the weakest bundle in its column does *not* suffice, for a mathematical reason rather than for convenience. An ι-entry means that the result is stated in that convention; only Section 4 *needs* its one — there an atom at t must lie inside the compensator up to t — whereas the $\iota = \mathrm{p}$ entries of the duality table mark the convention in which the proof is shortest, the conclusion being convention-agnostic by Remark 6.7. Example 7.36 needs $\iota = \mathrm{o}$ for a third reason, explained in Remark 7.37.

Result	Index	State	Clock
Definition 2.15, Definition 2.17	(T0)	(E0)	clock
Facts 2.16, 2.20, 2.21	(T2b)	(E0)	—
Facts 2.18, 2.19 (regularization)	(T2b)	(E2)	—
Section 2.6 entire (Skorokhod, weak convergence)	(T3)	(E3)	—

Result	Index	State	Clock
Setting 3.1, Definitions 3.2, 3.3	(T0)	(E0)	—
Definition 3.5, Proposition 3.8, Definition 3.15, (4)	(T0)	(E0)	clock
Lemma 3.16, Proposition 3.18	(T3)	(E0)	Lebesgue
Definition 3.20, Example 3.4	(T2b)	(E2)	—
Definition 4.2, Theorem 4.3 (càdlàg)	(T2b)	(E2)	$\boldsymbol{\iota} = \mathbf{o}$
Theorem 4.5 ([4] 4.3.6)	(T2b)	(E2)	clock, $\boldsymbol{\iota} = \mathbf{o}$
Lemma 5.2 (mixtures)	(T0)	(E0)	—
Lemma 5.3 (disintegration)	(T0)	**(E1)**	—
Lemma 2.2 ($r + \cdot$ is an order embedding)	(T0) + (T4)	—	—
Definition 5.10, Lemma 5.13 (restart)	(T0) + (T4)	(E0)	clock
Theorem 5.16(a) (Markov)	(T0) + (T4)	(E0)	clock
Definition 5.7, Proposition 5.9 (uniqueness)	**(T2a)**	(E0)	—
Lemma 5.15 (one-dimensional laws suffice)	(T0) + (T4)	(E0)	clock
Theorem 5.16(b) (uniqueness, via both)	**(T2a)** + (T4)	(E0)	clock
Theorem 5.19 (strong Markov)	(T2b) + (T4)	(E2)	clock
Definition 5.22, Lemmas 5.24, 5.26, 5.29, Theorem 5.30 (local theory; **(E1)** for Lemma 5.24(c))	(T2b) + (T4)	(E0)	clock
Definition 5.34, Definition 5.35, Lemma 5.36, Theorem 5.37 (local uniqueness)	(T2b)	(E0)	clock
Corollary 5.38 (its Markovian case)	(T2b) + (T4)	(E0)	clock

Result	Index	State	Clock
Lemma 6.1 (chain identity)	(T0)	(E0)	any
Proposition 6.3, Corollary 6.14	(T2a) + (T4)	(E0)	Haar, $\iota = \mathrm{p}$
Lemma 6.4, Theorem 6.5	**(T2b)**	(E0)	any, $\iota = \mathrm{p}$
Remark 6.7 (atomic duality)	(T0)	(E0)	atomic, any ι
Lemma 6.9, Theorem 6.10	(T3)	(E0)	Lebesgue
Corollary 6.18 (uniqueness via duality)	(T3)	(E2)	Lebesgue
Setting 7.2, Lemma 7.3, Proposition 7.4, Theorem 7.5 (existence from a dual)	(T0) + (T4)	**(E1)**	shift invariant
Proposition 7.7 (checking (D3))	(T0) + (T4)	(E3), E_1 compact	shift invariant
Setting 7.9, Theorem 7.12 (jump processes)	(T3)	(E0)	Lebesgue
Setting 7.50, Theorem 7.52, Example 7.55 (path-dependent rates, Hawkes)	(T3)	(E0)	Lebesgue
Theorem 7.57 (Hawkes to Volterra)	(T3)	(E3)	Lebesgue
Proposition 7.13, Remark 7.15	(T3)	(E2)	Lebesgue
Corollary 7.21 (from SDEs)	(T3)	$E = \mathbb{R}^d$	Lebesgue
Theorem 7.27 (abstract convergence)	(T0), or (T2b) for $D \subsetneq \mathbb{T}$	(E3)	—
Lemma 7.40, Theorem 7.41 (augmented form)	as Theorem 7.27	(E3)	—
Proposition 7.42 (atomic clocks)	(T3)	(E2)	any, atoms allowed
Theorem 7.34 (clock change)	(T3)	(E3)	q^n and q
Lemma 7.23, Theorem 7.30	(T3)	(E3)	Lebesgue

Two readings of this table are worth recording. First, the partially ordered case — and with it the multiparameter index $\mathbb{T} = \mathbb{R}^d_+$ — survives in Section 3, in the Markov half of Section 5, in the abstract convergence theorem and in Section 7.2, and dies everywhere else; the bold index entries are where it dies for a *mathematical* reason rather than for want of a Skorokhod space, and Remarks 5.18 and 6.8 say which — the second only for want of a proof, not because of a counterexample. Section 7.2 is the one *existence* result that survives a partially ordered index, which is worth noting because every other route in Section 7 is tied to $\mathbb{R}_+$. Second, (T1′) — the metric lattice of [4], Section 2.8, which the first draft of this manuscript treated as the base assumption — does not occur at all: every statement involving a stopping time already assumes (T2b), under which the lattice structure is automatic. See Remark 2.6.

3 Martingale problems

3.1 The abstract (local) martingale problem

Setting 3.1 (after [2], Section 3.1; (T0) + (E0))**.** Let $(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{t\in\mathbb{T}})$ be a filtered space carrying a family $\mathbb{X}$ of real-valued adapted processes (the *test processes*), right continuous if $\mathbb{T}$ satisfies (T2b). Let $(E,\mathcal{E})$ be a measurable space and let $F\subset E^{\mathbb{T}}$ be a *path space*: a set of maps $\mathbb{T}\to E$ equipped with a σ-field $\mathcal{S}$ for which the coordinates $\pi_t : F\to E$ are measurable and which they generate,

$$\mathcal{S} \;=\; \sigma\big(\pi_t \;:\; t\in\mathbb{T}\big). \tag{19}$$

Let $X=(X_t)_{t\in\mathbb{T}}$ be an E-valued process with $X(\omega)\in F$ for every ω and $\omega\mapsto X(\omega)$ measurable $\mathcal{F}/\mathcal{S}$.

Under (E3) we take F to carry a Polish topology and $\mathcal{S}=\mathcal{B}(F)$; (19) is then a theorem rather than an assumption for $F=D_E[0,\infty)$ (Fact 2.35), and it is what makes the finite-dimensional distributions determine the law.

Definition 3.2 ([2], Definition 3.2; (T0) + (E0))**.** A probability measure P on $(\Omega,\mathcal{F})$ is a *solution of the martingale problem* $\mathrm{MP}(\mathbb{X})$ if every $Y\in\mathbb{X}$ is an $(\mathbb{F},P)$-martingale, and a *solution of the local martingale problem* if every $Y\in\mathbb{X}$ is an $(\mathbb{F},P)$-local martingale. The corresponding sets of solutions are denoted $\mathcal{M}(\mathbb{X})$ and $\mathcal{M}_{\mathrm{loc}}(\mathbb{X})$.

Definition 3.3 ([2], Definitions 3.3 and 3.5; (T0) + (E0))**.** $\mathbb{X}$ is *canonical* for X if for every $t\in\mathbb{T}$ there is a set $\mathbb{X}_t^\circ$ of $\mathcal{B}(F)/\mathcal{B}(\mathbb{R})$-measurable maps $Y_t^\circ : F\to\mathbb{R}$ such that every $Y\in\mathbb{X}$ satisfies $Y_t=Y_t^\circ(X)$ for some $Y_t^\circ\in\mathbb{X}_t^\circ$. A family $\mathbb{Z}^\circ=\{\mathbb{Z}_t^\circ : t\in\mathbb{T}\}$ of sets of bounded measurable maps $F\to\mathbb{R}$ is *determining* for $\mathbb{X}$ if for every probability measure P, every $Y\in\mathbb{X}$ and all $s\le t$ with $Y_s,Y_t\in L^1(P)$,

$$E^P[Y_tZ_s^\circ(X)]=E^P[Y_sZ_s^\circ(X)]\quad\text{for all } Z_s^\circ\in\mathbb{Z}_s^\circ \qquad\Longrightarrow\qquad E^P[Y_t\mid\mathcal{F}_s]=Y_s\ \ P\text{-a.s.}$$

Example 3.4 ([2], Example 3.6(i); (T2b) + (E2))**.** If $\mathcal{F}_t=\sigma(X_s : s\le t)$, $F=D_E[0,\infty)$ or $F=C_E[0,\infty)$, and $D\subset\mathbb{T}$ is dense, then

$$\mathbb{Z}_t^\circ=\Big\{\textstyle\prod_{i=1}^n h_i(X_{t_i}) \;:\; n\in\mathbb{N},\ t_1,\dots,t_n\in D\cap[0,t],\ h_i\in C_b(E)\Big\}$$

is determining. This uses $\mathcal{B}(F)=\sigma(X_t : t\in\mathbb{T})$ (Fact 2.35) and the monotone class theorem.

3.2 The Markovian martingale problem

Definition 3.5 ((T0) + (E0) + clock)**.** Fix a clock q on $\mathbb{T}$ and a convention $\iota\in\{\mathrm{o},\mathrm{p}\}$ (Definition 2.3). Let $A\subset M(E)\times M(E)$ and let X be a jointly measurable E-valued process on $(\Omega,\mathcal{F},P)$. Put

$$\mathbb{X}_A^\iota(X) \;=\; \Big\{\big(f(X(t))-\int_{\langle 0,t\rangle_\iota} g(X(s))\,q(\mathrm{d}s)\big)_{t\in\mathbb{T}} \;:\; (f,g)\in A\Big\}, \tag{20}$$

and write $\mathbb{X}_A(X)$ when the convention is fixed by the context or immaterial. X is a *solution of the martingale problem for A* if every element of $\mathbb{X}_A(X)$ is a martingale with respect to $({}^*\mathcal{F}_t^X)_{t\in\mathbb{T}}$ of (4). If $(\mathcal{G}_t)$ is a filtration with $\mathcal{G}_t\supset{}^*\mathcal{F}_t^X$ for all t and every element of $\mathbb{X}_A(X)$ is a $(\mathcal{G}_t)$-martingale, X is a *solution of the martingale problem for A with respect to* $(\mathcal{G}_t)$. Given $\mu\in\mathcal{P}(E)$, X solves the martingale problem for (A,μ) if in addition $PX(0)^{-1}=\mu$. Replacing "martingale" by "local martingale" throughout defines the *local martingale problem* for A.

A probability measure $P\in\mathcal{P}(D_E[0,\infty))$ is called a solution of the $D_E[0,\infty)$ *martingale problem* for A, respectively for (A,μ), if the coordinate process $X(t,\omega)=\omega(t)$ on the space $(D_E[0,\infty),\mathcal{B}(D_E[0,\infty)),P)$ is one.

Definition 3.5 is the special case $\mathbb{X} = \mathbb{X}_A(X)$ of Definition 3.2.

Remark 3.6 (Provenance of Definition 3.2). The abstract formulation is older than [2]. [5], Definition III.1.3 already attaches a martingale problem to a family $\mathcal{X}$ of optional processes on a filtered space carrying no measure, and does so with *local* martingales as the default rather than as a variant. Two features of their formulation are worth importing. First, the initial condition is a measure $P_{\mathcal{H}}$ on an *initial σ-field* $\mathcal{H} \subset \mathcal{F}_0$, not merely an initial distribution $\mu = P\pi_0^{-1}$; everything below goes through with $\sigma(\pi_0)$ replaced by $\mathcal{H}$, and Lemma 5.3 in particular. Second, [5], Remark III.1.6 notes explicitly that the elements of $\mathcal{X}$ need be neither càdlàg nor even adapted — which is the observation that Setting 4.1 needs and that [2]'s standing assumptions exclude.

Before the characterization we record two measure-theoretic points that the proof uses and that are easy to pass over. Both concern the compensator alone.

Lemma 3.7 (The compensator is adapted, and Fubini is available; (T0) + (E0) + clock)**.** *Let X be jointly measurable, $g \in M(E)$, $\iota \in \{\mathrm{o}, \mathrm{p}\}$ and $u \in \mathbb{T}$, and write $I_u^g = \int_{\langle 0,u\rangle_\iota} g(X(v))\, q(\,\mathrm{d}v)$ on the set $N_u^g = \{\int_{\langle 0,u\rangle_\iota} |g(X(v))|\, q(\,\mathrm{d}v) < \infty\}$.*

(a) $N_u^g \in \mathcal{F}_u^X$ and I_u^g is ${}^*\mathcal{F}_u^X$-measurable on N_u^g. If $g \in B(E)$ then $N_u^g = \Omega$, I_u^g is one of the generators in (4), and $|I_u^g| \le \|g\|\, q(\mathbb{T}_{\le u})$.

(b) The restriction of q to $\mathbb{T}_{\le u}$ is a finite measure. Consequently finite products of such restrictions are well defined and Fubini's theorem is available on them, without any σ-finiteness assumption on q itself.

Proof. (a) For $g \in B(E)$ the bound is immediate from $q(\langle 0,u\rangle_\iota) \le q(\mathbb{T}_{\le u}) < \infty$ (Definition 2.3), and I_u^g appears in (4) by construction. For general g put $g_n = (g \wedge n) \vee (-n)$. Then $I_u^{|g_n|} \uparrow \int_{\langle 0,u\rangle_\iota} |g(X(v))| q(\,\mathrm{d}v)$ pointwise, so N_u^g is the set where that increasing limit is finite, hence lies in $\sigma(I_u^{|g_n|} : n) \subset {}^*\mathcal{F}_u^X$; and on N_u^g, $I_u^{g_n} \to I_u^g$ pointwise by dominated convergence, so I_u^g is a pointwise limit of ${}^*\mathcal{F}_u^X$-measurable functions.

(b) Immediate from $q(\mathbb{T}_{\le u}) < \infty$. Note that q itself need *not* be σ-finite — $\mathbb{T}$ need not be a countable union of down-sets — so the restriction is what makes the product measure legitimate. □

Proposition 3.8 ([4], (3.4); (T0) + (E0) + clock, and (E2) for the variant with $C_b(E)$)**.** *Let X be jointly measurable, fix $\iota \in \{\mathrm{o}, \mathrm{p}\}$, and let $(f, g) \in A$ be such that $Y_t^{f,g} = f(X(t)) - \int_{\langle 0,t\rangle_\iota} g(X(s))\, q(\,\mathrm{d}s)$ is defined and integrable for every $t \in \mathbb{T}$; by Lemma 3.7(a) it is then automatically ${}^*\mathcal{F}_t^X$-adapted. Then $Y^{f,g}$ is a $({}^*\mathcal{F}_t^X)$-martingale if and only if*

$$E\Big[\Big(f(X(t)) - f(X(s)) - \int_{\langle s,t\rangle_\iota} g(X(u))\, q(\,\mathrm{d}u)\Big) \prod_{k=1}^{n} h_k(X(t_k))\Big] \;=\; 0 \tag{21}$$

for all $s \le t$ in $\mathbb{T}$, all $n \ge 0$, all $t_1, \dots, t_n \in \mathbb{T}_{\le s}$ and all $h_1, \dots, h_n \in B(E)$ (equivalently, $h_k \in C_b(E)$). Consequently X solves the martingale problem for A if and only if (21) holds for every $(f, g) \in A$.

Remark 3.9 (The times need not form a chain). Earlier versions of this manuscript quantified (21) over *chains* $t_1 \le \dots \le t_n \le s$, following [4], (3.4), where $\mathbb{T} = \mathbb{R}_+$ is linearly ordered and the two formulations coincide. On a preordered index they do not, and the chain version is too weak: the proof below tests against the generators of $\mathcal{F}_s^X = \sigma(X_u : u \in \mathbb{T}_{\le s})$, and these are products over *arbitrary* finite subsets of $\mathbb{T}_{\le s}$. On $\mathbb{T} = \mathbb{R}_+^2$ the times $(1, 0)$

and $(0,1)$ lie in $\mathbb{T}_{\le(1,1)}$ without being comparable, and no chain reaches the product $h_1(X_{(1,0)})h_2(X_{(0,1)})$. This is the same phenomenon as in Remark 5.18, but with the opposite moral: here it is the *hypothesis* that has to be strengthened, and once it is, nothing is lost.

Proof. Throughout, $s \le t$ are fixed, $Y = Y^{f,g}$, and we abbreviate

$$I_u^h \;=\; \int_{\langle 0,u\rangle_\iota} h(X(v))\, q(\,\mathrm{d}v), \qquad u \in \mathbb{T},\ h \in B(E),$$

so that ${}^*\mathcal{F}_s^X = \sigma\big(X(u),\, I_u^h \;:\; u \in \mathbb{T}_{\le s},\, h \in B(E)\big)$ by (4). Note $|I_u^h| \le \|h\|\, q(\mathbb{T}_{\le u}) < \infty$, so each I_u^h is *bounded*, by Definition 2.3. By (3),

$$Y_t - Y_s \;=\; f(X(t)) - f(X(s)) - \int_{\langle s,t\rangle_\iota} g(X(u))\, q(\,\mathrm{d}u), \tag{22}$$

which is the bracket in (21); it lies in L^1 by hypothesis.

Necessity. Each $\prod_k h_k(X(t_k))$ with $t_k \le s$ is bounded and $\mathcal{F}_s^X \subset {}^*\mathcal{F}_s^X$-measurable, so the martingale property gives $E[(Y_t - Y_s)\prod_k h_k(X(t_k))] = 0$, which by (22) is (21).

Sufficiency. Assume (21) and put

$$\mathcal{H} \;=\; \big\{W : \Omega \to \mathbb{R} \text{ bounded},\ {}^*\mathcal{F}_s^X\text{-measurable},\ E[(Y_t - Y_s)W] = 0\big\}.$$

We must show that $\mathcal{H}$ contains *all* such W, for then $E[Y_t \mid {}^*\mathcal{F}_s^X] = Y_s$.

Step 1: $\mathcal{H}$ is a monotone vector space. It is a vector space because $W \mapsto E[(Y_t - Y_s)W]$ is linear, and it contains the constants by (21) with $n = 0$. If $W_m \in \mathcal{H}$, $0 \le W_m \uparrow W$ with W bounded, then $(Y_t - Y_s)W_m \to (Y_t - Y_s)W$ pointwise and $|(Y_t - Y_s)W_m| \le \|W\|\,|Y_t - Y_s| \in L^1$, so dominated convergence gives $W \in \mathcal{H}$.

Step 2: a multiplicative generating class. Let $\mathcal{K}$ be the set of finite products

$$Z \;=\; \prod_{k=1}^n h_k(X(t_k)) \cdot \prod_{j=1}^m I_{u_j}^{g_j}, \qquad n, m \ge 0,\ t_k, u_j \in \mathbb{T}_{\le s},\ h_k, g_j \in B(E). \tag{23}$$

Every such Z is bounded, by Lemma 3.7(a), and $\mathcal{K}$ is closed under multiplication, being a set of finite products of the two kinds of factor. It generates ${}^*\mathcal{F}_s^X$: the inclusion $\sigma(\mathcal{K}) \subset {}^*\mathcal{F}_s^X$ holds factor by factor, and the reverse holds because the cases $(n,m) = (1,0)$ and $(0,1)$ of (23) are the generators $h(X(t_1))$ and $I_{u_1}^{g_1}$ of (4) themselves.

Step 3: $\mathcal{K} \subset \mathcal{H}$. This is the only step with content, and it is Fubini rather than an approximation. Fix Z as in (23) and write the product of the m integrals as one integral over the product:

$$\prod_{j=1}^m I_{u_j}^{g_j} \;=\; \int_R \prod_{j=1}^m g_j(X(v_j))\, q^{\otimes m}(\,\mathrm{d}v), \qquad R \;=\; \langle 0,u_1\rangle_\iota \times\cdots\times \langle 0,u_m\rangle_\iota, \quad v = (v_1,\ldots,v_m),$$

The map $(v,\omega) \mapsto (Y_t - Y_s)(\omega)\prod_k h_k(X(t_k,\omega))\prod_j g_j(X(v_j,\omega))$ is $\mathcal{T}^{\otimes m} \otimes \mathcal{F}$-measurable, because X is jointly measurable, and it is dominated by $\|h\|\|g\|\,|Y_t - Y_s|$, which is integrable for $P \otimes q^{\otimes m}$ since $Y_t - Y_s \in L^1(P)$ and $q^{\otimes m}(R) \le \prod_j q(\mathbb{T}_{\le u_j}) < \infty$. The product measure is legitimate and Fubini applies by Lemma 3.7(b), giving

$$E\big[(Y_t - Y_s)Z\big] \;=\; \int_R E\Big[(Y_t - Y_s)\prod_{k=1}^n h_k(X(t_k))\prod_{j=1}^m g_j(X(v_j))\Big]\, q^{\otimes m}(\,\mathrm{d}v).$$

For each $v \in R$ the times $t_1, \dots, t_n$ and $v_1, \dots, v_m$ all lie in $\mathbb{T}_{\le s}$, and the functions $h_1, \dots, h_n$ and $g_1, \dots, g_m$ are bounded; so the inner expectation vanishes by (21), applied with $n+m$ in place of n. Hence $E[(Y_t - Y_s)Z] = 0$.

This is the step for which the times in (21) must be allowed to form an arbitrary finite subset of $\mathbb{T}_{\le s}$: the v_j are produced by the integration and cannot be ordered into a chain with the t_k.

Step 4: conclusion. By Steps 1–3 and the functional monotone class theorem (Fact 2.44), $\mathcal{H}$ contains every bounded $\sigma(\mathcal{K})$-measurable function, i.e. every bounded ${}^*\mathcal{F}_s^X$-measurable function. Since $s \le t$ were arbitrary, Y is a $({}^*\mathcal{F}_t^X)$-martingale.

The variant with $h_k \in C_b(E)$, and it alone needs (E2). On a separable metrizable E the class $C_b(E)$ generates $\mathcal{B}(E)$ — for a closed C the functions $x \mapsto (1 - m\,d(x, C))^+$ increase boundedly to $\mathbf{1}_C$ — so the class (23) built from $C_b(E)$ has the same generated σ-field, and Steps 1–4 are unchanged. Passing from $h_k \in C_b(E)$ back to $h_k \in B(E)$ in (21) is again Step 1 applied one coordinate at a time.

What was used. Only the preorder, through $\mathbb{T}_{\le u} \subset \mathbb{T}_{\le s}$ for $u \le s$ and through the additivity (3); the finiteness $q(\mathbb{T}_{\le u}) < \infty$, twice, for the boundedness of I_u^h and for Fubini; and the joint measurability of X. In particular no topology, no linear order, no property of q beyond Definition 2.3, *and no choice of ι*: the argument uses only that $\langle \cdot, \cdot \rangle_\iota$ is additive, which (3) gives for both conventions. □

Remark 3.10. Proposition 3.8 says that being a solution of the martingale problem for A is a statement about the *finite-dimensional distributions* of X only. Three consequences are used repeatedly:

(a) any measurable modification of a solution is a solution;

(b) a solution for A is a solution for the linear span A_S, by linearity of (21) in (f, g), and for a closure of A_S that Lemma 3.11 makes precise;

(c) if $A^{(1)} \subset A^{(2)}$, every solution for $A^{(2)}$ is one for $A^{(1)}$.

The closure in (b) deserves to be stated carefully, because [4] take it in the bounded pointwise sense and that is more than the argument uses.

Lemma 3.11 (Closure along a solution; (T0) + (E0) + clock)**.** *Let X solve the martingale problem for A with respect to $(\mathcal{G}_t)$, and let $(f, g) \in M(E) \times M(E)$ be such that $Y^{f,g}$ is integrable. Suppose there are $(f_n, g_n) \in A_S$ with, for all $s \le t$ in $\mathbb{T}$,*

$$\begin{aligned} f_n(X(t)) &\longrightarrow f(X(t)) && \text{in } L^1(P), \\ \int_{\langle s,t\rangle_\iota} g_n(X(u))\, q(\mathrm{d}u) &\longrightarrow \int_{\langle s,t\rangle_\iota} g(X(u))\, q(\mathrm{d}u) && \text{in } L^1(P). \end{aligned} \tag{24}$$

Then X solves the martingale problem for $A \cup \{(f, g)\}$.

Proof. We argue directly, which avoids Proposition 3.8 and gives the statement for the filtration $(\mathcal{G}_t)$ actually at hand rather than only for ${}^*\mathcal{F}^X$.

Adaptedness. $f(X(t))$ is $\mathcal{F}_t^X$-measurable, and $\int_{\langle 0,t\rangle_\iota} g(X(u))q(\mathrm{d}u)$ is ${}^*\mathcal{F}_t^X$-measurable by Lemma 3.7(a); both σ-fields are contained in $\mathcal{G}_t$, so $Y^{f,g}$ is $(\mathcal{G}_t)$-adapted.

The martingale property. Since $Y^{f',g'}$ is a $(\mathcal{G}_t)$-martingale for every $(f', g') \in A$ and $(f', g') \mapsto Y^{f',g'}$ is linear, the same holds for every $(f_n, g_n) \in A_S$; note also that Y^{f_n,g_n} is then integrable, being a finite linear combination of integrable processes. Let $s \le t$ and let W be bounded and $\mathcal{G}_s$-measurable. Then $E[(Y_t^{f_n,g_n} - Y_s^{f_n,g_n})W] = 0$, and by (3)

$$Y_t^{f_n,g_n} - Y_s^{f_n,g_n} = \big(f_n(X(t)) - f_n(X(s))\big) - \int_{\langle s,t\rangle_\iota} g_n(X(u))\, q(\,\mathrm{d}u).$$

Each of the three terms converges in $L^1(P)$ to its counterpart for (f,g), by (24); multiplying by the bounded W preserves L^1 convergence, so $E[(Y_t^{f,g} - Y_s^{f,g})W] = 0$. As W was arbitrary, $E[Y_t^{f,g} \mid \mathcal{G}_s] = Y_s^{f,g}$. □

Remark 3.12. The direct argument is worth the two extra lines. Routing through Proposition 3.8 would prove the conclusion only for ${}^*\mathcal{F}^X$, since that is the filtration the finite-dimensional characterization speaks of; the L^1 limit, by contrast, is taken against an arbitrary bounded $\mathcal{G}_s$-measurable test variable and therefore delivers the martingale property for whichever filtration the hypothesis was stated with.

Corollary 3.13 (Bounded pointwise convergence suffices; (T0) + (E0) + clock)**.** *Let $A \subset B(E) \times B(E)$ and let $(f_n, g_n) \in A_S$ with* bp-$\lim_n(f_n, g_n) = (f,g)$*. Then* (24) *holds — for* every *solution X, every P and every clock. Consequently a solution for A is a solution for the bp-closure of A_S, and if the bp-closures of $A_S^{(1)}$ and $A_S^{(2)}$ agree then the two martingale problems have the same solutions ([4], Proposition 4.3.1).*

Proof. $|f_n(X(t))| \le \sup_n \|f_n\| < \infty$ and $f_n(X(t)) \to f(X(t))$ pointwise, so the first convergence is dominated convergence under the probability measure P. For the second, $|g_n(X(u))| \le \sup_n \|g_n\|$ and $q(\langle s,t\rangle_\iota) \le q(\mathbb{T}_{\le t}) < \infty$ by Definition 2.3, so $\int_{\langle s,t\rangle_\iota} |g_n - g|(X(u))\, q(\,\mathrm{d}u) \to 0$ by dominated convergence under $q \otimes P$ — legitimate because $(u,\omega) \mapsto g_n(X(u,\omega))$ is jointly measurable, the constant $\sup_n \|g_n\|$ is $q \otimes P$-integrable on $\langle s,t\rangle_\iota \times \Omega$, and $|\int (g_n - g)(X(u)) q(\,\mathrm{d}u)| \le \int |g_n - g|(X(u)) q(\,\mathrm{d}u)$. Both dominating constants depend on the sequence alone. Finally f and g are again bounded, so $Y^{f,g}$ is integrable and Lemma 3.11 applies. □

Remark 3.14 (What bounded pointwise convergence is, and is not, for)*.* Corollary 3.13 isolates the rôle of the bp-topology, and it is worth stating, because the notion is easy to over-read.

It is the strongest hypothesis that is independent of X. The dominating constants in the proof depend on (f_n, g_n) alone — not on the law of X, not on the clock, not on the convention. That is exactly what makes Corollary 3.13 a statement about *operators*: two operators with the same bp-closure have the same solutions, with no solution mentioned. Lemma 3.11 is instead a statement about a *given* solution — weaker as an assertion about A, and stronger as a tool.

It costs generality that the rest of Section 3 does not spend. Definition 3.5 admits $A \subset M(E) \times M(E)$, and Proposition 3.8 assumes that $Y^{f,g}$ is integrable, not that f and g are bounded. The bp-closure lives in $B(E) \times B(E)$ and therefore silently narrows this; Lemma 3.11 does not.

For the formalization the difference is real. Lemma 3.11 is dominated convergence and nothing else. The bp apparatus, Definition 2.29 and Fact 2.30, is needed only for Corollary 3.13, and there only its first assertion.

Definition 3.15 ([4], Section 4.4; (T0) + (E0) + clock)**.** *Uniqueness* holds for the martingale problem for (A, μ) if any two solutions have the same finite-dimensional distributions. The martingale problem for (A, μ) is *well-posed* if a solution exists and uniqueness holds; the martingale problem for A is well-posed if it is well-posed for every $\mu \in \mathcal{P}(E)$. It is *well-posed in* $D_E[0,\infty)$ (resp. $C_E[0,\infty)$) if there is a unique $P \in \mathcal{P}(D_E[0,\infty))$ (resp. $\mathcal{P}(C_E[0,\infty))$) solving the $D_E[0,\infty)$ (resp. $C_E[0,\infty)$) martingale problem for (A,μ). Well-posedness in $D_E[0,\infty)$ does not imply well-posedness; Theorem 4.5 shows that this gap is rare.

The following two results connect Definition 3.5 with the operator-theoretic notions of Section 2.4.

Lemma 3.16 ([4], Lemma 4.3.2; (T3))**.** *Let X be a measurable process, $\mathcal{G}_t \supset {}^*\mathcal{F}_t^X$, and $f,g \in B(E)$. Then for fixed $\lambda \in \mathbb{R}$ the process $f(X(t)) - \int_0^t g(X(s))\,\mathrm{d}s$ is a $(\mathcal{G}_t)$-martingale if and only if*

$$e^{-\lambda t} f(X(t)) + \int_0^t e^{-\lambda u}\big(\lambda f(X(u)) - g(X(u))\big)\,\mathrm{d}u \tag{25}$$

is a $(\mathcal{G}_t)$-martingale. Consequently, if $(f,g) \in A$, $\lambda > 0$ and X solves the martingale problem for A with respect to $(\mathcal{G}_t)$, then

$$f(X(t)) \;=\; E\Big[\int_0^\infty e^{-\lambda s}\big(\lambda f(X(t+s)) - g(X(t+s))\big)\,\mathrm{d}s \;\Big|\; \mathcal{G}_t\Big]. \tag{26}$$

Proof. Write $Y_t = f(X(t)) - \int_0^t g(X(s))\,\mathrm{d}s$ and let Z_t denote (25).

Step 1: an identity. We claim

$$Z_t \;=\; e^{-\lambda t} Y_t + \lambda \int_0^t e^{-\lambda u} Y_u\,\mathrm{d}u. \tag{27}$$

Indeed, substituting $Y_u = f(X(u)) - \int_0^u g(X(v))\,\mathrm{d}v$ into the right-hand side and using Fubini,

$$\lambda \int_0^t e^{-\lambda u} \int_0^u g(X(v))\,\mathrm{d}v\,\mathrm{d}u = \lambda \int_0^t g(X(v)) \int_v^t e^{-\lambda u}\,\mathrm{d}u\,\mathrm{d}v = \int_0^t g(X(v))\big(e^{-\lambda v} - e^{-\lambda t}\big)\,\mathrm{d}v,$$

so the two terms involving g combine to $-e^{-\lambda t}\int_0^t g\,\mathrm{d}v - \int_0^t e^{-\lambda v} g\,\mathrm{d}v + e^{-\lambda t}\int_0^t g\,\mathrm{d}v = -\int_0^t e^{-\lambda v} g(X(v))\,\mathrm{d}v$, and (27) is (25). Solving (27) for Y — put $W_t = \int_0^t e^{-\lambda u} Y_u\,\mathrm{d}u$, so that $Z = W' + \lambda W$ and hence $(e^{\lambda t} W)' = e^{\lambda t} Z_t$ — gives the inverse relation

$$Y_t \;=\; e^{\lambda t} Z_t - \lambda \int_0^t e^{\lambda u} Z_u\,\mathrm{d}u. \tag{28}$$

Step 2: either relation transports the martingale property. Suppose Y is a $(\mathcal{G}_t)$-martingale. For $s \le t$ we use (27) together with Fubini for conditional expectations, which is legitimate because $\sup_{u\le t} E|Y_u| < \infty$:

$$\begin{aligned} E[Z_t \mid \mathcal{G}_s] &= e^{-\lambda t} Y_s + \lambda \int_0^s e^{-\lambda u} Y_u\,\mathrm{d}u + \lambda \int_s^t e^{-\lambda u} Y_s\,\mathrm{d}u \\ &= e^{-\lambda t} Y_s + \lambda \int_0^s e^{-\lambda u} Y_u\,\mathrm{d}u + Y_s\big(e^{-\lambda s} - e^{-\lambda t}\big) = Z_s. \end{aligned}$$

The converse is the same computation applied to (28), with $e^{\lambda u}$ in place of $e^{-\lambda u}$ and $-\lambda$ in place of λ. This proves the equivalence.

Step 3: the resolvent representation. Let $(f,g) \in A$ with f,g bounded and $\lambda > 0$, and let X solve the martingale problem for A with respect to $(\mathcal{G}_t)$, so that Z is a martingale by Step 2. For $r > 0$,

$$Z_t \;=\; E[Z_{t+r} \mid \mathcal{G}_t] \;=\; E\Big[e^{-\lambda(t+r)} f(X(t+r)) + \int_0^{t+r} e^{-\lambda u}(\lambda f - g)(X(u))\,\mathrm{d}u \;\Big|\; \mathcal{G}_t\Big].$$

As $r \to \infty$ the first term is bounded by $e^{-\lambda(t+r)}\|f\| \to 0$ and the second increases to an integral dominated by $\lambda^{-1}\|\lambda f - g\|$, so dominated convergence for conditional expectations gives

$$Z_t \;=\; E\Big[\int_0^\infty e^{-\lambda u}(\lambda f - g)(X(u))\,\mathrm{d}u \;\Big|\; \mathcal{G}_t\Big].$$

Subtracting $\int_0^t e^{-\lambda u}(\lambda f - g)(X(u))\,\mathrm{d}u$, which is $\mathcal{G}_t$-measurable, from both sides leaves $e^{-\lambda t} f(X(t)) = E[\int_t^\infty e^{-\lambda u}(\lambda f - g)(X(u))\,\mathrm{d}u \mid \mathcal{G}_t]$, and the substitution $u = t + s$ gives (26). $\square$

Remark 3.17. Step 1 is the only place where $\mathbb{T} = \mathbb{R}_+$ and $q =$ Lebesgue measure are used, and they are used twice: for the exponential $e^{-\lambda t}$, which is the solution of $\phi' = -\lambda\phi$, and for the Fubini rearrangement. On $\mathbb{T} = \mathbb{N}_0$ with q counting measure the analogue holds with $e^{-\lambda}$ replaced by $(1+\lambda)^{-1}$ and the integrals by sums; we do not need it, and record only that Lemma 3.16 is (T3) for a reason of calculus, not of probability.

Proposition 3.18 ([4], Proposition 4.3.5; (T3))**.** *Let A be a linear subset of $B(E) \times B(E)$. If for each $x \in E$ there exists a solution X_x of the martingale problem for (A, δ_x), then A is dissipative in the sense of Definition 2.11.*

Proof. Given $(f, g) \in A$ and $\lambda > 0$, (26) with $t = 0$ gives $f(x) = E[\int_0^\infty e^{-\lambda s}(\lambda f(X_x(s)) - g(X_x(s)))\,\mathrm{d}s]$, whence $|f(x)| \le \int_0^\infty e^{-\lambda s}\|\lambda f - g\|\,\mathrm{d}s = \lambda^{-1}\|\lambda f - g\|$ for every $x \in E$. $\square$

Remark 3.19. Only dissipative operators arise as generators of Markov processes. Indeed, if X is a Markov process with measurable transition semigroup $\{T(t)\}$ on $B(E)$ and full generator $\hat{A}$ as in Definition 2.12, then X solves the martingale problem for $\hat{A}$ ([4], Proposition 4.1.7), and $\hat{A}$ is dissipative by Fact 2.13. Proposition 3.18 is the converse direction.

Definition 3.20 ((T2b) + (E2))**.** Let $D \subset \mathbb{T}$ be the countable dense set of (T2b). A process X satisfies the *compact containment condition* if for all $\varepsilon > 0$ and $T \in \mathbb{T}$ there is a compact $K_{\varepsilon,T} \subset E$ with

$$P\big\{X(t) \in K_{\varepsilon,T} \text{ for all } t \in \mathbb{T}_{\le T} \cap D\big\} \;>\; 1 - \varepsilon. \tag{29}$$

A family $\{X_n\}$ satisfies the compact containment condition if (15) holds.

4 Sample path regularity: existence of a càdlàg modification

4.1 An abstract regularization theorem

[4] state Theorem 4.5 below for the Markovian martingale problem, that is, for the family $\mathbb{X}_A(X)$ of (20). Their proof, however, never uses the operator A, nor the special filtration (4), nor the particular form $\int_0^t g(X_s)\,\mathrm{d}s$ of the compensator. What it uses is only that $f(X)$ splits, for every f in a sufficiently rich class, into a martingale plus a process that is regular in time. Isolating that hypothesis gives a statement in the abstract setting of Definition 3.2, from which [4], Theorem 4.3.6 follows in one line.

[2] contain no counterpart of this result, and cannot: their Setting 3.1 *presupposes* a Polish path space F and right continuous test processes, which is exactly what a regularization theorem has to produce. Theorem 4.3 is therefore the statement that carries a solution *into* the [2] setting, with $F = D_E[0, \infty)$; it is the missing first step of their programme rather than a consequence of it.

Setting 4.1 ((T2b) + (E2) + clock, $\iota = \mathrm{o}$)**.** As in Setting 3.1, but with no path space and no path regularity assumed. Let $\mathbb{T}$ satisfy (T2b) with countable dense $D \subset \mathbb{T}$, let $(\Omega, \mathcal{F}, \mathbb{F}, P)$ be a filtered probability space, let E satisfy (E2), let $\mathbb{X}$ be a family of real-valued $\mathbb{F}$-adapted processes with $P \in \mathcal{M}(\mathbb{X})$, and let X be an E-valued, $\mathbb{F}$-adapted, jointly measurable process. Neither the elements of $\mathbb{X}$ nor X are assumed to be right continuous.

Definition 4.2 ((T2b) + (E2))**.** A set $\Phi \subset C_b(E)$ is a *regularizing class* for $(X, \mathbb{X})$ if for every $f \in \Phi$ there exist $Y^f \in \mathbb{X}$ and an adapted real-valued process C^f such that

(R1) $f(X(t)) = Y^f(t) + C^f(t)$ P-a.s., for every $t \in \mathbb{T}$;

(R2) P-a.s., $t \mapsto C^f(t)$ has a limit along D from the right at every $t \in \mathbb{T}$ and from the left at every $t > 0$;

(R3) C^f is right continuous in L^1, i.e. $E\big[|C^f(s) - C^f(t)|\big] \to 0$ as $s \downarrow t$, for every $t \in \mathbb{T}$.

Theorem 4.3 (Abstract càdlàg modification; (T2b) + (E2))**.** *In Setting 4.1, let Φ be a regularizing class for $(X, \mathbb{X})$ such that*

(a) Φ contains a countable subset separating the points of E, and

(b) Φ is separating (Definition 2.32),

and suppose that X satisfies the compact containment condition (29)*. Then X has a modification with sample paths in $D_E[0, \infty)$.*

Proof. Let $\{f_i\} \subset \Phi$ be countable and separating the points of E, and write $Y^i = Y^{f_i}$, $C^i = C^{f_i}$ for the associated data of Definition 4.2.

Step 1: regularity along the rationals. Each Y^i is an $(\mathbb{F}, P)$-martingale, so Fact 2.19, applied with the countable dense set D, provides for each i a set of full measure on which the limits of Y^i along D from above (and, for $t > 0$, from below) exist at every t. Intersect these over the countably many i; intersect further with the sets of full measure furnished by (R2), one for each i; and finally with the sets on which (R1) holds *at the times $t \in D$*, of which there are countably many. The result is $\Omega' \subset \Omega$ with $P(\Omega') = 1$. *Only countably many null sets have been discarded, and this is why (R1) may be assumed only t by t: the limits below are taken along D, so* (30) *is needed only for $t \in D$.* For $\omega \in \Omega'$, every i and every $t \in \mathbb{T}$ the one-sided limits of

$$f_i(X(t)) \;=\; Y^i(t) + C^i(t) \tag{30}$$

along D exist — as a sum of two summands each of which has them. This is the only place where (R2) is used, and it is weaker than continuity of C^i: the two summands are allowed to jump, as long as each does so in a càdlàg manner.

Step 2: compactness. First, D is cofinal in $\mathbb{T}$: if t is not maximal there is $u > t$, and (T2b) gives a point of D in (t, u); if t is maximal it is itself cofinal. So a countable cofinal set exists.

Fix such a T and let $A_{m,T}$ be the event in (29) for $\varepsilon = 1/m$, so $P(A_{m,T}) > 1 - 1/m$. On $A_{m,T}$ the set $\{X(t, \omega) : t \in \mathbb{T}_{\le T} \cap D\}$ is contained in the compact $K_{1/m,T}$, hence relatively compact. Now

$$P\Big(\bigcup_m A_{m,T}\Big) \;\ge\; \sup_m P(A_{m,T}) \;=\; 1,$$

so $\bigcup_m A_{m,T}$ has full measure, and $\Omega'' = \Omega' \cap \bigcap_T \bigcup_m A_{m,T}$ has full measure as a countable intersection. *The union over m is essential: the events $A_{m,T}$ are not nested and each has* *probability only $> 1 - 1/m$, so intersecting them would prove nothing.* For $\omega \in \Omega''$ and every T in the cofinal set, $\{X(t,\omega) : t \in \mathbb{T}_{\le T} \cap D\}$ is relatively compact.

Step 3: existence of one-sided limits. Fix $\omega \in \Omega''$ and a non-maximal $t \in \mathbb{T}$, and let $s_n \in D$ with $s_n \downarrow t$, which exist by (T2b). (If t is maximal there is nothing to prove and we set $Y(t,\omega) = X(t,\omega)$.) By Step 2 the sequence $(X(s_n,\omega))_n$ lies in a relatively compact set, so it has convergent subsequences. If x and x' are limits along two subsequences, then by Step 1 and the continuity of f_i,

$$f_i(x) \;=\; \lim_{s\to t+,\, s\in D} f_i(X(s,\omega)) \;=\; f_i(x') \qquad \text{for all } i,$$

and since $\{f_i\}$ separates points, $x = x'$. Hence

$$Y(t,\omega) \;\coloneqq\; \lim_{s\to t+,\, s\in D} X(s,\omega) \tag{31}$$

exists for all $t \in \mathbb{T}$, and similarly $Y^-(t,\omega) = \lim_{s\to t-,\, s\in D} X(s,\omega)$ exists for $t > 0$. By Fact 2.18, $Y(\cdot,\omega)$ is càdlàg, and lies in $D_E[0,\infty)$ when $\mathbb{T}$ satisfies (T3). Set $Y(\cdot,\omega) \equiv x_0$ for $\omega \notin \Omega''$ and some fixed $x_0 \in E$. Each $Y(t,\cdot)$ is a random variable: on Ω'' it is the pointwise limit of the measurable maps $X(s_n,\cdot)$ along a fixed sequence $s_n \downarrow t$ in D, and a pointwise limit of measurable maps into a metrizable space is measurable where it exists.

Step 4: Y is a modification of X. Let $f \in \Phi$ and $t \in \mathbb{T}$. For $s > t$, (R1) and the martingale property of Y^f give

$$E\big[f(X(s)) \,\big|\, \mathcal{F}_t\big] \;=\; E\big[Y^f(s) \,\big|\, \mathcal{F}_t\big] + E\big[C^f(s) \,\big|\, \mathcal{F}_t\big] \;=\; Y^f(t) + E\big[C^f(s) \,\big|\, \mathcal{F}_t\big],$$

and by (R3) and conditional Jensen, $E[C^f(s) \mid \mathcal{F}_t] \to C^f(t)$ in L^1 as $s \downarrow t$. Hence, using (R1) once more, $E[f(X(s)) \mid \mathcal{F}_t] \to Y^f(t) + C^f(t) = f(X(t))$ in L^1. On the other hand $X(s) \to Y(t)$ a.s. along D by (31), so $f(X(s)) \to f(Y(t))$ a.s. and boundedly, whence $E[f(X(s)) \mid \mathcal{F}_t] \to E[f(Y(t)) \mid \mathcal{F}_t]$ in L^1. Therefore

$$E\big[f(Y(t)) \,\big|\, \mathcal{F}_t\big] \;=\; f(X(t)) \qquad \text{for all } f \in \Phi,\ t \in \mathbb{T}. \tag{32}$$

Since $X(t)$ is $\mathcal{F}_t$-measurable and Φ is separating, Fact 2.46 yields $P\{Y(t) = X(t)\} = 1$ for every $t \in \mathbb{T}$. $\square$

Remark 4.4 (What the abstract form buys). Three hypotheses of [4], Theorem 4.3.6 disappear, and each of them is one that [2] would want gone.

- (i) *No operator.* The compensator C^f is an arbitrary adapted process subject to (R2) and (R3). In particular it may be *path dependent*: the [2] prototype $C^f(t) = \int_{(0,t]} g(s,L,X)\, q(\mathrm{d}s)$, with a control variable L and a locally finite measure q, satisfies (R2) because it is of finite variation, and (R3) because $E|C^f(s) - C^f(t)| \le \int_{(t,s]} E|g(u,L,X)|\, q(\mathrm{d}u) \to 0$ under the integrability condition that [2] impose anyway. A statement of this generality cannot be made in the language of an operator $A \subset C_b(E) \times B(E)$.
- (ii) *Atoms are harmless.* Nothing above requires q to be atomless. (R2) asks for one-sided limits, not for continuity; the classical statement "$t \mapsto \int_0^t g(X_s)\,\mathrm{d}s$ is Lipschitz" is the special case $q =$ Lebesgue measure. In (R3) an atom at t contributes to $(0,t]$ but not to $(t,s]$, and so does not obstruct the limit — which is precisely why the half-open convention $(0,t]$ is the right one here.
- (iii) *No special filtration.* (4) plays no rôle; $\mathbb{F}$ is any filtration making X adapted and the Y^f martingales. The filtration ${}^*\mathcal{F}^X$ is needed for Proposition 3.8, not here.

What is *not* gained is a local version; see Remark 4.7, whose argument is unaffected.

4.2 The Markovian case

Theorem 4.5 ([4], Theorem 4.3.6; (T2b) + (E2) + clock, $\iota = \mathrm{o}$)**.** *Assume (E2). Let $A \subset C_b(E) \times B(E)$ and suppose that $\mathcal{D}(A)$ is separating and contains a countable subset that separates points. Let X be a solution of the martingale problem for A satisfying the compact containment condition (29). Then X has a modification with sample paths in $D_E[0,\infty)$.*

Proof. Apply Theorem 4.3 with $\mathbb{X} = \mathbb{X}_A(X)$, $\mathbb{F} = ({}^*\mathcal{F}_t^X)$ and $\Phi = \mathcal{D}(A)$: for $f \in \mathcal{D}(A)$ pick g with $(f,g) \in A$ and put $Y^f = f(X(\cdot)) - \int_{(0,\cdot]} g(X(s))\, q(\,\mathrm{d}s) \in \mathbb{X}_A^{\mathrm{o}}(X)$ and $C^f(t) = \int_{(0,t]} g(X(s))\, q(\,\mathrm{d}s)$ — here, and only here, the convention must be $\iota = \mathrm{o}$. Then (R1) holds by definition. Since g is bounded, C^f is of finite variation with $|C^f(t) - C^f(s)| \le \|g\|\, q((s,t])$, so it has one-sided limits, which is (R2); and $q((t,s]) \downarrow q(\emptyset) = 0$ as $s \downarrow t$ by continuity from above, which is (R3). For $q =$ Lebesgue measure this is the familiar statement that C^f is Lipschitz, but no atomlessness is needed: an atom of q at t lies in $(0,t]$ and not in $(t,s]$, which is exactly why the convention of Remark 2.5 is the right one here. Hypotheses (a) and (b) are the assumptions on $\mathcal{D}(A)$. $\square$

Remark 4.6. The two hypotheses on $\mathcal{D}(A)$ — hypotheses (a) and (b) of Theorem 4.3 — play different rôles and should be kept apart in the formalization: "contains a countable subset separating points" is used in Step 3 of the proof of Theorem 4.3 to identify limit points, "separating" is used in Step 4 to identify $Y(t)$ with $X(t)$. They are independent: neither implies the other.

Remark 4.7. Theorem 4.5 is stated for genuine martingales and this is not an artefact of the proof. For a local martingale problem the localizing sequence consists of stopping times, whose usefulness (via optional stopping) presupposes right continuity of the filtration, and typically of the paths — which is exactly the conclusion being sought. If one *postulates* a localizing sequence of *deterministic* times, or — equivalently for the present purpose — if A is replaced by $A^K = \{(f,g) \in A : f, g \text{ bounded on } K\}$ for the compact sets K of (29), then the proof goes through verbatim. This is the honest local version of Theorem 4.5.

Remark 4.8 (Verifying compact containment). Condition (29) is the price paid for allowing non-locally-compact Polish E. If E is locally compact and separable and $\mathcal{D}(A)$ is dense in $\hat{C}(E)$ (continuous functions vanishing at infinity) in the norm topology, then (29) may be dispensed with (see also Fact 2.33): pass to the one-point compactification E^Δ, apply Theorem 4.5 there (where compact containment is trivial), and conclude that the modification has paths in $D_{E^\Delta}[0,\infty)$; see [4], Corollary 4.3.7.

5 Uniqueness, the Markov property, and the strong Markov property

The theorem of this section is the workhorse of the whole theory: it reduces uniqueness for a martingale problem, which is a statement about all finite-dimensional distributions, to a statement about *one-dimensional* distributions only, and it produces the Markov property for free. No semigroup is assumed to exist, and $\mathcal{D}(A)$ is not assumed to be separating.

As with Theorem 4.5, the operator plays no rôle in the proof. Section 5.1 extracts what does, which turns out to be a single lemma applied four times; Section 5.6 then recovers [4], Theorem 4.4.2. [2] have no counterpart of this result either — they prove no uniqueness theorem at all — but, unlike the situation in Section 4, the obstruction is not structural: the statement fits their setting once one adds the one ingredient a uniqueness theory needs and a convergence theory does not, namely a way of *restarting* the martingale problem.

5.1 Uniqueness without a Markov structure

Throughout this subsection we work on the path space, as in Setting 3.1; $\mathbb{T}$ satisfies (T0) and E satisfies (E0) unless stated otherwise. Let $F \subset E^{\mathbb{T}}$ be a path space with σ-field $\mathcal{S} = \sigma(\pi_t : t \in \mathbb{T})$, coordinate process $\pi = (\pi_t)_{t\in\mathbb{T}}$ and canonical filtration $\mathcal{F}_t^\circ = \sigma(\pi_u : u \le t)$, and let $\mathbb{X}^\circ$ be a family of $(\mathcal{F}_t^\circ)$-adapted real-valued functionals on F. Write

$$\mathcal{M}(\mathbb{X}^\circ) \;=\; \big\{P \in \mathcal{P}(F) \;:\; \text{every } Y^\circ \in \mathbb{X}^\circ \text{ is a } (\mathcal{F}_t^\circ, P)\text{-martingale}\big\},$$

and $\mathcal{M}(\mathbb{X}^\circ, \mu) = \{P \in \mathcal{M}(\mathbb{X}^\circ) : P\pi_0^{-1} = \mu\}$.

Remark 5.1. Nothing is lost by using the canonical filtration. If X is a process with paths in F solving the martingale problem with respect to *some* filtration $(\mathcal{G}_t)$ to which it is adapted — for instance (4) — then, since each $Y_t^\circ(X)$ is $\sigma(X_u : u \le t)$-measurable and $\sigma(X_u : u \le t) \subset \mathcal{G}_t$, the tower property makes $Y^\circ(X)$ a martingale for the smaller filtration as well. Hence $PX^{-1} \in \mathcal{M}(\mathbb{X}^\circ)$, and $\mathcal{M}(\mathbb{X}^\circ)$ is the set of laws of all solutions, for all filtrations at once.

The martingale property is an identity that is *linear* in the measure. This trivial observation is what carries the mixing steps of the classical proof, and it deserves to be recorded once.

Lemma 5.2 (Mixture stability; (T0) + (E0)). *Let $(\Theta, \mathcal{A}, \nu)$ be a probability space and $\vartheta \mapsto P_\vartheta \in \mathcal{M}(\mathbb{X}^\circ)$ a measurable family, and put $P = \int P_\vartheta\, \nu(\mathrm{d}\vartheta)$. If $\int E^{P_\vartheta}|Y_t^\circ|\, \nu(\mathrm{d}\vartheta) < \infty$ for every $Y^\circ \in \mathbb{X}^\circ$ and $t \in \mathbb{T}$, then $P \in \mathcal{M}(\mathbb{X}^\circ)$. In particular $\mathcal{M}(\mathbb{X}^\circ)$ is convex.*

Proof. For $G \in \mathcal{F}_s^\circ$ the identity $E^{P_\vartheta}[Y_t^\circ \mathbf{1}_G] = E^{P_\vartheta}[Y_s^\circ \mathbf{1}_G]$ holds for every ϑ; integrate it $\nu(\mathrm{d}\vartheta)$ and use Fubini, which is licensed by the integrability hypothesis. □

Its converse holds as well, and is what actually extends well-posedness from degenerate to general initial distributions. It is worth separating from Theorem 5.16, because — unlike the argument used in Corollary 6.18 — it does not go through the Markov property.

Lemma 5.3 (Disintegration; (T0) + (E1)). *Suppose $\mathbb{X}^\circ$ admits a determining set $\mathbb{Z}^\circ$ and that membership in $\mathcal{M}(\mathbb{X}^\circ)$ can be tested along countably many data, i.e. there are a countable $\mathbb{X}_0^\circ \subset \mathbb{X}^\circ$, a countable $D \subset \mathbb{T}$ and countable sets $\mathbb{Z}_s^\circ$ such that $P \in \mathcal{M}(\mathbb{X}^\circ)$ if and only if*

$$E^P\big[(Y_t^\circ - Y_s^\circ)Z_s^\circ\big] = 0 \qquad \textit{for all } Y^\circ \in \mathbb{X}_0^\circ,\ s \le t \textit{ in } D,\ Z_s^\circ \in \mathbb{Z}_s^\circ. \tag{33}$$

Let $P \in \mathcal{M}(\mathbb{X}^\circ)$ with $P\pi_0^{-1} = \mu$. Then $P(\cdot \mid \pi_0 = x) \in \mathcal{M}(\mathbb{X}^\circ, \delta_x)$ for μ-a.e. x. In particular, if $\mathcal{M}(\mathbb{X}^\circ, \delta_x) = \{P_x\}$ for μ-a.e. x, then $P = \int P_x\, \mu(\mathrm{d}x)$, and $\mathcal{M}(\mathbb{X}^\circ, \mu)$ is a singleton.

Proof. Fix $Y^\circ \in \mathbb{X}_0^\circ$, $s \le t$ in D and $Z_s^\circ \in \mathbb{Z}_s^\circ$. For bounded measurable $h : E \to \mathbb{R}$ the variable $Z_s^\circ h(\pi_0)$ is bounded and $\mathcal{F}_s^\circ$-measurable, so the martingale property of Y° under P gives $E^P[(Y_t^\circ - Y_s^\circ)Z_s^\circ h(\pi_0)] = 0$. As h is arbitrary, $E^P[(Y_t^\circ - Y_s^\circ)Z_s^\circ \mid \pi_0] = 0$ P-a.s. The data being countable, one null set serves all of them at once; off it, $P(\cdot \mid \pi_0 = x)$ satisfies (33) and has initial law δ_x. □

Remark 5.4. Lemma 5.3 together with Lemma 5.2 is [7], Theorem 32.10, in abstract form; [7] attributes it to Stroock and Varadhan and uses it exactly this way, to pass from P_x to $P_\mu = \int P_x \mu(\mathrm{d}x)$ for the local martingale problem of a diffusion. The countability hypothesis is his too: he replaces $\hat{C}^\infty$ by a countable class of truncated coordinate functions and their products. Note that Corollary 6.18 currently justifies the same passage via Theorem 5.16(a); Lemma 5.3 is the better route, since it needs no Markov property.

Restarting a solution is available at once, and — this is worth stating separately, because it is easy to attribute to the Markov property what belongs to the tower property — it needs no shift and no Markov structure whatever.

Lemma 5.5 (Restart with memory; (T0) + (E0))**.** *Let $P \in \mathcal{M}(\mathbb{X}^\circ)$, let $r \in \mathbb{T}$ and let $Z \geq 0$ be bounded and $\mathcal{F}_r^\circ$-measurable with $E^P[Z] = 1$. Then $Z \cdot P$ is again a solution* from r onwards*: for every $Y^\circ \in \mathbb{X}^\circ$ and all $r \leq s \leq t$,*

$$E^{Z \cdot P}\big[Y_t^\circ \mid \mathcal{F}_s^\circ\big] \;=\; Y_s^\circ.$$

Proof. Let $G \in \mathcal{F}_s^\circ$. Since $r \leq s$, the variable $Z\mathbf{1}_G$ is bounded and $\mathcal{F}_s^\circ$-measurable, so the martingale property of Y° under P gives $E^{Z\cdot P}[(Y_t^\circ - Y_s^\circ)\mathbf{1}_G] = E^P[(Y_t^\circ - Y_s^\circ)Z\mathbf{1}_G] = 0$. □

Remark 5.6 (Restarting is not where the Markov structure enters). Lemma 5.5 says that one may condition on any event of positive probability observed up to r, or reweight by any bounded $\mathcal{F}_r^\circ$-measurable density, and what remains after r is again a solution. Nothing is forgotten and nothing is assumed. If one wants the conditional laws themselves rather than a reweighting, Lemma 5.3 supplies them: for standard Borel F the family $P(\,\cdot \mid \mathcal{F}_r^\circ)$ exists, and it consists of solutions from r onwards by the same two lines.

What a shift adds is therefore not the restart but a *re-indexing*. A hypothesis stated about *initial* distributions — as (39) is, and as [4], Theorem 4.4.2 is — can be applied to the restarted object only after that object has been turned into a process starting at time 0, and the map that does this is θ_r, which discards the path before r. That is the whole of the Markov structure in Section 5.2: not the ability to begin again, but the demand that beginning again should look like beginning.

Definition 5.7 is the same idea without the re-indexing: the two measures are compared at the time s where they already agree, so nothing has to be transported back to 0 and no shift is needed. This is why Proposition 5.9 is Markov-free while Lemma 5.15 is not.

So far nothing beyond a preorder has been used, and nothing Markovian. It is worth pushing that as far as it goes, because uniqueness of the finite-dimensional distributions is *not* a Markovian fact. What is Markovian is only the reduction of its hypothesis to the *one-dimensional* distributions, and that reduction is the subject of the next subsection. The condition the uniqueness argument actually consumes is the following.

Definition 5.7 (Propagation of agreement; (T0) + (E0))**.** A set $\mathcal{N} \subset \mathcal{P}(F)$ *propagates agreement* if for all $P, Q \in \mathcal{N}$, all $s \leq t$ in $\mathbb{T}$ and every bounded $\mathcal{F}_s^\circ$-measurable $Z \geq 0$,

$$\begin{aligned} & E^P\big[Z\,h(\pi_s)\big] = E^Q\big[Z\,h(\pi_s)\big] \quad \text{for all } h \in B(E) \\ \Longrightarrow\quad & E^P\big[Z\,f(\pi_t)\big] = E^Q\big[Z\,f(\pi_t)\big] \quad \text{for all } f \in B(E). \end{aligned} \tag{34}$$

Remark 5.8 (Why the weight Z cannot be dropped). It is tempting to state (34) in the simpler unconditional form

$$P = Q \ \text{ on } \mathcal{F}_s^\circ \quad\Longrightarrow\quad P = Q \ \text{ on } \mathcal{F}_s^\circ \vee \sigma(\pi_t), \tag{35}$$

and an earlier version of this manuscript did. That is a mistake, and it is worth recording where the induction of Proposition 5.9 breaks under it. The induction produces agreement of the finite-dimensional distributions, i.e. agreement on $\sigma(\pi_{t_1}, \dots, \pi_{t_n})$ for finite chains; (35) demands agreement on all of $\mathcal{F}_{t_n}^\circ = \sigma(\pi_u : u \leq t_n)$, which involves *all* times below t_n and is therefore strictly more than any stage of the induction delivers. Iterating (35) adds one future coordinate at a time but never closes the gap, since after adding $\pi_{t_{n+1}}$ one is still short of $\mathcal{F}_{t_{n+1}}^\circ$.

Form (34) repairs this by carrying the past in the weight Z instead of in a σ-field: at stage n one takes $Z = \prod_{k\le n} f_k(\pi_{t_k})$, and the hypothesis of (34) is then exactly the induction hypothesis. It is also exactly what the restart of Lemma 5.13 supplies, which is no accident: the two are the same statement read from opposite ends.

Proposition 5.9 (Uniqueness; (T2a) + (E0))**.** *Let $\mathcal{N} \subset \mathcal{P}(F)$ propagate agreement. Then $\mathcal{N}$ contains at most one element with a given law of π_0. In particular, if $\mathcal{M}(\mathbb{X}^\circ)$ propagates agreement then $\mathcal{M}(\mathbb{X}^\circ, \mu)$ has at most one element for every μ, and the same holds for $\mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$.*

Proof. Let $P, Q \in \mathcal{N}$ with $P\pi_0^{-1} = Q\pi_0^{-1}$.

Step 1: the finite-dimensional distributions. We show by induction on m that

$$E^P\Big[\prod_{k=1}^m f_k(\pi_{t_k})\Big] = E^Q\Big[\prod_{k=1}^m f_k(\pi_{t_k})\Big] \tag{36}$$

for every chain $t_1 \le \cdots \le t_m$ in $\mathbb{T}$ and all bounded measurable $f_k \ge 0$.

For $m = 0$ both sides are 1. For $m = 1$, apply (34) with $s = t = t_1$ and $Z = 1$: the hypothesis is $P\pi_{t_1}^{-1} = Q\pi_{t_1}^{-1}$ — which for $t_1 = 0$ is the assumption on the initial laws, and for general t_1 follows once the case $m = 1$ is known for $s = 0$, i.e. from (34) with $s = 0$, $t = t_1$, $Z = 1$.

Assume (36) for $m = n$, fix a chain $t_1 \le \cdots \le t_{n+1}$ and bounded $f_1, \ldots, f_{n+1} \ge 0$, and put

$$Z \;=\; \prod_{k=1}^n f_k(\pi_{t_k}),$$

which is bounded, nonnegative and $\mathcal{F}^\circ_{t_n}$-measurable because $t_k \le t_n$ for $k \le n$. For every bounded measurable $h \ge 0$ the identity $E^P[Z\,h(\pi_{t_n})] = E^Q[Z\,h(\pi_{t_n})]$ is (36) for $m = n$ with f_n replaced by $f_n h$, hence holds; by linearity it holds for all $h \in B(E)$. Now (34), applied with $s = t_n$ and $t = t_{n+1}$, gives $E^P[Zf_{n+1}(\pi_{t_{n+1}})] = E^Q[Zf_{n+1}(\pi_{t_{n+1}})]$, which is (36) for $m = n + 1$.

Step 2: from the finite-dimensional distributions to the law. By linearity, (36) holds for all $f_k \in B(E)$. Under (T2a) every finite subset of $\mathbb{T}$ is a chain, so the collection

$$\mathcal{K} \;=\; \Big\{\textstyle\prod_{k=1}^m f_k(\pi_{t_k}) \;:\; m \ge 0,\; t_k \in \mathbb{T},\; f_k \in B(E)\Big\}$$

is closed under multiplication and generates $\sigma(\pi_t : t \in \mathbb{T}) = \mathcal{S}$ by (19). Two probability measures agreeing on a multiplicative generating class agree, by Dynkin's lemma. Hence $P = Q$. ☐

5.2 The Markov layer: shift systems

Everything so far is free of any Markov structure. What follows adds it, in the one form the arguments need, and it does two things: it makes (34) checkable from the one-dimensional distributions (Lemma 5.15), and it produces the Markov property itself (Theorem 5.16). It is what a convergence theory does not need and a uniqueness theory cannot do without.

Definition 5.10 (Shift system; (T4) + (E0))**.** A family of measurable maps $\theta_r : F \to F$, $r \in \mathbb{T}$, is a *shift* on F if $\pi_t \circ \theta_r = \pi_{r+t}$ for all $r, t \in \mathbb{T}$.

A *shift system* for $\mathbb{X}^\circ$ is a family $(\mathbb{X}^\circ_r)_{r\in\mathbb{T}}$ of families of $(\mathcal{F}^\circ_t)$-adapted real-valued processes on F, with $\mathbb{X}^\circ_0 = \mathbb{X}^\circ$, such that for every $r \in \mathbb{T}$ and every $\hat{Y}^\circ \in \mathbb{X}^\circ_r$ there are $Y^\circ \in \mathbb{X}^\circ$ and an $\mathcal{F}^\circ_r$-measurable κ with

$$\hat{Y}^\circ_t \circ \theta_r \;=\; Y^\circ_{r+t} - Y^\circ_r + \kappa, \qquad t \in \mathbb{T}. \tag{37}$$

We call $\mathbb{X}_r^\circ$ the *problem shifted to* r, and say that $\mathbb{X}^\circ$ is *shift stable* in the narrow sense if $\mathbb{X}_r^\circ = \mathbb{X}^\circ$ for every r is a shift system.

The system is *measurable* if there is a set I and, for each $i \in I$ and $r \in \mathbb{T}$, an element $\hat{Y}^{\circ,r,i} \in \mathbb{X}_r^\circ$ exhausting $\mathbb{X}_r^\circ$, such that $(\omega, r) \mapsto \hat{Y}_t^{\circ,r,i}(\omega)$ is $\mathcal{B}(F) \otimes \mathcal{T}$-measurable for every i and t. For an a.s. finite stopping time τ one then writes $\mathbb{X}_\tau^\circ = \{\hat{Y}^{\circ,\tau(\cdot),i} : i \in I\}$.

Remark 5.11. Requiring $\mathbb{X}_r^\circ = \mathbb{X}^\circ$ — which is what the first version of this manuscript did — is a real restriction and not a normalization: it forces the martingale problem to look the same from every instant, hence forces the clock to be shift invariant and the resulting Markov process to be *homogeneous*. Definition 5.10 follows [5], III.2.39 instead, where a whole family of shifted triplets $(\rho_r B, \rho_r C, \rho_r \nu)$ is carried along and $(\rho_r B)_t(\theta_r \omega) = B_{r+t}(\omega) - B_r(\omega)$; the measurability clause is their III.2.39(i). The cost is nil: the proofs below use only that the *shifted* process is a P-martingale, never that it belongs to the original family. The gain is the time-inhomogeneous case, and with it the removal of shift invariance of the clock from the hypotheses; see Remark 5.21(ii) and Example 5.12.

Example 5.12 (The shifted problem for $\mathbb{X}_A$; (T0) + (T4))**.** Let q be a clock on $\mathbb{T}$, let $A \subset M(\mathbb{T} \times E) \times M(E)$ allow a time-dependent coefficient, and put

$$\mathbb{X}^\circ \;=\; \Big\{ \big(f(\pi_t) - \int_{\langle 0,t\rangle_\iota} g(s, \pi_s)\, q(\,\mathrm{d}s) \big)_{t \in \mathbb{T}} \;:\; (f,g) \in A \Big\}.$$

Define the *pulled-back clock* q_r by $q_r(A) = q(r + A)$, i.e. the image of q restricted to $\{v : v \geq r\}$ under $v \mapsto v - r$, and set

$$\mathbb{X}_r^\circ \;=\; \Big\{ \big(f(\pi_t) - \int_{\langle 0,t\rangle_\iota} g(r+u, \pi_u)\, q_r(\,\mathrm{d}u) \big)_{t \in \mathbb{T}} \;:\; (f,g) \in A \Big\}.$$

Then $(\mathbb{X}_r^\circ)$ is a shift system, with $\kappa = f(\pi_r)$: substituting $v = r + u$,

$$\begin{aligned} \hat{Y}_t^\circ(\theta_r \omega) &= f(\omega(r+t)) - \int_{\langle 0,t\rangle_\iota} g(r+u, \omega(r+u))\, q_r(\,\mathrm{d}u) \\ &= f(\omega(r+t)) - \int_{\langle r,\, r+t\rangle_\iota} g(v, \omega(v))\, q(\,\mathrm{d}v), \end{aligned}$$

which is $Y_{r+t}^\circ(\omega) - Y_r^\circ(\omega) + f(\omega(r))$. Note $q_0 = q$ and $\mathbb{X}_0^\circ = \mathbb{X}^\circ$.

The shifted problem always exists. It coincides with the original one, i.e. $\mathbb{X}_r^\circ = \mathbb{X}^\circ$ for all r, precisely when $q_r = q$ and g does not depend on time — that is, when the clock is shift invariant in the sense of Definition 2.3. The system is measurable as soon as $(r, u, x) \mapsto g(r+u, x)$ is, which for $q =$ Lebesgue measure and measurable g is automatic.

Lemma 5.13 (Restart; (T0) + (T4) + (E0))**.** *Let $(\mathbb{X}_r^\circ)$ be a shift system for $\mathbb{X}^\circ$ and let $\mathbb{Z}^\circ$ be a determining set for every $\mathbb{X}_r^\circ$ (Definition 3.3). Let X be a process with paths in F on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{G}, P)$, solving the martingale problem for $\mathbb{X}^\circ$ with respect to $\mathbb{G}$; let $r \in \mathbb{T}$ and let $Z \geq 0$ be bounded and $\mathcal{G}_r$-measurable with $E^P[Z] = 1$. Then*

$$R \;:=\; (Z \cdot P) \circ \big(X(r + \cdot) \big)^{-1} \;\in\; \mathcal{M}(\mathbb{X}_r^\circ). \tag{38}$$

In the canonical case $\Omega = F$, $\mathbb{G} = \mathbb{F}^\circ$, $X = \pi$ this reads $R = (Z \cdot P) \circ \theta_r^{-1}$ with Z being $\mathcal{F}_r^\circ$-measurable.

Remark 5.14. The two levels matter and cannot be collapsed. In Theorem 5.16(a) the weight is $Z = \mathbf{1}_{F_0}$ with $F_0 \in \mathcal{G}_r$, and $\mathcal{G}_r$ may be strictly larger than $\sigma(X(u) : u \leq r)$; such a Z is *not* a functional of the path and cannot be transported to F. What can be transported is the *law* of $X(r + \cdot)$ under $Z \cdot P$, which is what (38) asserts.

Proof. Let $\hat{Y}^\circ \in \mathbb{X}^\circ_r$, let $s \le t$ and let $Z^\circ_s \in \mathbb{Z}^\circ_s$. Choose $Y^\circ \in \mathbb{X}^\circ$ and κ as in (37), and evaluate everything along X; to keep the notation readable write $\tilde{V} = V(X)$ for a functional V on F, so that $\tilde{Y}^\circ_u$ is a $\mathbb{G}$-martingale by hypothesis.

Since Z°_s is $\mathcal{F}^\circ_s$-measurable and $\pi_u \circ \theta_r = \pi_{r+u}$, the variable $(Z^\circ_s \circ \theta_r)(X) = Z^\circ_s\big(X(r+\cdot)\big)$ is $\sigma(X(u) : u \le r+s)$-measurable, hence $\mathcal{G}_{r+s}$-measurable; so is Z, because $\mathcal{G}_r \subset \mathcal{G}_{r+s}$, and so is $\tilde{\kappa}$. Hence $W := Z \cdot (Z^\circ_s \circ \theta_r)(X)$ is bounded and $\mathcal{G}_{r+s}$-measurable, and the martingale property gives $E^P[(\tilde{Y}^\circ_{r+t} - \tilde{Y}^\circ_{r+s})W] = 0$. Therefore

$$\begin{aligned} E^R\big[\hat{Y}^\circ_t Z^\circ_s\big] &= E^P\big[Z\,(\hat{Y}^\circ_t \circ \theta_r)(X)\,(Z^\circ_s \circ \theta_r)(X)\big] = E^P\big[(\tilde{Y}^\circ_{r+t} - \tilde{Y}^\circ_r + \tilde{\kappa})W\big] \\ &= E^P\big[(\tilde{Y}^\circ_{r+s} - \tilde{Y}^\circ_r + \tilde{\kappa})W\big] = E^R\big[\hat{Y}^\circ_s Z^\circ_s\big], \end{aligned}$$

the second equality by (37) and the third by the display above. As $Z^\circ_s \in \mathbb{Z}^\circ_s$ was arbitrary, part (ii) of Definition 3.3 yields $E^R[\hat{Y}^\circ_t \mid \mathcal{F}^\circ_s] = \hat{Y}^\circ_s$; integrability of $\hat{Y}^\circ_t$ under R holds because Z is bounded and $\tilde{Y}^\circ_{r+t}, \tilde{Y}^\circ_r, \tilde{\kappa} \in L^1(P)$. □

Note where the two hypotheses act. Shift stability is used only to rewrite $\hat{Y}^\circ_t \circ \theta_r$; the determining set is used only in the last step, and it is exactly the rôle that Proposition 3.8 plays in [4]. Everything else is the tower property.

Lemma 5.15 (Agreement propagates under a shift system; (T0) + (T4) + (E0))**.** *Let $(\mathbb{X}^\circ_r)$ be a shift system for $\mathbb{X}^\circ$ with $\mathbb{Z}^\circ$ determining for each $\mathbb{X}^\circ_r$, and suppose that for every $r \in \mathbb{T}$*

$$P, Q \in \mathcal{M}(\mathbb{X}^\circ_r),\ P\pi_0^{-1} = Q\pi_0^{-1} \quad \Longrightarrow \quad P\pi_t^{-1} = Q\pi_t^{-1} \ \ \text{for all } t \in \mathbb{T}. \tag{39}$$

Then $\mathcal{M}(\mathbb{X}^\circ)$ propagates agreement.

Proof. Let $P, Q \in \mathcal{M}(\mathbb{X}^\circ)$, let $s \le t$, and let $Z \ge 0$ be bounded and $\mathcal{F}^\circ_s$-measurable with

$$E^P[Z\,h(\pi_s)] = E^Q[Z\,h(\pi_s)] \qquad \text{for all } h \in B(E). \tag{40}$$

Taking $h \equiv 1$ gives $c := E^P[Z] = E^Q[Z]$. If $c = 0$ then $Z = 0$ both P- and Q-a.s. and the conclusion is trivial, so assume $c > 0$ and replace Z by Z/c, i.e. assume $E^P[Z] = E^Q[Z] = 1$.

By Lemma 5.13, $R_P = (Z \cdot P) \circ \theta_s^{-1}$ and $R_Q = (Z \cdot Q) \circ \theta_s^{-1}$ both lie in $\mathcal{M}(\mathbb{X}^\circ_s)$ — the *same* shifted problem, which is what makes (39) applicable. Since $\pi_0 \circ \theta_s = \pi_s$,

$$E^{R_P}[h(\pi_0)] = E^P[Z\,h(\pi_s)] = E^Q[Z\,h(\pi_s)] = E^{R_Q}[h(\pi_0)]$$

for every $h \in B(E)$ by (40), so R_P and R_Q have the same initial law. Hence (39) gives $E^{R_P}[f(\pi_u)] = E^{R_Q}[f(\pi_u)]$ for every $u \in \mathbb{T}$ and $f \in B(E)$. Taking $u = t - s$, which exists by (T4), and using $\pi_u \circ \theta_s = \pi_{s+u} = \pi_t$, this reads $E^P[Zf(\pi_t)] = E^Q[Zf(\pi_t)]$. That is (34). □

Theorem 5.16 (Abstract uniqueness and Markov property; (T0) + (T4), and (T2a) for (b))**.** *Let $(\mathbb{X}^\circ_r)$ be a shift system for $\mathbb{X}^\circ$, let $\mathbb{Z}^\circ$ be determining for every $\mathbb{X}^\circ_r$, and suppose (39) holds for every $r \in \mathbb{T}$. Then, subject to the integrability provisos of Lemmas 5.2 and 5.13:*

(a) every solution is Markov — in general time inhomogeneously *so: if X has paths in F and solves the martingale problem with respect to $(\mathcal{G}_t)$, then for $f \in B(E)$, $r, t \in \mathbb{T}$,*

$$E\big[f(X(r+t)) \bigm| \mathcal{G}_r\big] \;=\; E\big[f(X(r+t)) \bigm| X(r)\big];$$

(b) if $\mathbb{T}$ satisfies (T2a), then $\mathcal{M}(\mathbb{X}^\circ, \mu)$ contains at most one element, for every $\mu \in \mathcal{P}(E)$.

Proof. *(a).* Fix $r \in \mathbb{T}$ and $F_0 \in \mathcal{G}_r$ with $P(F_0) > 0$, and put

$$Z_1 = \frac{\mathbf{1}_{F_0}}{P(F_0)}, \qquad Z_2 = \frac{E[\mathbf{1}_{F_0} \mid X(r)]}{P(F_0)}.$$

Both are nonnegative, bounded by $1/P(F_0)$, of mean 1, and $\mathcal{G}_r$-measurable — Z_2 is even $\sigma(X(r))$-measurable, while Z_1 need not be a functional of the path, which is why Lemma 5.13 is stated on $(\Omega, \mathcal{F}, \mathbb{G}, P)$ and not on F; see Remark 5.14. Applying Lemma 5.13 at r, the two laws $R_i = (Z_i \cdot P) \circ (X(r+\cdot))^{-1}$ lie in $\mathcal{M}(\mathbb{X}_r^\circ)$ — the *same* shifted problem for $i = 1$ and $i = 2$, which is all that (39) needs. A direct computation gives, for $B \in \mathcal{B}(E)$,

$$R_1\pi_0^{-1}(B) = \frac{E[\mathbf{1}_{F_0}\mathbf{1}_B(X(r))]}{P(F_0)} = R_2\pi_0^{-1}(B),$$

the middle expression for $i = 2$ because $E[\mathbf{1}_{F_0}\mathbf{1}_B(X(r))] = E\big[E[\mathbf{1}_{F_0} \mid X(r)]\mathbf{1}_B(X(r))\big]$. So (39) applies and yields $E^{R_1}[f(\pi_t)] = E^{R_2}[f(\pi_t)]$, i.e.

$$E\big[\mathbf{1}_{F_0}\, E[f(X(r+t)) \mid \mathcal{G}_r]\big] \;=\; E\big[\mathbf{1}_{F_0}\, E[f(X(r+t)) \mid X(r)]\big],$$

where on the right we used the $\sigma(X(r))$-measurability of Z_2 once more. Since $F_0 \in \mathcal{G}_r$ was arbitrary, the Markov property follows.

(b). Lemma 5.15 and Proposition 5.9. □

Remark 5.17 (Uniqueness is not a Markovian fact)*.* The decomposition above is worth reading off, because the classical statement hides it.

	needs	**Markovian?**
Uniqueness (Proposition 5.9)	(34)	**no**
(34) from one-dimensional laws (Lemma 5.15)	shift system	yes
Markov property (Theorem 5.16(a))	shift system	yes

Two consequences. First, Proposition 5.9 uses no shift, no θ_r, no re-indexing and *no* bundle (T4): only (T2a), for the chains of Remark 5.18. [4], Theorem 4.4.2 looks Markovian because its hypothesis is stated *unconditionally*, on the one-dimensional distributions; and an unconditional hypothesis can be applied at a later time only after the restarted process has been re-indexed to start at 0, which is what θ_r does and where the shift enters. Restarting *itself* is free, by Lemma 5.5. State the hypothesis conditionally, as (34) does, and the Markov structure disappears from the uniqueness half.

Second, (34) is available in genuinely non-Markovian situations, where no shift system exists:

- *Path-dependent equations.* Pathwise uniqueness gives more than uniqueness in law: it gives uniqueness in law of the equation *restarted from a prescribed past*, and that is what (34) asks for. Plain uniqueness in law is *not* enough — (34) is a conditional statement, and Remark 5.8 explains why the unconditional one does not suffice. The coefficients may depend on the whole past.

- *Change of measure.* If $\mathbb{X}^\circ$ arises from a family with (34) by an absolutely continuous change of measure with a path-dependent density, the property is inherited.

- *Semimartingale problems.* [5], Chapter III prove uniqueness for triplets (B, C, ν) that are predictable functionals of the path, with no Markov structure in sight.

None of these is developed here; the point is only that Proposition 5.9 is the statement they would use, and Theorem 5.16(b) is not.

Remark 5.18 (Why (b) needs a linear order). Part (a) uses nothing about $\mathbb{T}$ beyond (T0) and (T4): the proof needs θ_r, the inclusion $\mathcal{F}^\circ_r \subset \mathcal{F}^\circ_{r+s}$ and the martingale property at $r+s \le r+t$, all of which hold in a preordered monoid. There is no topology and no linear order in it.

Part (b) is different, and the reason is the induction of Proposition 5.9, not the shift. That induction advances along $\mathcal{F}^\circ_{t_1} \subset \mathcal{F}^\circ_{t_2} \subset \dots$, so the times must form a *chain*. On a linearly ordered $\mathbb{T}$ every finite set of times is a chain and nothing is lost. On a merely partially ordered $\mathbb{T}$ one obtains equality of the finite-dimensional distributions along chains only, and these do not determine the law: on $\mathbb{T} = \mathbb{R}^2_+$ the times $(1,0)$ and $(0,1)$ are incomparable, and no chain sees their joint distribution. Replacing t_n by $t_1 \vee \dots \vee t_n$, which (T1) would allow, does not repair this, since the next time t_{n+1} need not dominate that join.

So the slogan "uniqueness of the one-dimensional distributions implies uniqueness" is *false* for a multiparameter index, and the obstruction is the induction, not the martingale problem. This is the sharpest instance of the general pattern of Section 2.2: the Markov property is algebraic and survives on a poset, uniqueness is order-theoretic and does not.

For the strong Markov property one restarts at a stopping time. The only extra input is optional sampling, which is where path regularity finally enters.

Theorem 5.19 (Abstract strong Markov property; (T2b) + (T4))**.** *In the situation of Theorem 5.16, assume in addition that $F \subset D_E[0,\infty)$, that (θ_r) extends to a measurable map $(\omega, r) \mapsto \theta_r\omega$, that the shift system is measurable in the sense of Definition 5.10, and that every $Y^\circ \in \mathbb{X}^\circ$ has right continuous paths and uniformly integrable increments on bounded intervals. Let X have paths in F, solve the martingale problem with respect to $(\mathcal{G}_t)$, and let τ be an a.s. finite $(\mathcal{G}_t)$-stopping time. Then*

$$E\big[f(X(\tau+t)) \,\big|\, \mathcal{G}_\tau\big] \;=\; E\big[f(X(\tau+t)) \,\big|\, X(\tau)\big], \qquad f \in B(E),\ t \in \mathbb{T}.$$

If moreover τ takes countably many values and there is a measurable family $(P_{x,r})_{x\in E,\, r\in\mathbb{T}}$ with $P_{x,r} \in \mathcal{M}(\mathbb{X}^\circ_r, \delta_x)$, then

$$E\big[f(X(\tau+t)) \,\big|\, \mathcal{G}_\tau\big] \;=\; \big(T_{\tau,\tau+t}f\big)(X(\tau)), \qquad \big(T_{r,r+t}f\big)(x) \;=\; \int f(\pi_t)\,\mathrm{d}P_{x,r},$$

a two-parameter family of kernels satisfying the Chapman–Kolmogorov relation $T_{r,s}T_{s,t} = T_{r,t}$ for $r \le s \le t$.

Proof. Step 1: the restart identity at a stopping time. The proof of Lemma 5.13 rests on the single identity $E^P[(\tilde{Y}^\circ_{r+t} - \tilde{Y}^\circ_{r+s})W] = 0$ for bounded $\mathcal{G}_{r+s}$-measurable W. We need it with r replaced by τ. Now $\tau+s$ and $\tau+t$ are $(\mathcal{G}_u)$-stopping times: by (T4) the map $u \mapsto u-s$ is defined on $\{u \ge s\}$ and order preserving, so $\{\tau+s \le u\} = \{\tau \le u-s\} \in \mathcal{G}_{u-s} \subset \mathcal{G}_u$ for $u \ge s$, and the event is empty otherwise. Since $\tilde{Y}^\circ$ is a right continuous martingale with uniformly integrable increments on bounded intervals, optional sampling (Fact 2.20) gives $E^P[\tilde{Y}^\circ_{\tau+t} \mid \mathcal{G}_{\tau+s}] = \tilde{Y}^\circ_{\tau+s}$, hence the identity for every bounded $\mathcal{G}_{\tau+s}$-measurable W.

Step 2: the first assertion. With $\mathbb{X}^\circ_\tau$ in place of $\mathbb{X}^\circ_r$ — meaningful because the shift system is measurable, so that $\hat{Y}^{\circ,\tau(\omega)}$ is jointly measurable — the remainder of the proof of Lemma 5.13 and of Theorem 5.16(a) is unchanged.

Step 3: the second assertion. Let τ take the countably many values $r_1, r_2, \dots$, and fix j. It suffices to prove the identity on $\{\tau = r_j\}$, since these events partition Ω and lie in $\mathcal{G}_\tau$.

Let $F_0 \in \mathcal{G}_\tau$ with $F_0 \subset \{\tau = r_j\}$ and $P(F_0) > 0$, write $\mu_{F_0} = P\{X(\tau) \in \cdot \mid F_0\}$, and put

$$R_1 = (Z_1 \cdot P) \circ (X(\tau + \cdot))^{-1} \text{ with } Z_1 = \mathbf{1}_{F_0}/P(F_0), \qquad R_2 = \int P_{x,r_j}\, \mu_{F_0}(\mathrm{d}x).$$

Both lie in $\mathcal{M}(\mathbb{X}^\circ_{r_j})$: the first by Steps 1–2, the second by Lemma 5.2, and it is here that F_0 must be carried by a single value of τ — a mixture of solutions of *different* shifted problems need solve none of them. Since $R_1\pi_0^{-1} = \mu_{F_0} = R_2\pi_0^{-1}$, (39) applies and gives the assertion on F_0. The Chapman–Kolmogorov relation follows by applying the first assertion twice. □

Remark 5.20 (On the two restrictions). Two hypotheses in Theorem 5.19 deserve comment.

Countably many values. In the time-homogeneous case $\mathbb{X}^\circ_r = \mathbb{X}^\circ$ the restriction is vacuous, because then all the measures $P_{x,r} = P_x$ solve one and the same problem and Step 3 needs no partition. It is only in the inhomogeneous case that the mixture must be confined to a single r. A general τ is then reached by approximation from above by countably-valued stopping times $\tau_n \downarrow \tau$; this requires right continuity of $\mathbb{G}$ and of $r \mapsto P_{x,r}$ in a suitable sense, and is not developed here.

Uniformly integrable increments. This is what Step 1 needs in order to apply optional sampling at the unbounded times $\tau + s$; for the family $\mathbb{X}_A$ with bounded f and g the increments on $\mathbb{T}_{\le T}$ are bounded by $2\|f\| + \|g\|q(\mathbb{T}_{\le T})$, so the hypothesis holds there.

Remark 5.21 (What the abstract form buys). (i) *One lemma, four applications.* In [4] the conditioning construction in the proof of the Markov property, its variant in the induction step of the uniqueness proof, and the two constructions in parts (b) and (c) look like four separate arguments. They are one: Lemma 5.13 with, in turn,

$$\begin{aligned} &Z = \mathbf{1}_{F_0}, && Z = E[\mathbf{1}_{F_0} \mid X(r)], \\ &Z = \mathbf{1}_G \text{ for } G \in \mathcal{F}^\circ_s, && Z = \mathbf{1}_{F_0} \text{ at a stopping time,} \end{aligned}$$

all normalized to mean 1. The only property of Z that is ever used is that it is nonnegative and $\mathcal{F}^\circ_r$-measurable. The third of these is Lemma 5.15; [4] use $Z = \prod_k f_k(\pi_{t_k})$ there, which is the same lemma applied to a tilted measure instead of a conditioned one, and is unnecessary once the induction is arranged as in Proposition 5.9.

(ii) *The shift system is the real hypothesis, and it costs nothing.* Definition 5.10 is where the Markov property comes from, and it is checkable. For $\mathbb{X}_A$ it never fails: whatever the clock q and however g depends on time, the shifted problem exists and is given explicitly by Example 5.12, with the pulled-back clock $q_r = q(r + \cdot)$ and the shifted coefficient $g(r + \cdot, \cdot)$. What *does* depend on q and g is whether the shifted problem is the *same* problem: $\mathbb{X}^\circ_r = \mathbb{X}^\circ$ holds if and only if q is shift invariant and g is time independent, and only then is the Markov process homogeneous. The first version of this manuscript required $\mathbb{X}^\circ_r = \mathbb{X}^\circ$ and therefore had to assume shift invariance of the clock; carrying the shifted family instead — as [5], III.2.39 do — removes that hypothesis and yields the inhomogeneous case for free. [4] never have to say any of this because their clock is Lebesgue measure and their A does not depend on time.

(iii) *No operator, no boundedness, no path regularity.* Parts (a) and (b) use neither $C_b(E)$ nor $D_E[0, \infty)$; F may be any Polish path space, for instance L^p_{loc}. Path regularity enters only in Theorem 5.19, and there only through optional sampling.

(iv) *Mixtures.* Lemma 5.2 says $\mathcal{M}(\mathbb{X}^\circ)$ is convex and stable under measurable mixtures. This is what makes the passage from δ_x to a general μ legitimate — in [4] it is invoked silently in part (c) and in Corollary 6.18.

5.3 The local martingale problem

Everything in Section 5.1 is stated for $\mathcal{M}(\mathbb{X}^\circ)$. The local analogue is not a corollary, and the reason is worth stating plainly: *"Y is a local martingale" is not a condition linear in* P, because the localizing sequence may depend on P. Lemma 5.2 and Lemma 5.3 rest on that linearity, and Lemma 5.13 rests on a martingale identity that a local martingale does not supply. All three therefore fail for $\mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$ without a further hypothesis.

The hypothesis that repairs them is due to Stroock and Varadhan and is used in this form by [7], Theorems 32.10 and 32.11: insist that the localization be performed by stopping times that are *functionals of the path*, hence the same for every P, and that the family of such times be stable under the shift.

Definition 5.22 (Localizing system; (T2b) + (T4))**.** A *localizing system* for $\mathbb{X}^\circ$ is a family Σ of *strict* $(\mathcal{F}^\circ_t)$-stopping times on F — functionals of the path, not of any measure — such that the following hold. If the solution is presented on an ambient space $(\Omega, \mathcal{F}, \mathbb{G}, P)$ rather than on F, all three conditions are to be read with $\mathcal{F}^\circ$ replaced by $\mathbb{G}$ and Y° by $\tilde{Y} = Y^\circ(X)$; strictness of the times in Σ is what makes $\sigma(X)$ a $\mathbb{G}$-stopping time for every such presentation.

(L1) *Uniform localization.* Σ contains a sequence $\tau_1 \le \tau_2 \le \dots$ with $\tau_n \uparrow \infty$ pointwise, and

$$P \in \mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ) \iff Y^{\circ,\tau_n} := Y^\circ_{\cdot\wedge\tau_n} \text{ is a } P\text{-martingale for all } Y^\circ \in \mathbb{X}^\circ,\ n \in \mathbb{N}.$$

(L2) *Shift covariance.* $\sigma \in \Sigma$ and $r \in \mathbb{T}$ imply $r + \sigma \circ \theta_r \in \Sigma$.

(L3) *Integrable increments after a restart.* For every $P \in \mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$, $Y^\circ \in \mathbb{X}^\circ$, $r \in \mathbb{T}$ and $\sigma \in \Sigma$ with $\sigma \ge r$, the process $t \mapsto Y^\circ_{(r+t)\wedge\sigma} - Y^\circ_r$ is a martingale for the filtration $(\mathcal{F}^\circ_{r+t})_{t\in\mathbb{T}}$ under P.

Σ is *strongly* shift covariant if (L2) and (L3) hold with r replaced by any a.s. finite $(\mathcal{F}^\circ_t)$-stopping time τ, the time $\tau + \sigma \circ \theta_\tau$ being required to be a stopping time.

Remark 5.23. Three comments on Definition 5.22.

(L1) is the whole point. It turns membership in $\mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$ into a family of identities $E^P[(Y^{\circ,\tau_n}_t - Y^{\circ,\tau_n}_s)\mathbf{1}_G] = 0$, each of which is linear in P. The implication "$\Leftarrow$" is automatic because $\tau_n \uparrow \infty$; "$\Rightarrow$" is the assumption, and it says that *one* sequence localizes every test process under every solution.

(L3) is not implied by (L1). Stopping a P-local martingale at $\sigma \in \Sigma$ leaves a local martingale, not a martingale: $Y^{\circ,\sigma}$ is unstopped before r, where nothing controls it. What (L3) asks is only that the *increments after* r be integrable, which is the same kind of hypothesis as in Theorem 5.19 and is usually visible at once.

The diffusion case. For $\mathbb{X}^\circ = \{f(\pi) - f(\pi_0) - \int Af(\pi)\}$ on $F = C_{\mathbb{R}_+,\mathbb{R}^d}$ take Σ generated by the exit times $\tau_n = \inf\{t : |\pi_t| \ge n\}$ and their shifts. Then (L1) is [7], Theorem 32.10, (L2) holds because $r + \tau_n \circ \theta_r = \inf\{u \ge r : |\pi_u| \ge n\}$ is again an exit time, and (L3) holds because between r and σ the path stays in $\overline{B}_n$, so that the increments are bounded by $2\|f\|_{\overline{B}_n} + q(\mathbb{T}_{\le T})\|Af\|_{\overline{B}_n}$. [7] uses $\tau^f_n = \inf\{t : |M^f_t| \ge n\}$ instead, which serves equally well and has the advantage of not referring to the state space.

With Definition 5.22 in hand the three lemmas of Section 5.1 all go through, and their proofs are the earlier ones with Y° replaced by Y°,τ_n}. We give them in full, since the point of the exercise is that nothing is being waved through.

Lemma 5.24 (Local mixtures and disintegration; (T0) + (E1))**.** *Parts (a) and (b) need only (E0); (E1) is used in (c).*

(a) $\mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$ *is convex.* No hypothesis is needed for this.

(b) If Σ *satisfies (L1), then* $\mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$ *is stable under measurable mixtures* $P = \int P_\vartheta \nu(\mathrm{d}\vartheta)$ *subject to* $\int E^{P_\vartheta}|Y_t^{\circ,\tau_n}|\,\nu(\mathrm{d}\vartheta) < \infty$.

(c) If in addition membership can be tested along countably many data as in (33)*, then for* $P \in \mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$ *one has* $P(\cdot \mid \pi_0 = x) \in \mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ, \delta_x)$ *for* $P\pi_0^{-1}$*-a.e.* x.

Proof. (a) Let $P, P' \in \mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$, $Q = \alpha P + (1-\alpha)P'$, and let $Y^\circ \in \mathbb{X}^\circ$ be localized by (τ_n) under P and by (τ_n') under P'. Put $\sigma_n = \tau_n \wedge \tau_n'$. Then Y°,σ_n} is a martingale under P and under P' — stopping a martingale at a further stopping time leaves a martingale — hence under Q, and $\sigma_n \to \infty$ Q-a.s. because $\tau_n \to \infty$ P-a.s. and $\tau_n' \to \infty$ P'-a.s. Here the two localizing sequences are allowed to differ; it is only for (b) that this ceases to be possible.

(b), (c) By (L1), $\mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ) = \mathcal{M}(\mathbb{X}^\circ_\bullet)$ with $\mathbb{X}^\circ_\bullet = \{Y^{\circ,\tau_n} : Y^\circ \in \mathbb{X}^\circ, n \in \mathbb{N}\}$, a family of $(\mathcal{F}_t^\circ)$-adapted processes — stopping a process at a stopping time preserves adaptedness once the process is progressively measurable, which under (T2b) a càdlàg adapted process is. Now apply Lemma 5.2 respectively Lemma 5.3 to $\mathbb{X}^\circ_\bullet$; neither proof uses anything about the family beyond adaptedness, and the countability hypothesis for (c) is inherited because n ranges over $\mathbb{N}$. ☐

Remark 5.25. The contrast between (a) and (b) is the whole point of (L1), and it is easy to get wrong. A *finite* convex combination costs nothing, because two localizing sequences can be combined by $\tau_n \wedge \tau_n'$. A mixture over a continuum cannot: one would need $\inf_\vartheta \tau_n^\vartheta$, which need not tend to infinity and need not even be measurable. That is exactly where a localizing sequence common to all P becomes indispensable, and it is why [7] states his Theorem 32.10 for a *kernel* $x \mapsto P_x$ and takes the trouble to exhibit one.

The hypothesis (L1) is less exotic than it looks: in the two places in the literature where it is used it is not assumed but *constructed*, and always by the same device — localizing along the test process itself rather than along the state space.

Lemma 5.26 (Uniform localization for bounded jumps; (T2b))**.** *Suppose every* $Y^\circ \in \mathbb{X}^\circ$ *is càdlàg with* $Y_0^\circ = 0$ *and has jumps bounded by a constant* c_{Y°*. Write* $S_t^{Y^\circ} = \sup_{s \le t}|Y_s^\circ|$ *for the running supremum and put*

$$\tau_n^{Y^\circ} \;=\; \inf\{t \in \mathbb{T} : S_t^{Y^\circ} \ge n\}, \qquad n \in \mathbb{N}.$$

Then each $\tau_n^{Y^\circ}$ *is a* strict $(\mathcal{F}_t^\circ)$*-stopping time, and* $\Sigma_0 = \{\tau_n^{Y^\circ} : Y^\circ \in \mathbb{X}^\circ,\ n \in \mathbb{N}\}$ *satisfies (L1).*

Proof. Fix $Y = Y^\circ$ and drop the superscript.

Step 1: S is càdlàg, nondecreasing and adapted. Monotonicity is clear. Right continuity: given t and $\varepsilon > 0$, right continuity of Y gives $u > t$ with $|Y_s| \le |Y_t| + \varepsilon \le S_t + \varepsilon$ for $s \in [t, u)$, whence $S_s \le S_t + \varepsilon$ there. Left limits exist because S is monotone. For adaptedness note that, again by right continuity of Y and the density hypothesis in (T2b),

$$S_t \;=\; \sup\bigl\{|Y_s| : s \in (D \cap \mathbb{T}_{\le t}) \cup \{t\}\bigr\}, \tag{41}$$

a countable supremum of $\mathcal{F}_t^\circ$-measurable functions. Indeed "$\ge$" is trivial, and for "$\le$" let $s \le t$; if $s = t$ there is nothing to prove, and if $s < t$ then s is non-maximal in $\mathbb{T}_{\le t}$, so by

(T2b) there are $s_k \in D$ with $s < s_k$, $s_k \to s$ from the right and $s_k \le t$ (replace s_k by $s_k \wedge t$, which lies in D or equals t), and $|Y_{s_k}| \to |Y_s|$.

Step 2: τ_n is a strict stopping time. We claim

$$\{\tau_n \le t\} \;=\; \{S_t \ge n\}, \tag{42}$$

which lies in $\mathcal{F}^\circ_t$ by (41). If $S_t \ge n$ then t belongs to the set whose infimum is τ_n, so $\tau_n \le t$. Conversely let $\tau_n \le t$. If t is maximal, then $S_u \ge n$ for some $u \le t$ and monotonicity gives $S_t \ge n$. Otherwise, for every $u > t$ there is $v < u$ with $S_v \ge n$, hence $S_u \ge n$; letting $u \downarrow t$ along D — possible by (T2b) — right continuity of S gives $S_t \ge n$. This is where the *closed* threshold $S_t \ge n$ matters: $\inf\{t : |Y_t| > n\}$, the choice of [5], satisfies (42) only with $\{S_t > n\}$ replaced by an event that is not $\mathcal{F}^\circ_t$-measurable, and is a stopping time for $\mathcal{F}^\circ_{t+}$ only. See Remark 5.27.

Step 3: the stopped process is bounded. On $\{s < \tau_n\}$ we have $S_s < n$, hence $|Y_s| < n$; consequently $|Y_{\tau_n -}| \le n$, and $|Y_{\tau_n}| \le |Y_{\tau_n -}| + |\Delta Y_{\tau_n}| \le n + c_Y$. Thus $|Y^{\tau_n}| \le n + c_Y$ everywhere.

Step 4: (L1). A bounded local martingale is a uniformly integrable martingale, so for every $P \in \mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$ the process Y^{τ_n} is a P-martingale; this is the implication "$\Rightarrow$". For "$\Leftarrow$" note that $\tau_n \uparrow \infty$ because a càdlàg path is bounded on bounded intervals, so $S_t < \infty$ for each t and hence $\tau_n > t$ for $n > S_t$. $\square$

Remark 5.27 (Strict versus right continuous débuts). The distinction in Step 2 is not pedantry, and it is the one place in the local theory where the raw filtration bites. For a càdlàg adapted Y and an *open* set G the début $\inf\{t : Y_t \in G\}$ satisfies $\{\tau < t\} = \bigcup_{s \in D,\, s<t}\{Y_s \in G\}$ and is therefore a stopping time for $\mathcal{F}^\circ_{t+}$, but in general not for $\mathcal{F}^\circ_t$: a path with $|Y_s| \le n$ for $s \le t$ and $|Y_s| > n$ for $s > t$ has $\tau = t$, yet it agrees on $\mathbb{T}_{\le t}$ with a path that never exceeds n. Replacing the threshold by the running supremum repairs this, because $\{S_t \ge n\}$ is decided by the path up to and including t. For *continuous* paths and a closed set the début is strict without any such device, which is why the exit times of Remark 5.23 cause no trouble. In the formalization this is the difference between `IsStoppingTime f` and `IsStoppingTime (f.rightLimit)` and has to be tracked.

Remark 5.28. Lemma 5.26 is the argument of [5], III.2.8, where it proves that the set of solutions of a semimartingale martingale problem is convex: they take $T_n = \inf\{t : |Y_t| > n\}$ and observe that $Y_0 = 0$ together with bounded $|\Delta Y|$ makes Y^{T_n} bounded. [5] work throughout with a filtration satisfying the usual conditions, so for them this T_n is a stopping time; here the raw filtration forces the running-supremum variant of Lemma 5.26, and the boundedness argument is unaffected. [7] uses the same device, $\tau^f_n = \inf\{t : |M^f_t| \ge n\}$, in his Theorem 32.10. Both localize along the *test process*, not along the state space; the exit times from balls in Remark 5.23 are the less canonical choice, since they refer to E and presuppose that f and g are bounded on compacts.

Lemma 5.29 (Local restart; (T2b) + (T4))**.** *Let $(\mathbb{X}^\circ_r)$ be a shift system for $\mathbb{X}^\circ$ with $\mathbb{Z}^\circ$ determining for each $\mathbb{X}^\circ_r$, and let Σ be a localizing system. Let X be a process on $(\Omega, \mathcal{F}, \mathbb{G}, P)$ solving the local martingale problem for $\mathbb{X}^\circ$ with respect to $\mathbb{G}$ and Σ, let $r \in \mathbb{T}$, and let $Z \ge 0$ be* bounded *and $\mathcal{G}_r$-measurable with $E^P[Z] = 1$. Then*

$$R \;\coloneqq\; (Z \cdot P) \circ \big(X(r + \cdot)\big)^{-1} \;\in\; \mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ_r).$$

On the canonical space this is $R = (Z \cdot P) \circ \theta_r^{-1}$. The same holds with r replaced by an a.s. finite $\mathbb{G}$-stopping time τ and $\mathcal{G}_r$ by $\mathcal{G}_\tau$, provided Σ is strongly shift covariant.

Proof. By (L1) it suffices to show that $\hat{Y}^{\circ,\tau_n}$ is an R-martingale for every $\hat{Y}^\circ \in \mathbb{X}_r^\circ$ and $n \in \mathbb{N}$, and by the determining property it suffices to show, for $s \le t$ and $Z_s^\circ \in \mathbb{Z}_s^\circ$,

$$E^R\big[(\hat{Y}_t^{\circ,\tau_n} - \hat{Y}_s^{\circ,\tau_n})Z_s^\circ\big] \;=\; 0. \tag{43}$$

Step 1: the shifted stopped process. Let $Y^\circ \in \mathbb{X}^\circ$ and the $\mathcal{F}_r^\circ$-measurable κ be as in (37), and write $\tilde{Y} = Y^\circ(X)$, $\tilde{\kappa} = \kappa(X)$, $\tilde{\tau}_n = \tau_n(X(r+\cdot))$; these are defined because the elements of Σ are path functionals, and $\tilde{\tau}_n$ is a $(\mathcal{G}_{r+t})_t$-stopping time because they are strict. Put $\tilde{\sigma} := r + \tilde{\tau}_n$, the ambient realization of $\sigma = r + \tau_n \circ \theta_r \in \Sigma$ (by (L2)), so that $\tilde{\sigma} \ge r$. For $\omega \in \Omega$ write $u = t \wedge \tilde{\tau}_n(\omega)$. Then, by (37),

$$\hat{Y}_t^{\circ,\tau_n}\big(X(r+\cdot)(\omega)\big) = \hat{Y}_u^\circ\big(X(r+\cdot)(\omega)\big) = \tilde{Y}_{r+u}(\omega) - \tilde{Y}_r(\omega) + \tilde{\kappa}(\omega),$$

and, $r + \cdot$ being an order embedding that commutes with $\wedge$ (Lemma 2.2, using (T2a), which (T2b) supplies),

$$r + u = r + \big(t \wedge \tilde{\tau}_n(\omega)\big) = (r+t) \wedge \big(r + \tilde{\tau}_n(\omega)\big) = (r+t) \wedge \tilde{\sigma}(\omega).$$

Hence

$$\hat{Y}_t^{\circ,\tau_n} \circ X(r+\cdot) \;=\; \tilde{Y}_{(r+t)\wedge\tilde{\sigma}} - \tilde{Y}_r + \tilde{\kappa}, \qquad t \in \mathbb{T}. \tag{44}$$

This is [7]'s display (17) in the proof of his Theorem 32.11, and it is the only computation in the local theory that is not already in Section 5.1.

Step 2: conclusion. Put $W := Z \cdot \big(Z_s^\circ \circ X(r+\cdot)\big)$. As in the proof of Lemma 5.13, W is bounded and $\mathcal{G}_{r+s}$-measurable: Z_s° is $\mathcal{F}_s^\circ$-measurable and $\pi_u \circ X(r+\cdot) = X(r+u)$ is $\mathcal{G}_{r+u}$-measurable, while Z is bounded, $\mathcal{G}_r$-measurable and $\mathcal{G}_r \subset \mathcal{G}_{r+s}$. *Boundedness of Z is used here and cannot be dropped*: (L3) supplies a martingale identity against bounded test variables only, and the stopped increments need not be in any L^p with $p > 1$. Using (44) twice, the terms $-\tilde{Y}_r + \tilde{\kappa}$ cancel, and

$$E^R\big[(\hat{Y}_t^{\circ,\tau_n} - \hat{Y}_s^{\circ,\tau_n})Z_s^\circ\big] = E^P\Big[\big(\tilde{Y}_{(r+t)\wedge\tilde{\sigma}} - \tilde{Y}_{(r+s)\wedge\tilde{\sigma}}\big)W\Big] = 0$$

by (L3), which says exactly that $t \mapsto \tilde{Y}_{(r+t)\wedge\tilde{\sigma}} - \tilde{Y}_r$ is a $(\mathcal{G}_{r+t})$-martingale under P. This is (43).

For a stopping time τ the argument is unchanged, with $\mathcal{G}_{\tau+s}$ in place of $\mathcal{G}_{r+s}$ and with $\tilde{\sigma} = \tau + \tau_n(X(\tau+\cdot))$, a stopping time by hypothesis. $\square$

Theorem 5.30 (Local uniqueness, Markov and strong Markov property)**.** *Let $(\mathbb{X}_r^\circ)$ be a shift system for $\mathbb{X}^\circ$ with $\mathbb{Z}^\circ$ determining for each $\mathbb{X}_r^\circ$, let Σ be a localizing system, and suppose that for every $r \in \mathbb{T}$*

$$P, Q \in \mathcal{M}_{\mathrm{loc}}(\mathbb{X}_r^\circ),\ P\pi_0^{-1} = Q\pi_0^{-1} \quad\Longrightarrow\quad P\pi_t^{-1} = Q\pi_t^{-1} \ \text{ for all } t \in \mathbb{T}. \tag{45}$$

Then Theorem 5.16 holds verbatim with $\mathcal{M}$ replaced by $\mathcal{M}_{\mathrm{loc}}$ throughout — for the uniqueness half nothing has to be redone, since Proposition 5.9 is a statement about an arbitrary family of measures and applies to $\mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$ as it stands; and if Σ is strongly shift covariant and the Y°,τ_n} are right continuous with uniformly integrable increments on bounded intervals, so does Theorem 5.19.

Proof. The proof of Theorem 5.16 uses exactly three things: Lemma 5.13, applied with the densities of Remark 5.21(i); the hypothesis (39); and (19). Replace the first by Lemma 5.29 and the second by (45); the third is a statement about F and is untouched. Every density used is an indicator or a conditional expectation of one, hence nonnegative, bounded by

$1/P(F_0)$ and $\mathcal{G}_r$-measurable, so Lemma 5.29 applies to each; and Proposition 5.9 is about $\mathcal{M}$ only, so it carries over with $\mathcal{M}_{\mathrm{loc}}$ unchanged.

For the strong Markov property, the proof of Theorem 5.19 consists of the stopping time version of Lemma 5.13, which is the second assertion of Lemma 5.29, together with Lemma 5.2 for the mixture $\int P_x \mu_{F_0}(\,\mathrm{d}x)$; the latter is Lemma 5.24(b). Optional sampling (Fact 2.20) is applied to the Y°,τ_n}, which are genuine martingales by (L1). □

Remark 5.31 (What [5] do differently). Three points where [5], Chapter III go beyond the account above.

(i) The shifted problem — adopted. [5], III.2.39 carry a whole family of shifted triplets $(\rho_t B, \rho_t C, \rho_t \nu)$ with $(\rho_t B)_s(\theta_t\omega) = B_{t+s}(\omega) - B_t(\omega)$ and likewise for C and ν, rather than demanding that the shifted problem be the original one. Definition 5.10 is that idea in the present notation, and Example 5.12 is the corresponding computation for $\mathbb{X}_A$. The earlier version of this manuscript required $\mathbb{X}^\circ_r = \mathbb{X}^\circ$ and consequently had to assume that the clock be shift invariant and g time independent; both hypotheses are now gone, and the conclusion is the Markov property in its time-inhomogeneous form. The change costs nothing, because Lemmas 5.13 and 5.29 use only that the *shifted* process is a P-martingale.

(ii) Local uniqueness. [5], Definition III.2.37 isolate a notion strictly stronger than the uniqueness of (45): *local uniqueness* holds if for every strict stopping time T any two solutions of the *stopped* martingale problem agree on $\mathcal{F}_T$. It implies uniqueness ($T \equiv \infty$) and is what absolute-continuity and limit-theorem arguments actually require. Their Theorem III.2.40 shows that uniqueness *does* imply local uniqueness once the problem is of Markovian type in the sense of (i) — precisely, once there is a kernel $P_{x,t}$ solving the shifted problem. This is carried out in Section 5.4 — and, as it turns out, without the Markovian type: Theorem 5.37 needs only a restart kernel, which their construction supplies but which does not presuppose a shift.

(iii) Strict stopping times. The canonical filtration $\mathcal{F}^\circ_t = \sigma(\pi_u : u \le t)$ is not right continuous, and [5], III.2.35 therefore distinguish *strict* stopping times — those with $\{T \le t\} \in \mathcal{F}^\circ_t$ for the raw filtration — from stopping times for the completed right continuous filtration, with $\mathcal{F}^\circ_T \subset \mathcal{F}_T$ and $\mathcal{F}_{T-} \subset \mathcal{F}^\circ_T$ on $\{T > 0\}$. Definition 5.22 *demands* that the times in Σ be strict, and this is a genuine restriction, not a consequence of their being path functionals: the début $\inf\{t : |Y_t| > n\}$ is a path functional and is not strict (Remark 5.27). Strictness is what Lemma 5.26 has to work for, and it is also what allows a localizing system defined on F to be transported to an ambient filtration $\mathbb{G}$, as Lemma 5.29 does. The distinction should be made explicit in the formalization, where Mathlib's `IsStoppingTime` is relative to whatever filtration it is given.

Remark 5.32. Two things are worth extracting.

First, the entire local theory rests on the single identity (44), which says that shifting a stopped test process produces a test process stopped at the *shifted* time. That is (L2), and it is the local form of shift stability. Everything else is either the global argument verbatim or the linearity that (L1) restores.

Second, this route localizes the *test processes* and leaves the operator alone. [4], Section 4.6 — and hence Remark 5.45 — instead replaces A by $A^{(m)} = \{(f\mathbf{1}_{K_m}, g\mathbf{1}_{K_m})\}$. In the abstract setting the present route is the right one, because $\mathbb{X}^\circ$ is primitive here and A is not; it is also the one that [7] takes.

What is *not* obtained is a local version of Theorem 4.3: Definition 5.22 presupposes stopping times, which presuppose the path regularity that Theorem 4.3 exists to produce. See Remark 4.7.

5.4 Local uniqueness

Theorem 5.30 concludes that two solutions with the same initial law have the same one-dimensional — hence all — distributions. [5], Definition III.2.37 isolate a strictly stronger property, and it is the one that absolute-continuity and limit arguments actually use.

Definition 5.33 (Local uniqueness; (T2b) + (T4))**.** Write $\mathbb{X}^{\circ,T} = \{Y^{\circ,T} : Y^\circ \in \mathbb{X}^\circ\}$ for the family stopped at a stopping time T. *Local uniqueness* holds for $\mathbb{X}^\circ$ if for every strict stopping time T (Remark 5.31(iii)) and any two $P, P' \in \mathcal{M}(\mathbb{X}^{\circ,T})$ with $P\pi_0^{-1} = P'\pi_0^{-1}$ one has $P = P'$ on $\mathcal{F}_T^\circ$.

Taking $T \equiv \infty$ recovers uniqueness, so local uniqueness is at least as strong; [5] note that it is strictly stronger in general. [5], Theorem III.2.40 derives it from uniqueness for problems of Markovian type, and it is natural to expect a shift system to be needed. It is not. What is needed are two properties of the path space — one of the stopping time, one supplying solutions after it — and neither mentions a shift.

Definition 5.34 (Strict stopping times; (T2b))**.** A stopping time T on F is *strict* if F is stable under the stopping operator $a_T : F \to F$, $\pi_t \circ a_T = \pi_{t\wedge T}$, and $\mathcal{F}_T^\circ = a_T^{-1}(\mathcal{S})$. Equivalently, a set lies in $\mathcal{F}_T^\circ$ exactly when membership depends on the path only through a_T. This is [5], Lemma III.2.43 for the canonical space, and it is what "strict" buys. Note the consequence $T \circ a_T = T$: a strict stopping time is determined by the stopped path.

The second ingredient is a supply of solutions that continue a given past. In the Markovian case one takes the restarted process to depend on the current *state*; the whole point of what follows is that it may depend on the whole *stopped path*, and that nothing is lost by letting it.

Definition 5.35 (Restart kernel; (T2b))**.** Let T be a strict stopping time. A *restart kernel at* T is a kernel $\alpha \mapsto Q_\alpha$ from $(F, \mathcal{F}_T^\circ)$ to $(F, \mathcal{S})$ such that

(R1) $Q_\alpha\{\beta \in F : a_T\beta = a_T\alpha\} = 1$ — the past is kept;

(R2) for every $Y^\circ \in \mathbb{X}^\circ$ and all $T(\alpha) \le s \le t$, $E^{Q_\alpha}[Y_t^\circ \mid \mathcal{F}_s^\circ] = Y_s^\circ$ — it is a solution from T onwards, in the sense of Lemma 5.5.

By (R1) and Definition 5.34, T is Q_α-a.s. equal to the constant $T(\alpha)$, so the times in (R2) are deterministic: indeed $T \circ a_T = T$ gives $T(\beta) = T(a_T\beta) = T(a_T\alpha) = T(\alpha)$ on the set of full measure in (R1). Two further points. First, the event in (R1) is measurable: $\{(\alpha,\beta) : a_T\alpha = a_T\beta\}$ is the preimage of the diagonal of $F \times F$ under $a_T \times a_T$, and the diagonal is measurable because $\mathcal{S}$ is countably generated and separating — which is (E1). Second, (R2) is a martingale property only *from* $T(\alpha)$ *onwards*; nothing is asked of Y° before that time, where the path is frozen at $a_T\alpha$ anyway.

Lemma 5.36 (Pasting with memory; (T2b))**.** *Let T be a strict stopping time, let $P \in \mathcal{M}(\mathbb{X}^{\circ,T})$ and let (Q_α) be a restart kernel at T. Assume the integrability condition*

$$\int_F E^{Q_\alpha}\big[|Y_t^\circ|\big]\, P(\mathrm{d}\alpha) \;<\; \infty \qquad \textit{for every } Y^\circ \in \mathbb{X}^\circ \textit{ and } t \in \mathbb{T}.$$

Then

$$Q \;\coloneqq\; \int_F Q_\alpha\, P(\mathrm{d}\alpha) \;\in\; \mathcal{M}(\mathbb{X}^\circ), \qquad \textit{and} \qquad Q = P \quad \textit{on } \mathcal{F}_T^\circ.$$

Proof. Agreement on $\mathcal{F}_T^\circ$. Let $A \in \mathcal{F}_T^\circ$. By Definition 5.34, $\mathbf{1}_A = \mathbf{1}_A \circ a_T$, and by (R1), $Q_\alpha(A) = \mathbf{1}_A(\alpha)$. Hence $Q(A) = P(A)$.

The martingale property. Fix $Y^\circ \in \mathbb{X}^\circ$, $s \le t$ and a bounded $\mathcal{F}_s^\circ$-measurable W, and split as before,

$$Y_t^\circ - Y_s^\circ = \underbrace{\left(Y_t^{\circ,T} - Y_s^{\circ,T}\right)}_{\text{(I)}} + \underbrace{\left(Y_t^\circ - Y_{t\wedge T}^\circ\right) - \left(Y_s^\circ - Y_{s\wedge T}^\circ\right)}_{\text{(II)}}.$$

The term (I). We claim that for every α

$$E^{Q_\alpha}\left[(\mathrm{I})W\right] \;=\; (\mathrm{I})(\alpha)\, W(\alpha)\, \mathbf{1}_{\{T(\alpha)>s\}}. \tag{46}$$

Fix α and recall from Definition 5.35 that Q_α-a.s. $a_T\beta = a_T\alpha$ and $T(\beta) = T(\alpha)$.

If $T(\alpha) \le s$, then Q_α-a.s. $(\mathrm{I})(\beta) = Y_{T(\beta)}^\circ(\beta) - Y_{T(\beta)}^\circ(\beta) = 0$ and both sides of (46) vanish.

If $T(\alpha) > s$, then Q_α-a.s. $W(\beta) = W(\alpha)$, because W is $\mathcal{F}_s^\circ$-measurable, hence a function of β restricted to $\mathbb{T}_{\le s} \subset \mathbb{T}_{\le T(\beta)}$, on which β and α agree. Likewise $(\mathrm{I})(\beta) = (\mathrm{I})(\alpha)$, because Y° is adapted, so $Y_{u\wedge T}^\circ$ is a function of the path on $\mathbb{T}_{\le T}$, i.e. of a_T. This proves (46).

Integrating (46) over $P(\,\mathrm{d}\alpha)$ and using that $\mathbf{1}_{\{T>s\}}W$ is bounded and $\mathcal{F}_s^\circ$-measurable — $\{T > s\}$ lies in $\mathcal{F}_s^\circ$ because T is a stopping time — we get $E^Q[(\mathrm{I})W] = E^P[(Y_t^{\circ,T} - Y_s^{\circ,T})\mathbf{1}_{\{T>s\}}W] = 0$ because $Y^{\circ,T}$ is a P-martingale.

The term (II). Fix α and write $r = T(\alpha)$, which by Definition 5.35 is the Q_α-a.s. value of T.

- If $r \le s$, then $(\mathrm{II}) = Y_t^\circ - Y_s^\circ$ and W is bounded and $\mathcal{F}_s^\circ$-measurable with $s \ge r$, so (R2) gives $E^{Q_\alpha}[(\mathrm{II})W] = 0$.

- If $r > s$, then $(\mathrm{II}) = Y_t^\circ - Y_{t\wedge r}^\circ$, which vanishes for $t \le r$; for $t > r$ it is $Y_t^\circ - Y_r^\circ$, and W is $\mathcal{F}_r^\circ$-measurable on this event because $s < r$ and $\mathcal{F}_s^\circ \subset \mathcal{F}_r^\circ$. Again (R2), now at the time r, gives $E^{Q_\alpha}[(\mathrm{II})W] = 0$.

Integrating over $P(\,\mathrm{d}\alpha)$ gives $E^Q[(\mathrm{II})W] = 0$. □

Theorem 5.37 (Uniqueness implies local uniqueness; [5], Theorem III.2.40)**.** *Suppose that a restart kernel exists at every strict stopping time, and that $\mathcal{M}(\mathbb{X}^\circ, \mu)$ contains at most one element for every $\mu \in \mathcal{P}(E)$. Then local uniqueness holds in the sense of Definition 5.33.*

Proof. Let T be a strict stopping time and $P \in \mathcal{M}(\mathbb{X}^{\circ,T})$ with $P\pi_0^{-1} = \mu$. Lemma 5.36 produces $Q \in \mathcal{M}(\mathbb{X}^\circ)$ with $Q = P$ on $\mathcal{F}_T^\circ$; in particular $Q\pi_0^{-1} = \mu$, since π_0 is $\mathcal{F}_T^\circ$-measurable. By hypothesis Q is the unique element of $\mathcal{M}(\mathbb{X}^\circ, \mu)$, hence depends on μ alone; therefore so does $P|_{\mathcal{F}_T^\circ} = Q|_{\mathcal{F}_T^\circ}$. □

Corollary 5.38 (The Markovian case)**.** *Let $(\mathbb{X}_r^\circ)$ be a shift system, measurable in the sense of Definition 5.10 and* full*, i.e. satisfying in addition that for every $Y^\circ \in \mathbb{X}^\circ$ and r there are $\hat{Y}^\circ \in \mathbb{X}_r^\circ$ and an $\mathcal{F}_r^\circ$-measurable κ with $\hat{Y}_u^\circ \circ \theta_r = Y_{r+u}^\circ - Y_r^\circ + \kappa$. Let $(P_{x,r})$ be a measurable family with $P_{x,r} \in \mathcal{M}(\mathbb{X}_r^\circ, \delta_x)$, and suppose F admits concatenation at T: a measurable $\gamma(\alpha, \beta)$ agreeing with α before $T(\alpha)$, with β shifted after it, and with $\theta_T\gamma(\alpha,\beta) = \beta$. Then*

$$Q_\alpha \;=\; \textit{law of } \gamma(\alpha,\beta), \qquad \beta \sim P_{\alpha_{T(\alpha)},T(\alpha)},$$

is a restart kernel at T, and Theorem 5.37 applies.

Proof. (R1) holds because $a_T\gamma(\alpha,\beta) = a_T\alpha$. For (R2) write $r = T(\alpha)$. Fullness gives $\hat{Y}^\circ \in \mathbb{X}_r^\circ$ and an $\mathcal{F}_r^\circ$-measurable κ with $Y_{r+u}^\circ - Y_r^\circ = \hat{Y}_u^\circ \circ \theta_r - \kappa$. Under Q_α the shifted path θ_r has law $P_{\alpha_r,r}$, under which $\hat{Y}^\circ$ is a martingale for $(\mathcal{F}_u^\circ)_u$, and κ is Q_α-a.s. the constant $\kappa(\alpha)$, being $\mathcal{F}_r^\circ$-measurable and hence a function of a_T.

It remains to transport the filtrations. Under Q_α the path on $\mathbb{T}_{\le r}$ is Q_α-a.s. the deterministic $a_T\alpha$, so for $u \in \mathbb{T}$ the σ-field $\mathcal{F}_{r+u}^\circ$ coincides Q_α-a.s. with $\theta_r^{-1}\mathcal{F}_u^\circ$: the generators π_v with $v \le r+u$ are either deterministic ($v \le r$) or of the form $\pi_w \circ \theta_r$ with $w = v-r \le u$, the latter by Lemma 2.2. Hence for $r \le s \le t$, writing $s = r+u$ and $t = r+v$,

$$E^{Q_\alpha}\left[Y_t^\circ - Y_s^\circ \mid \mathcal{F}_s^\circ\right] = E^{P_{\alpha_r,r}}\left[\hat{Y}_v^\circ - \hat{Y}_u^\circ \mid \mathcal{F}_u^\circ\right] \circ \theta_r = 0,$$

which is (R2). □

Remark 5.39 (What memory buys, and what it does not). Three comments.

The shift has left the statement. Definition 5.35, Lemma 5.36 and Theorem 5.37 mention no θ_r, no shifted family and no concatenation; the restarted measures are indexed by the *stopped path*, not by the state, and condition (R2) is exactly the conclusion of Lemma 5.5. The Markovian version, with a state-indexed kernel and concatenation, is now the corollary it should be. The proof is also shorter: because a strict stopping time satisfies $T \circ a_T = T$, it is Q_α-a.s. *deterministic*, so no optional sampling and no case analysis with θ_T are needed.

What is genuinely assumed. The existence of a restart kernel is a hypothesis and cannot be otherwise. Unlike the restart of Lemma 5.5, which merely reweights a solution one already has, pasting must *supply* solutions after T for every possible past; and the conditional distributions of a solution of the full problem are of no use here, since producing such a solution is the point. This is the one place in Section 5 where something must be given from outside.

Why strict stopping times. Definition 5.34 is used twice: to know that $\mathcal{F}_T^\circ$-measurable functions see only a_T, and to know that T is determined by the stopped path. For the raw filtration $\mathcal{F}_t^\circ = \sigma(\pi_u : u \le t)$ this is [5], Lemma III.2.43; for a completed or right continuous filtration it is false, which is the practical reason for [5]'s insistence on the distinction.

5.5 Non-Markovian test objects

The point of the arrangement above is that it can be tested. We run it against the two examples of [2] that are not Markovian, and record for each which results apply and which fail.

Example 5.40 (Volterra equations; [2], Example 3.13)**.** Let $F = L^p_{\mathrm{loc}}(\mathbb{R}_+,\mathbb{R}^d)\times D_{\mathbb{R}^k}[0,\infty)$ with coordinate process (π^X,π^Z), let $K : \mathbb{R}_+ \to \mathbb{R}^{d\times k}$ be a convolution kernel and $g_0 : \mathbb{R}_+ \to \mathbb{R}^d$, and let

$$Lf(x,z) = \langle b(x),\nabla f(z)\rangle + \tfrac{1}{2}\operatorname{tr}\big(a(x)\nabla^2 f(z)\big) + \int \big(f(z+y) - f(z) - \langle h(y),\nabla f(z)\rangle\big)\nu(x,\mathrm{d}y).$$

Take for $\mathbb{X}^\circ$ the processes $f(\pi_t^Z) - \int_0^t Lf(\pi_s^X,\pi_s^Z)\,\mathrm{d}s$, $f \in C_b^2(\mathbb{R}^k)$, together with the requirement that $\int_0^t \pi_s^X\,\mathrm{d}s = \int_0^t g_0(s)\,\mathrm{d}s + \int_0^t K_{t-s}\pi_s^Z\,\mathrm{d}s$ for all t. Then $\mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$ is the set of laws of solutions of the Volterra equation.

This is not a Markov problem, and not by accident: the kernel K_{t-s} makes π_t^X depend on the whole past of π^Z. In the language of Section 5.2, *there is no shift system*: shifting by r turns K_{t-s} into a kernel that also sees π^Z before r, which no member of a family $\mathbb{X}_r^\circ$ of the form required by Definition 5.10 can reproduce. What survives is exactly what Section 5.1 proves:

- Definitions 3.2 and 3.3 apply as they stand, on a path space that is not $D_E[0,\infty)$ and on which π^X is not even right continuous;

- Lemma 5.2 (mixtures) and Lemma 5.3 (disintegration) apply, the latter because F is Polish, hence (E1);

- **Proposition 5.9 applies**, provided the Volterra equation is uniquely solvable in law *from any prescribed past* — which under Lipschitz conditions on b, a, ν it is, by the Picard iteration, since the kernel term $\int_{[0,t)} K_{t-s}\,\mathrm{d}Z_s$ splits into a known contribution from the past and a contraction on the future. That is (34), and Proposition 5.9 turns it into uniqueness of the whole law with no Markov structure. Plain uniqueness in law would not suffice; see Remark 5.8;

- Theorem 7.27 applies, and [2], Section 4.2 is precisely a stability theorem for Volterra equations obtained from it;

- Theorem 5.16 does *not* apply, and the conclusion is false: a Volterra process is not Markov. Nor do Theorem 5.19 or Section 6, both of which use the shift. Section 5.4 applies as soon as a restart kernel is available.

Example 5.41 (Semimartingales with path-dependent characteristics)**.** Let $F = D_E[0,\infty)$ and let (B, C, ν) be a triplet of *predictable functionals of the path*, as in [5], Chapter III. The associated $\mathbb{X}^\circ$ consists of the processes of [5], III.2.7, and $\mathcal{M}_{\mathrm{loc}}(\mathbb{X}^\circ)$ is the set of laws of semimartingales with these characteristics. Again there is no shift system unless the triplet happens to be shift covariant in the sense of [5], III.2.39, which for a genuinely path-dependent triplet it is not.

Here Section 5.3 is available in full — the localizing system of Lemma 5.26 is constructed from the test processes themselves and needs no shift, which is exactly the argument [5] give in their III.2.8 — and so is Section 5.4, provided a restart kernel (Definition 5.35) can be exhibited. For a path-dependent triplet that is a genuine requirement rather than a formality: one must produce, for every stopped path, a semimartingale continuing it with the prescribed characteristics. But it is a requirement on the *data*, not a Markov hypothesis, and [5] verify it in their setting.

Remark 5.42 (Where the Markov structure is used, in one place)*.* Collecting the two examples with the results of Sections 4–7:

Needs no Markov structure	**Needs a shift system**
Definitions 3.2, 3.3, 4.2	Definition 5.10
Theorem 4.3 (càdlàg modification)	Lemma 5.15
Lemma 5.2, Lemma 5.3	Example 5.12
Lemma 5.5 (**restart with memory**)	Theorem 5.16 (Markov property)
Definition 5.7, Proposition 5.9 (**uniqueness**)	Lemma 5.13 (restart with re-indexing)
Definition 5.22, Lemma 5.26, Lemma 5.24	Lemma 5.29, Theorem 5.30
Lemma 6.1 (chain identity)	Corollary 5.38
Definition 5.35, Lemma 5.36, Theorem 5.37 (**local uniqueness**)	—
Theorem 7.27, Theorem 7.34	Proposition 6.3 and Section 6.2

The left column is the whole of Sections 3 and 7, all of Section 4, the uniqueness half of Section 5 and the analytic core of Section 6. The right column is: the shift system and what is built on it. Every entry there produces a statement whose *conclusion* is Markovian

— the Markov property, the reduction to one-dimensional distributions, pasting, duality — so nothing in it could have been avoided by a better argument.

5.6 The Markovian case

Theorem 5.43 ([4], Theorem 4.4.2; (T2b) + (T4) + shift invariant clock, for homogeneity)**.** *Assume (E2) and let $A \subset B(E) \times B(E)$. Suppose that for each $\mu \in \mathcal{P}(E)$ any two solutions X, Y of the martingale problem for (A, μ) have the same one-dimensional distributions, that is, for each $t \in \mathbb{T}$,*

$$P\{X(t) \in \Gamma\} \;=\; P\{Y(t) \in \Gamma\}, \qquad \Gamma \in \mathcal{B}(E). \tag{47}$$

Then the following hold.

(a) Any solution of the martingale problem for A with respect to a filtration $(\mathcal{G}_t)$ is a Markov process with respect to $(\mathcal{G}_t)$, and any two solutions of the martingale problem for (A, μ) have the same finite-dimensional distributions. In particular, (47) implies uniqueness.

(b) If moreover $A \subset C_b(E) \times B(E)$, X is a solution of the martingale problem for A with respect to $(\mathcal{G}_t)$, and X has sample paths in $D_E[0, \infty)$, then for every a.s. finite $(\mathcal{G}_t)$-stopping time τ,

$$E\big[f(X(\tau + t)) \bigm| \mathcal{G}_\tau\big] \;=\; E\big[f(X(\tau + t)) \bigm| X(\tau)\big], \qquad f \in B(E),\ t \in \mathbb{T}. \tag{48}$$

(c) If, in addition to the conditions of (b), for each $x \in E$ there exists a solution $P_x \in \mathcal{P}(D_E[0, \infty))$ of the $D_E[0, \infty)$ martingale problem for (A, δ_x) such that $x \mapsto P_x(B)$ is Borel measurable for each $B \in \mathcal{B}(D_E[0, \infty))$, then, setting $T(t)f(x) = \int f(\omega(t)) P_x(\mathrm{d}\omega)$,

$$E\big[f(X(\tau + t)) \bigm| \mathcal{G}_\tau\big] \;=\; T(t)f(X(\tau)) \tag{49}$$

for all $f \in B(E)$, $t \in \mathbb{T}$ and a.s. finite $(\mathcal{G}_t)$-stopping times τ; that is, X is strong Markov.

Proof. We check the hypotheses of Theorems 5.16 and 5.19 for $\mathbb{X}^\circ = \mathbb{X}_A$, i.e. for

$$\mathbb{X}^\circ = \Big\{ \big(f(\pi_t) - \int_{\langle 0,t\rangle_\iota} g(\pi_s)\, q(\mathrm{d}s) \big)_{t \in \mathbb{T}} \;:\; (f, g) \in A \Big\}$$

on the path space $F = E^{\mathbb{T}}$ (respectively $F = D_E[0, \infty)$ in (b)–(c)).

Determining set. Proposition 3.8 says precisely that the products $\prod_k h_k(\pi_{t_k})$, over arbitrary finite subsets $\{t_k\} \subset \mathbb{T}_{\le s}$, form a determining set for $\mathbb{X}_A$ in the sense of Definition 3.3: (21) for all such products is equivalent to the martingale property. Note that this is stronger than what Definition 3.3 asks — it delivers the martingale property for ${}^*\mathcal{F}^X$, not merely for the canonical filtration.

Shift system. Example 5.12 exhibits one, with $\theta_r\omega = \omega(r + \cdot)$, the pulled-back clock q_r and $\kappa = f(\pi_r)$. Since $A \subset C_b(E) \times B(E)$ does not depend on time and q is assumed shift invariant, $q_r = q$ and $\mathbb{X}^\circ_r = \mathbb{X}^\circ$ for every r; the shift system is the constant one, and the Markov process obtained is homogeneous. Dropping either assumption leaves Theorem 5.16 intact and yields a time-inhomogeneous Markov process instead — this is the only point at which [4], Theorem 4.4.2 is genuinely less general than Theorem 5.16.

Integrability. f and g are bounded, so every $Y^\circ \in \mathbb{X}^\circ$ satisfies $|Y^\circ_t| \le \|f\| + \|g\|\, q(\mathbb{T}_{\le t})$, which is finite by Definition 2.3; all the integrability provisos are automatic, and the increments of Y° on $\mathbb{T}_{\le T}$ are bounded, hence uniformly integrable.

Conclusion. Hypothesis (47) is (39) by Remark 5.1, so Theorem 5.16(a) gives the Markov property and (b) the equality of the finite-dimensional distributions, which is (a). For (b) and (c), $A \subset C_b(E) \times B(E)$ and paths in $D_E[0,\infty)$ make each $Y^\circ(X)$ a right continuous martingale with bounded increments, so Theorem 5.19 applies; its second assertion is (49) with $T(t)f(x) = \int f(\omega(t))P_x(\mathrm{d}\omega)$. Its restriction to stopping times with countably many values is vacuous here: the shift system is the constant one, so all the measures $P_{x,r} = P_x$ solve one and the same problem and the partition of Step 3 there is unnecessary. See Remark 5.20. □

Corollary 5.44 ([4], Corollary 4.4.3; (T3))**.** *Assume (E3) and let $A \subset B(E) \times B(E)$. Suppose that for each $\mu \in \mathcal{P}(E)$ any two solutions X, Y of the martingale problem for (A, μ) with sample paths in $D_E[0,\infty)$ (respectively $C_E[0,\infty)$) satisfy* (47)*. Then for each $\mu \in \mathcal{P}(E)$ any two such solutions have the same distribution on $D_E[0,\infty)$ (respectively $C_E[0,\infty)$).*

Proof. The processes $\tilde{X}$, $\tilde{Y}$ constructed in the proof of Theorem 5.43 again have sample paths in $D_E[0,\infty)$ (resp. $C_E[0,\infty)$) whenever X and Y do, so the induction goes through and X, Y have the same finite-dimensional distributions. By Fact 2.35 these determine the law on $D_E[0,\infty)$ (resp. $C_E[0,\infty)$). □

Remark 5.45. For the *local* martingale problem, Theorem 5.43 holds in the following localized form. Suppose $A \subset M(E) \times M(E)$ and let (K_m) be an increasing sequence of compact sets with $\tau_m = \inf\{t : X(t) \notin K_m \text{ or } X(t-) \notin K_m\}$ and $\tau_m \to \infty$ a.s. for every solution X. If for each μ and each m any two solutions of the local martingale problem for (A, μ) agree in the one-dimensional distributions of the *stopped* processes $X(\cdot \wedge \tau_m)$, then the conclusions (a)–(c) hold. The proof is the above one applied to the stopped martingale problem for $A^{(m)} = \{(f\mathbf{1}_{K_m}, g\mathbf{1}_{K_m}) : (f,g) \in A\}$, followed by $m \to \infty$. This is the content of [4], Section 4.6. It is worth stating separately in the formalization, since the reduction "local $\rightsquigarrow$ global via stopping" is reused in Section 7.

This route modifies the operator. Section 5.3 gives the alternative, which leaves A alone and stops the test process instead; it is the one that fits the abstract setting, and the one [7] uses.

6 Uniqueness via duality

Duality reduces the computation of $E[f(X(t), y)]$ for a process X to the computation of $E[f(x, Y(t))]$ for a second, typically simpler, process Y. Combined with Theorem 5.43 it is one of the main practical routes to uniqueness.

Here, unlike in Sections 4 and 5, there is nothing to strip away on the probabilistic side: Theorem 6.10 below already assumes only that X and Y are *measurable* processes, uses no path regularity, no Polish structure and no boundedness, and would be stated verbatim in the [2] setting. What can be generalized is the *clock*. Section 6.1 therefore isolates the analytic core as a statement about a function of two time variables, valid for an arbitrary measure q, and reads off exactly which q make the duality relation true. The answer turns out to be sharp, and it is not the answer one gets in Sections 4 and 5.

6.1 The anti-diagonal lemma

Duality is the assertion that a certain function of two time variables is constant on anti-diagonals. The following identity is that assertion stripped of all analysis: it is an exact telescoping, with no hypothesis beyond the two increment representations, and every version of the duality lemma is obtained from it by choosing a chain.

Lemma 6.1 (Chain identity; (T0) + (E0) + clock)**.** *Let $\mathbb{T}$ be a preordered set with a least element 0, let q be a measure on $\mathbb{T}$ for which the sets $[s,s') := \mathbb{T}_{<s'} \setminus \mathbb{T}_{<s}$ are measurable, and let $\Phi, \gamma_1, \gamma_2 : \mathbb{T} \times \mathbb{T} \to \mathbb{R}$ satisfy, for all $s \le s'$ and $t \le t'$,*

$$\Phi(s',t) - \Phi(s,t) = \int_{[s,s')} \gamma_1(r,t)\, q(\mathrm{d}r), \qquad \Phi(s,t') - \Phi(s,t) = \int_{[t,t')} \gamma_2(s,r)\, q(\mathrm{d}r), \quad (50)$$

all integrals being assumed to exist. Let $t \in \mathbb{T}$ and let

$$0 = s_0 \le s_1 \le \cdots \le s_m = t, \qquad t = t_0 \ge t_1 \ge \cdots \ge t_m = 0$$

be a staircase *from $(0,t)$ to $(t,0)$. Then*

$$\Phi(t,0) - \Phi(0,t) = \sum_{k=0}^{m-1} \Big[\int_{[s_k,s_{k+1})} \gamma_1(r,\, t_{k+1})\, q(\mathrm{d}r) \;-\; \int_{[t_{k+1},\, t_k)} \gamma_2(s_k,\, r)\, q(\mathrm{d}r) \Big]. \qquad (51)$$

Proof. Telescope along the staircase: $\Phi(t,0) - \Phi(0,t) = \sum_k \big[\Phi(s_{k+1},t_{k+1}) - \Phi(s_k,t_k)\big]$, the sum collapsing because $(s_0,t_0) = (0,t)$ and $(s_m,t_m) = (t,0)$. Split each summand through the corner (s_k,t_{k+1}):

$$\big[\Phi(s_{k+1},t_{k+1}) - \Phi(s_k,t_{k+1})\big] - \big[\Phi(s_k,t_k) - \Phi(s_k,t_{k+1})\big].$$

Now apply the first representation in (50) to the first bracket, which is legitimate because $s_k \le s_{k+1}$, and the second to the second, legitimate because $t_{k+1} \le t_k$. $\square$

Remark 6.2 (What the staircase form buys)*.* The version of (51) one meets in the literature takes $t_k = t - s_k$, the *anti-diagonal* staircase, and therefore needs (T4) so that $t - s_k$ is defined. Nothing in the proof requires this: the two coordinates may be stepped through independently, and the identity is then a statement about an arbitrary preorder with a clock. Which staircase one chooses is the whole art of the section, and the point of Lemma 6.4 below is that the anti-diagonal one is rarely the right choice: the correct staircase is the one that is anti-diagonal *in clock time*.

The duality relation is the case in which the two brackets in (51) cancel. Under the balance condition $\gamma_1 = \gamma_2 = \gamma$ the k-th summand is

$$\int_{[s_k,s_{k+1})} \gamma(r,\, t_{k+1})\, q(\mathrm{d}r) \;-\; \int_{[t_{k+1},\, t_k)} \gamma(s_k,\, r)\, q(\mathrm{d}r), \qquad (52)$$

in which the two integrals run over the block $[s_k, s_{k+1})$ and over the partner block $[t_{k+1}, t_k)$. So the duality relation holds as soon as the staircase can be chosen so that a block and its partner receive the same q-mass, at every scale. On the anti-diagonal staircase $t_k = t - s_k$ that is translation invariance of q; but the anti-diagonal staircase is not forced, and it is not the right one.

Proposition 6.3 (Haar clocks; (T2a) + (T4), $\iota = \mathrm{p}$)**.** *Assume (50) with $\gamma_1 = \gamma_2 = \gamma$.*

- *(a) If $\mathbb{T} = \mathbb{N}_0$ and q is counting measure, then $\Phi(t,0) = \Phi(0,t)$ for every t, for every γ whatsoever, and no integrability hypothesis is needed.*
- *(b) If $\mathbb{T} = \mathbb{R}_+$ and q is Lebesgue measure, then $\Phi(t,0) = \Phi(0,t)$ for almost every t, under the integrability hypothesis (55); this is Lemma 6.9 below.*

Proof. (a) Take the anti-diagonal staircase $s_k = k$, $t_k = t-k$. Then $[s_k, s_{k+1}) = \{k\}$ and $[t_{k+1}, t_k) = \{t-k-1\}$, so (52) reads $\gamma(k, t-k-1) - \gamma(k, t-k-1) = 0$. Every summand vanishes identically; no regularity of γ is used.

(b) Refine the anti-diagonal staircase; see Lemma 6.9. $\square$

The two cases of Proposition 6.3 are the ones in the literature, and they are the ones for which the anti-diagonal staircase works. It is tempting to conclude — as an earlier version of this manuscript did — that duality *requires* a translation invariant clock, on the grounds that (52) does not cancel otherwise. That inference is invalid: the failure of one staircase to cancel says nothing about $\Phi(t,0) - \Phi(0,t)$, and in fact no clock is excluded.

Lemma 6.4 (Rectification of the clock; (T2b) + (E0) + clock, $\iota = \mathrm{p}$)**.** *Assume (T2b), let q be a clock, fix $t^* \in \mathbb{T}$ and put*

$$Q(s) \;=\; q(\mathbb{T}_{<s}) \quad (s \in \mathbb{T}_{\le t^*}), \qquad L \;=\; Q(t^*) \;<\; \infty.$$

Let Φ, γ satisfy (50) with $\gamma_1 = \gamma_2 = \gamma$ on $\mathbb{T}_{\le t^*}$. Then there are functions $\Psi, \psi : [0,L]^2 \to \mathbb{R}$ with

$$\Psi(x', y) - \Psi(x, y) = \int_x^{x'} \psi(z, y)\,\mathrm{d}z, \qquad \Psi(x, y') - \Psi(x, y) = \int_y^{y'} \psi(x, z)\,\mathrm{d}z \tag{53}$$

for $0 \le x \le x' \le L$ and $0 \le y \le y' \le L$, and with

$$\Psi\big(Q(s), Q(t)\big) \;=\; \Phi(s,t), \qquad s, t \in \mathbb{T}_{\le t^*}. \tag{54}$$

In particular $\Phi(t^*, 0) - \Phi(0, t^*) = \Psi(L, 0) - \Psi(0, L)$.

Proof. Step 1: Q is well behaved. Q is nondecreasing with $Q(0) = 0$ and $Q(t^*) = L$, and $q([s,s')) = Q(s') - Q(s)$ for $s \le s'$, because under (T2a) $[s,s') = \mathbb{T}_{<s'} \setminus \mathbb{T}_{<s}$ and the down-sets are nested. If $Q(s) = Q(s')$ with $s \le s'$, then $q([s,s')) = 0$, so (50) gives $\Phi(s,t) = \Phi(s',t)$ and $\Phi(t,s) = \Phi(t,s')$ for every t. Hence

$$\Psi_0\big(Q(s), Q(t)\big) := \Phi(s,t), \qquad \psi_0\big(Q(r), Q(t)\big) := \gamma(r,t)$$

are well defined on $R \times R$, $R := Q(\mathbb{T}_{\le t^*}) \subset [0,L]$ — for ψ_0 up to a q-null set of r, which is immaterial since ψ_0 will only be integrated.

Step 2: filling the gaps. Q is left continuous, because under (T2b) $\mathbb{T}_{<s} = \bigcup_{u<s,\, u\in D} \mathbb{T}_{\le u}$ and q is continuous from below; and $Q(a+) = q(\mathbb{T}_{\le a})$, because $\mathbb{T}_{\le a} = \bigcap_{u>a,\, u\in D} \mathbb{T}_{<u}$ and q restricted to $\mathbb{T}_{\le t^*}$ is finite, so continuity from above applies. Hence the jump of Q at a has size exactly $q(\{a\})$, and since R is the range of a monotone map, $[0,L] \setminus R$ is the countable union of the corresponding open jump intervals; write $G_a = [Q(a), Q(a) + q(\{a\})]$ for the corresponding closed interval and note that R meets G_a only in its endpoints. Extend Ψ_0 and ψ_0 by

$$\psi(x,y) \;=\; \psi_0\big(\rho(x), \rho(y)\big), \qquad \rho(x) \;=\; \begin{cases} Q(a), & x \in G_a, \\ x, & x \in R, \end{cases}$$

and $\Psi(x,y) = \Psi_0(\rho(x), \rho(y)) + (x - \rho(x))\,\psi(\rho(x), y) + (y - \rho(y))\,\psi(x, \rho(y))$, i.e. by affine interpolation across each gap in each variable, bilinear on a gap $\times$ gap rectangle. Note that ψ is a *single* function of two variables, as required for the balance condition: on $G_a \times G_b$ it is the constant $\psi_0(Q(a), Q(b)) = \gamma(a,b)$, whichever variable one arrives from.

Step 3: (53). For $x \le x'$ both in R the identity is (50) together with $q([s,s')) = Q(s') - Q(s)$; across a gap G_a it holds by construction, the integrand being constant there with value $\gamma(a,y)$ and the increment of Ψ being $q(\{a\})\gamma(a,y)$, which is what (50) assigns to the atom a. The general case follows by additivity. (54) is Step 1, and the final assertion is (54) at $(t^*, 0)$ and $(0, t^*)$ together with $Q(0) = 0$. $\square$

Theorem 6.5 (Every clock admits duality; (T2b) + (E0) + clock, $\iota = \mathrm{p}$)**.** *Assume (T2b), let q be a clock and let Φ, γ satisfy (50) with $\gamma_1 = \gamma_2 = \gamma$. Suppose the rectified ψ of Lemma 6.4 satisfies the integrability hypothesis (55). Then*

$$\Phi(t,0) \;=\; \Phi(0,t) \qquad \textit{for } Q\textit{-almost every } t \in \mathbb{T},$$

that is, for every t outside a set whose image under Q is Lebesgue null.

Proof. Fix t^* and rectify. Extend Ψ and ψ to $[0,\infty)^2$ by $\Psi(x,y) = \Psi(x \wedge L, y \wedge L)$ and $\psi = 0$ outside $[0,L]^2$; (53) persists. By (53) the function Ψ is absolutely continuous in each variable separately with $\nabla\Psi = (\psi,\psi)$, so Lemma 6.9 applies to $f = \Psi$ and gives, for almost every $\ell \in [0,L]$,

$$\Psi(\ell,0) - \Psi(0,\ell) = \int_0^\ell \big(\psi(z,\ell-z) - \psi(z,\ell-z)\big)\,\mathrm{d}z \;=\; 0,$$

the two partial derivatives being the *same* function ψ. Now apply the last assertion of Lemma 6.4. □

Remark 6.6 (The role of Haar measure, correctly stated)*.* Theorem 6.5 says that translation invariance of q is not a hypothesis of duality but a property of a particular *proof*: it is what makes the anti-diagonal staircase $t_k = t - s_k$ coincide with the staircase that is anti-diagonal in clock time,

$$Q(s_k) + Q(t_k) \;=\; Q(t) \qquad \text{for all } k,$$

which is the one (52) actually wants. Equivalently: a martingale problem with clock q is a time change of one with Lebesgue clock, and duality is invariant under that time change. This is the same phenomenon as in Section 7.8, seen from the analytic side.

Two consequences worth recording. First, the conclusion is genuinely "Q-almost every t": on an interval where q vanishes, Φ is constant in both arguments and there is nothing to prove, and where q has an atom the conclusion holds at the atom by Step 2 of Lemma 6.4. Second, for a purely atomic clock the conclusion holds at *every* t and without any integrability hypothesis, exactly as in Proposition 6.3(a); see Remark 6.7.

Remark 6.7 (Atomic clocks, and why the conventions do not clash here)*.* For a purely atomic clock there is a second, purely algebraic reason for the conclusion, and it is instructive because it does not go through Lemma 6.9 at all.

Let $q = \sum_k m_k \delta_{a_k}$ with finitely many atoms in each $\mathbb{T}_{\le t}$. Then $\Phi(s,t)$ depends on s only through $A_s = \{k : a_k \in \mathbb{T}_{<s}\}$, and (50) pins the values $\gamma(a_k,t)$ and $\gamma(s,a_l)$ individually:

$$m_k\gamma(a_k,t) = \Phi(s',t) - \Phi(s,t) \quad \text{whenever } A_{s'} \setminus A_s = \{k\},$$

and symmetrically. At a pair of atoms (a_k,a_l) *both* identities describe the same number $\gamma(a_k,a_l)$, and equating them is one linear relation per pair. Solving the resulting system — an elementary but tedious computation, verified symbolically for up to five atoms and for the two-dimensional index $\{0,1,2\}^2$ — gives $\Phi(t,0) = \Phi(0,t)$ for every t. For equal masses the relations say precisely that Φ is constant on anti-diagonals of the atom grid.

Two things follow. First, the argument uses no order structure beyond a preorder, which is why duality is not confined to a linearly ordered index even though Theorem 6.5 is proved only there. Second — and this corrects a claim of an earlier version — the two conventions of Definition 2.3 *agree* here. With $\iota = \mathrm{o}$ the k-th summand of (52) on the anti-diagonal staircase becomes $\gamma(k+1,t-k-1) - \gamma(k,t-k)$ and the sum telescopes to $\gamma(t,0) - \gamma(0,t)$ rather than to 0; but the relations above force $\gamma(t,0) = \gamma(0,t)$, so the conclusion is the same. Section 6 is therefore convention-agnostic, and the bundle table records it as such.

Remark 6.8 (Scope of Section 6). It is worth saying plainly what Section 6 needs, since the matter is easy to get wrong.

It needs the *balance condition* $\gamma_1 = \gamma_2$ — in the probabilistic form of Theorem 6.10, that the two generators act on the same duality function f so as to produce the same $g = h$ — and it needs integrability. It does *not* need a translation invariant clock, and it does not need a linearly ordered index: (T2a) enters only through the rectification of Lemma 6.4, which is a device of proof, and the atomic case of Remark 6.7 dispenses with it. In this respect Section 6 is no narrower than Sections 3 and 5; the opposite claim, made in an earlier version of this manuscript on the strength of the non-cancellation of (52) for $\mathbb{T} = \mathbb{R}^d_+$, was a fallacy — the non-cancellation of a particular staircase is not the failure of the conclusion.

What genuinely remains open is a proof of Theorem 6.5 for a non-atomic clock on a non-linearly ordered index, where neither the rectification nor the atom-by-atom relations are available. Nothing in this manuscript depends on it.

6.2 The continuous-time theory

Lemma 6.9 ([4], Lemma 4.4.10; (T3)). *Let $f : [0,\infty)\times[0,\infty) \to \mathbb{R}$ be absolutely continuous in s for each fixed t and absolutely continuous in t for each fixed s, and set $(f_1, f_2) = \nabla f$. Suppose that*

$$\int_0^T \int_0^T |f_i(s,t)|\,\mathrm{d}s\,\mathrm{d}t < \infty, \qquad i = 1,2, \quad T > 0. \tag{55}$$

Then for almost every $t \geq 0$,

$$f(t,0) - f(0,t) \; = \; \int_0^t \big(f_1(s,t-s) - f_2(s,t-s)\big)\,\mathrm{d}s. \tag{56}$$

Proof. By (55) and Fubini,

$$\begin{aligned}
&\int_0^T \int_0^t \big(f_1(t-s,s) - f_2(s,t-s)\big)\,\mathrm{d}s\,\mathrm{d}t \\
&\quad = \int_0^T \int_0^t f_1(t-s,s)\,\mathrm{d}s\,\mathrm{d}t - \int_0^T \int_0^t f_2(s,t-s)\,\mathrm{d}s\,\mathrm{d}t \\
&\quad = \int_0^T \int_s^T f_1(t-s,s)\,\mathrm{d}t\,\mathrm{d}s - \int_0^T \int_s^T f_2(s,t-s)\,\mathrm{d}t\,\mathrm{d}s \\
&\quad = \int_0^T \big(f(T-s,s) - f(0,s)\big)\,\mathrm{d}s - \int_0^T \big(f(s,T-s) - f(s,0)\big)\,\mathrm{d}s \\
&\quad = \int_0^T \big(f(s,0) - f(0,s)\big)\,\mathrm{d}s,
\end{aligned}$$

where the third equality uses absolute continuity in the first resp. second variable, and the fourth is the substitution $s \mapsto T - s$ in the first two integrals. Differentiating with respect to T gives (56) for a.e. T. ☐

Theorem 6.10 ([4], Theorem 4.4.11; (T3)). *Let X and Y be independent measurable processes with values in E_1 and E_2 respectively. Let $f, g, h \in M(E_1 \times E_2)$, $\alpha \in M(E_1)$ and $\beta \in M(E_2)$. Suppose that for each $T > 0$ there exist an integrable random variable Γ_T and* *a constant C_T such that*

$$\begin{aligned}
\sup_{r,s,t\le T}\big(|\alpha(X(r))|+1\big)|f(X(s),Y(t))| &\le \Gamma_T,\\
\sup_{r,s,t\le T}\big(|\beta(Y(r))|+1\big)|f(X(s),Y(t))| &\le \Gamma_T,\\
\sup_{r,s,t\le T}\big(|\alpha(X(r))|+1\big)|g(X(s),Y(t))| &\le \Gamma_T,\\
\sup_{r,s,t\le T}\big(|\beta(Y(r))|+1\big)|h(X(s),Y(t))| &\le \Gamma_T,
\end{aligned} \tag{57}$$

and

$$\int_0^T |\alpha(X(u))|\,\mathrm{d}u + \int_0^T |\beta(Y(u))|\,\mathrm{d}u \;\le\; C_T. \tag{58}$$

Suppose that

$$f(X(t),y) - \int_0^t g(X(s),y)\,\mathrm{d}s \tag{59}$$

is a $({}^*\mathcal{F}_t^X)$-martingale for each $y\in E_2$, and

$$f(x,Y(t)) - \int_0^t h(x,Y(s))\,\mathrm{d}s \tag{60}$$

is a $({}^*\mathcal{F}_t^Y)$-martingale for each $x\in E_1$ (the integrals in (59) and (60) being assumed to exist). Then for almost every $t\ge 0$,

$$\begin{aligned}
&E\Big[f(X(t),Y(0))\exp\Big\{\int_0^t \alpha(X(u))\,\mathrm{d}u\Big\}\Big] - E\Big[f(X(0),Y(t))\exp\Big\{\int_0^t \beta(Y(u))\,\mathrm{d}u\Big\}\Big]\\
&= E\Big[\int_0^t \big\{g(X(s),Y(t-s)) - h(X(s),Y(t-s)) + \big(\alpha(X(s))-\beta(Y(t-s))\big)f(X(s),Y(t-s))\big\}\\
&\qquad\qquad \times\exp\Big\{\int_0^s \alpha(X(u))\,\mathrm{d}u + \int_0^{t-s}\beta(Y(u))\,\mathrm{d}u\Big\}\,\mathrm{d}s\Big].
\end{aligned} \tag{61}$$

Proof. Set

$$F(s,t) = E\Big[f(X(s),Y(t))\exp\Big\{\int_0^s \alpha(X(u))\,\mathrm{d}u\Big\}\exp\Big\{\int_0^t \beta(Y(u))\,\mathrm{d}u\Big\}\Big]. \tag{62}$$

Step 1: freezing Y. Write $e_\alpha(s)=\exp\{\int_0^s\alpha(X(u))\,\mathrm{d}u\}$ and $e_\beta(t)=\exp\{\int_0^t\beta(Y(u))\,\mathrm{d}u\}$, and fix t. Both $e_\beta(t)$ and $Y(t)$ are $\sigma(Y)$-measurable, and $\sigma(Y)$ is independent of $\sigma(X)$; so every expectation below may be computed by conditioning on $\sigma(Y)$, freezing $Y(t)=y$ and $e_\beta(t)$, and using that $M_r^y = f(X(r),y)-\int_0^r g(X(v),y)\,\mathrm{d}v$ is a $({}^*\mathcal{F}_r^X)$-martingale under the conditioned measure — which it is, because conditioning on an independent σ-field leaves the law of X and the filtration ${}^*\mathcal{F}^X$ unchanged. We suppress the conditioning from the notation.

Step 2: the increment identity. For $h>0$ split

$$\begin{aligned}
&f(X(s{+}h),y)e_\alpha(s{+}h) - f(X(s),y)e_\alpha(s)\\
&\qquad = \underbrace{\big[f(X(s{+}h),y)-f(X(s),y)\big]e_\alpha(s)}_{(A)} + \underbrace{f(X(s{+}h),y)\big[e_\alpha(s{+}h)-e_\alpha(s)\big]}_{(B)}.
\end{aligned} \tag{63}$$

For (A), write $f(X(s{+}h),y)-f(X(s),y)=M^y_{s+h}-M^y_s+\int_s^{s+h} g(X(v),y)\,\mathrm{d}v$. Since $e_\alpha(s)$ is ${}^*\mathcal{F}^X_s$-measurable and bounded by e^{C_T}, the martingale term contributes 0, so

$$\begin{aligned}E[(A)] &= \int_s^{s+h} E\big[g(X(r),y)e_\alpha(s)\big]\,\mathrm{d}r\\ &= \int_s^{s+h} E\big[g(X(r),y)e_\alpha(r)\big]\,\mathrm{d}r\\ &\quad+\int_s^{s+h} E\big[g(X(r),y)(e_\alpha(s)-e_\alpha(r))\big]\,\mathrm{d}r.\end{aligned}$$

For (B), the chain rule gives $e_\alpha(s{+}h)-e_\alpha(s)=\int_s^{s+h}\alpha(X(r))e_\alpha(r)\,\mathrm{d}r$. For each $r\in[s,s{+}h]$ write

$$f(X(s{+}h),y) \;=\; f(X(r),y)+\big(M^y_{s+h}-M^y_r\big)+\int_r^{s+h} g(X(v),y)\,\mathrm{d}v.$$

The martingale term again contributes 0, since $\alpha(X(r))e_\alpha(r)$ is ${}^*\mathcal{F}^X_r$-measurable, and one obtains

$$\begin{aligned}E[(B)] &= \int_s^{s+h} E\big[f(X(r),y)\alpha(X(r))e_\alpha(r)\big]\,\mathrm{d}r\\ &\quad+\int_s^{s+h}\int_r^{s+h} E\big[g(X(v),y)\alpha(X(r))e_\alpha(r)\big]\,\mathrm{d}v\,\mathrm{d}r.\end{aligned}$$

Multiplying by $e_\beta(t)$, integrating over the law of Y and using Fubini — legitimate because all integrands are dominated by $\Gamma_T e^{C_T}$ by (57)–(58) — we obtain

$$\begin{aligned}F(s+h,t)-F(s,t) \;=\; &\underbrace{\int_s^{s+h} E\big[f\alpha(X(r),Y(t))e_\alpha(r)e_\beta(t)\big]\,\mathrm{d}r}_{T_1}\\ &+\underbrace{\int_s^{s+h}\int_r^{s+h} E\big[g(X(v),Y(t))\alpha(X(r))e_\alpha(r)e_\beta(t)\big]\,\mathrm{d}v\,\mathrm{d}r}_{T_2}\\ &+\underbrace{\int_s^{s+h} E\big[g(X(r),Y(t))e_\alpha(r)e_\beta(t)\big]\,\mathrm{d}r}_{T_3}\\ &+\underbrace{\int_s^{s+h} E\big[g(X(r),Y(t))\big(e_\alpha(s)-e_\alpha(r)\big)e_\beta(t)\big]\,\mathrm{d}r}_{T_4},\end{aligned}\tag{64}$$

where $f\alpha$ abbreviates $(x,y)\mapsto f(x,y)\alpha(x)$.

Step 3: T_2 and T_4 are $O(h^2)$. Let $t,s+h\le T$. The third line of (57) gives $|\alpha(X(r))|\,|g(X(v),Y(t))|\le\Gamma_T$, and (58) gives $e_\alpha(r)e_\beta(t)\le e^{C_T}$. Hence

$$|T_2| \;\le\; e^{C_T}E[\Gamma_T]\int_s^{s+h}(s+h-r)\,\mathrm{d}r \;=\; \tfrac12 h^2 e^{C_T}E[\Gamma_T].$$

For T_4 use $e_\alpha(s)-e_\alpha(r)=-\int_s^r\alpha(X(u))e_\alpha(u)\,\mathrm{d}u$ and the same two bounds:

$$|T_4| \;\le\; \int_s^{s+h}\int_s^r E\big[|g(X(r),Y(t))|\,|\alpha(X(u))|\,e_\alpha(u)e_\beta(t)\big]\,\mathrm{d}u\,\mathrm{d}r \;\le\; \tfrac12 h^2 e^{C_T}E[\Gamma_T].$$

Step 4: passing to the limit. Since $T_1+T_3 = \int_s^{s+h} E[(f\alpha+g)(X(r),Y(t))e_\alpha(r)e_\beta(t)]\,\mathrm{d}r$, summing (64) over a partition $0 = s_0 < \cdots < s_m = s$ gives

$$\Big|F(s,t)-F(0,t)-\int_0^s E\big[(f\alpha+g)(X(r),Y(t))e_\alpha(r)e_\beta(t)\big]\,\mathrm{d}r\Big| \;\le\; \tfrac{1}{2}e^{C_T}E[\Gamma_T]\sum_{i=1}^m (s_i-s_{i-1})^2$$

— the main terms telescoping *exactly*, with no limit needed — and the right-hand side is at most $\frac{1}{2}e^{C_T}E[\Gamma_T]\,s\max_i(s_i-s_{i-1}) \to 0$. This is

$$F(s,t)-F(0,t) = \int_0^s E\big[\big(f(X(r),Y(t))\alpha(X(r))+g(X(r),Y(t))\big)e_\alpha(r)e_\beta(t)\big]\,\mathrm{d}r. \quad (65)$$

Symmetrically, Symmetrically,

$$F(s,t)-F(s,0) = \int_0^t E\big[\big(f(X(s),Y(r))\beta(Y(r))+h(X(s),Y(r))\big)e_\alpha(s)e_\beta(r)\big]\,\mathrm{d}r. \quad (66)$$

Thus F satisfies the hypotheses of Lemma 6.9, with F_1 and F_2 the integrands in (65) and (66); the integrability (55) follows from (57) and (58). Now (56) is exactly (61). □

Remark 6.11 ([4], Remark 4.4.12). Equation (61) can frequently be extended to more general α and β by approximating them by bounded functions.

Corollary 6.12 (Duality relation; [4], Corollary 4.4.13; (T3))**.** *If, in addition to the conditions of Theorem 6.10,*

$$g(x,y)+\alpha(x)f(x,y) \;=\; h(x,y)+\beta(y)f(x,y), \qquad (x,y)\in E_1\times E_2, \quad (67)$$

then for all $t \ge 0$

$$E\Big[f(X(t),Y(0))\exp\Big\{\int_0^t \alpha(X(u))\,\mathrm{d}u\Big\}\Big] = E\Big[f(X(0),Y(t))\exp\Big\{\int_0^t \beta(Y(u))\,\mathrm{d}u\Big\}\Big]. \quad (68)$$

Proof. Under (67) the right-hand side of (61) vanishes, so (68) holds for almost every t. It extends to all t because $t\mapsto F(t,0)$ and $t\mapsto F(0,t)$ are continuous, which follows from (65) and (66). □

Remark 6.13. Corollary 6.12 is Proposition 6.3(b) applied to $\Phi = F$ of (62), with γ_1 and γ_2 the integrands of (65) and (66); the balance condition (67) is exactly $\gamma_1 = \gamma_2$. The whole of Theorem 6.10 is thus Lemma 6.1 plus the computation (64) that produces the increment representations (50) from the martingale hypotheses. Only the latter uses probability; and it uses no path regularity, no topology on E_1, E_2 and no boundedness, so it is already stated at the level of generality of [2]. What it does use, and what Remark 6.8 isolates, is the balance condition and integrability — not, as one might guess from the shape of (52), translation invariance of the clock; see Theorem 6.5.

The discrete counterpart, which [4] do not state, now costs nothing.

Corollary 6.14 (Duality for Markov chains; $\mathbb{T} = \mathbb{N}_0$, (E0), $\iota = \mathrm{p}$)**.** *Let P and Q be transition kernels on measurable spaces E_1 and E_2, and let $f \in B(E_1\times E_2)$ satisfy the duality relation*

$$\int f(x',y)\,P(x,\mathrm{d}x') \;=\; \int f(x,y')\,Q(y,\mathrm{d}y'), \qquad (x,y)\in E_1\times E_2. \quad (69)$$

Let X and Y be independent Markov chains with kernels P and Q and $X(0)=x$, $Y(0)=y$. Then

$$E\big[f(X(n),y)\big] \;=\; E\big[f(x,Y(n))\big], \qquad n\in\mathbb{N}_0. \quad (70)$$

Proof. Put $\Phi(n,m) = E[f(X(n),Y(m))]$. Writing P for the action of the kernel on the first variable and Q for its action on the second, the Markov property and independence give $\Phi(n+1,m) - \Phi(n,m) = E[((P-I)f)(X(n),Y(m))]$ and $\Phi(n,m+1) - \Phi(n,m) = E[((Q-I)f)(X(n),Y(m))]$, which is (50) for q = counting measure with $\gamma_1(n,m) = E[((P-I)f)(X(n),Y(m))]$ and γ_2 its analogue. By (69), $(P-I)f = (Q-I)f$, so $\gamma_1 = \gamma_2$, and Proposition 6.3(a) applies. ☐

Remark 6.15. In the language of Section 3, (69) says that $(f(\cdot,y), g(\cdot,y)) \in A$ and $(f(x,\cdot), g(x,\cdot)) \in B$ hold with the *same* $g = (P-I)f = (Q-I)f$, where A and B denote the one-step generators; the associated test process is then the Doob decomposition

$$f(X(n),y) - \sum_{k<n} g(X(k),y),$$

i.e. the compensator in the predictable convention $[0,t)$. So Corollary 6.14 is the martingale-problem duality theorem for $\mathbb{T} = \mathbb{N}_0$; the predictable convention is the one in which its proof is shortest, not the only one in which it is true (Remark 6.7).

Corollary 6.16 (Stopped version; [4], Corollary 4.4.14; (T3))**.** *Let $(\mathcal{F}_t)$ and $(\mathcal{G}_t)$ be independent filtrations, let X be $(\mathcal{F}_t)$-progressive with $(\mathcal{F}_t)$-stopping time τ, and let Y be $(\mathcal{G}_t)$-progressive with $(\mathcal{G}_t)$-stopping time σ. Suppose (57) and (58) hold with X, Y replaced by $X(\cdot\wedge\tau)$, $Y(\cdot\wedge\sigma)$, and that $f(X(t\wedge\tau),y) - \int_0^{t\wedge\tau} g(X(s),y)\,\mathrm{d}s$ is an $(\mathcal{F}_t)$-martingale for each y, and $f(x,Y(t\wedge\sigma)) - \int_0^{t\wedge\sigma} h(x,Y(s))\,\mathrm{d}s$ is a $(\mathcal{G}_t)$-martingale for each x. Then the analogue of (61) holds with X, Y replaced by the stopped processes and the integrand multiplied by the indicators $\mathbf{1}_{\{s\le\tau\}}$ resp. $\mathbf{1}_{\{t-s\le\sigma\}}$.*

Proof. Rewrite the martingale in the hypothesis as $f(X(t\wedge\tau),y) - \int_0^t \mathbf{1}_{\{s\le\tau\}} g(X(s),y)\,\mathrm{d}s$ and repeat the proof of Theorem 6.10. ☐

Remark 6.17. Corollary 6.16 is the practically relevant form: the estimates (57) are often unavailable globally but easy on the exit times τ from compact sets. One then lets $\tau,\sigma\to\infty$ along an exhausting sequence; the integrand on the right of the stopped version of (61) tends to zero, and the only difficulty is to justify the interchange of limits and expectations ([4], Remark 4.4.16). This is also the route by which duality is applied to *local* martingale problems.

We can now close the circle with Section 5.

Corollary 6.18 (Uniqueness via duality; (T3))**.** *Assume (E2), let $A \subset B(E)\times B(E)$, let $(E_2,\mathcal{E}_2)$ be a measurable space, and let $B \subset M(E_2)\times M(E_2)$. Suppose there is $f \in M(E\times E_2)$ such that:*

(i) $\{f(\cdot,y) : y\in E_2\} \subset B(E)$ is separating for $\mathcal{P}(E)$;

(ii) $(f(\cdot,y), g(\cdot,y)) \in A$ for every $y \in E_2$, and $(f(x,\cdot), h(x,\cdot)) \in B$ for every $x\in E$, with $g = h$;

(iii) for every $y\in E_2$ there is a solution Y^y of the martingale problem for (B,δ_y), and for every solution X of the martingale problem for A with X independent of Y^y the hypotheses (57)–(58) hold with $\alpha=\beta=0$.

Then for every $\mu\in\mathcal{P}(E)$ the martingale problem for (A,μ) has at most one solution, every solution is Markov, and under the additional hypotheses of Theorem 5.43(b)–(c) it is strong Markov.

Proof. Let X solve the martingale problem for (A, δ_x) and let Y^y be as in (iii), independent of X. By (ii), (67) holds with $\alpha = \beta = 0$, so Corollary 6.12 gives

$$E\big[f(X(t), y)\big] \;=\; E\big[f(x, Y^y(t))\big], \qquad t \geq 0. \tag{71}$$

The right-hand side does not depend on X. Hence $\int f(\cdot, y)\,\mathrm{d}\nu_t$ is the same for the one-dimensional distributions $\nu_t = PX(t)^{-1}$ of any two solutions, and by (i) these coincide. For general μ, integrate (71) over $\mu(\mathrm{d}x)$, which is legitimate because the one-dimensional distributions of a solution for (A, μ) are mixtures of those for (A, δ_x) by Theorem 5.43(a). Now apply Theorem 5.43. □

7 Existence

Sections 5 and 6 are a uniqueness theory, and by themselves they are vacuous: nothing so far exhibits a single solution of a single martingale problem. This section collects the five routes to existence, in increasing order of cost.

(a) *From a transition semigroup* (Section 7.1). Immediate, and correspondingly weak: it presupposes the object one usually wants to construct.

(b) *From a dual process* (Section 7.2). Produces the semigroup that (a) presupposes, out of a *known* process on a second state space and a duality function. Cheap, because Section 6 has already done the analytic work; conditional on one hypothesis, kernel representability, which is where all the difficulty is concentrated.

(c) *Explicit construction* (Sections 7.3 and 7.10). Jump processes are built by hand out of a Markov chain and exponential holding times, and the martingale property is verified by a two-line computation with the exponential distribution; the same computation survives a *path-dependent* rate, which gives Hawkes processes. These are the only places in the manuscript where a solution is written down.

(d) *From stochastic differential equations* (Section 7.4). The most important route in practice and the most expensive here, since it presupposes stochastic integration; we state the chain of results and prove none of it.

(e) *By convergence* (Sections 7.5–7.7). Weak limits of solutions of approximating problems. Conditional, like (a), but on much less.

Routes (c) and (d) are the two that produce a solution out of nothing, and it is worth noticing before we start that they are the same argument at two levels of difficulty: both are Picard–Lindelöf. In (c) it is the Picard iteration for the linear equation $\dot{v} = Av$ in $B(E)$, with A bounded (Remark 7.14); in (d) it is the Picard iteration for a Lipschitz stochastic differential equation. What differs is the space on which the contraction acts. Route (b) is of a different nature: it produces nothing out of nothing either, but what it presupposes lives on a *different* state space, which in the applications is a simpler one — a finite particle system dual to a measure-valued diffusion, say.

7.1 From a transition semigroup

Proposition 7.1 ([4], Proposition 4.1.7; (T3) + (E0))**.** *Let X be a Markov process with measurable transition semigroup $\{T(t)\}$ on $B(E)$ and full generator $\hat{A}$ in the sense of Definition 2.12. Then X solves the martingale problem for $\hat{A}$.*

This is Remark 3.19, restated as an existence statement, and it is recorded only for completeness: it turns a Markov process into a solution, and producing the Markov process is the whole difficulty. The next section produces one; the three after that do not presuppose one.

7.2 From a dual process

There is a route that produces exactly the object Proposition 7.1 presupposes, and it produces it out of the material that Section 6 has already assembled. It is due to [3], and its point is that no approximating sequence of processes of a *different kind* — jump processes to approximate diffusions, and the like — is needed: the dual process, which one has anyway for uniqueness, delivers existence as well.

The idea in one line: if Y is a known process on E_2 and H is a duality function, then the right-hand side $E^y[H(x,Y_t)]$ of the duality relation is a candidate for the left-hand side $E^x[H(X_t,y)]$ of a process X that does not yet exist; if that candidate is *represented by a kernel* in x, the kernels form a transition semigroup and X exists. Everything else is bookkeeping, and all of it survives an arbitrary clock.

Setting 7.2 (Duality data; (T4) + shift invariant clock + (E1) on E_1, (E0) on E_2)**.** Let $\mathbb{T}$ satisfy (T4), let q be a clock which is *shift invariant*, $q(r+\cdot)=q(\cdot)$, and fix ι. Let $(E_1,\mathcal{E}_1)$ satisfy (E1) and $(E_2,\mathcal{E}_2)$ satisfy (E0). Let

$$H, g : E_1\times E_2\to\mathbb{R}, \qquad \beta : E_2\to\mathbb{R}$$

be bounded and measurable, and let $(P^y)_{y\in E_2}$ be a time-homogeneous Markov family on a path space F_2 over E_2 with $P^y\{\pi_0=y\}=1$. Write

$$V_t \;=\; \exp\Big\{-\int_{\langle 0,t\rangle_\iota}\beta(\pi_u)\,q(\,\mathrm{d}u)\Big\}, \tag{72}$$

the *Feynman–Kac factor*, and note $0<V_t\le e^{\|\beta\|q(\mathbb{T}_{\le t})}<\infty$, the finiteness being Definition 2.3. Define, for $\varphi\in\{H,g\}$,

$$(P_t\varphi)(x,y) \;\coloneqq\; E^y\big[\varphi(x,\pi_t)V_t\big], \qquad x\in E_1,\ y\in E_2,\ t\in\mathbb{T}. \tag{73}$$

We assume throughout:

(D1) *Balance.* For every $x\in E_1$ the process

$$t\;\longmapsto\; H(x,\pi_t)V_t-\int_{\langle 0,t\rangle_\iota} g(x,\pi_u)V_u\,q(\,\mathrm{d}u)$$

is a P^y-martingale, for every $y\in E_2$.

(D2) *Separation.* $\operatorname{span}\{H(\cdot,y):y\in E_2\}$ is separating on $\mathcal{P}(E_1)$.

(D3) *Kernel representability.* There is a family $(\mu_t)_{t\in\mathbb{T}}$ of probability kernels on E_1 with $(t,x)\mapsto\mu_t(x,\Gamma)$ jointly measurable for every $\Gamma\in\mathcal{E}_1$, such that

$$(P_t\varphi)(x,y) \;=\; \int_{E_1}\mu_t(x,\,\mathrm{d}x')\,\varphi(x',y), \qquad \varphi\in\{H,g\},$$

for all x,y,t.

Three comments before the statements. (D1) is the balance condition of Section 6 in integrated form: it says that the pair $(H(x,\cdot), g(x,\cdot))$, twisted by the Feynman–Kac factor, is a test process for Y — in the differential notation of [3] it reads $G_1 H(\cdot,y)(x) = G_2 H(x,\cdot)(y) + \beta(y)H(x,y)$ with $g = G_1 H(\cdot,y)$. (D2) is a mild non-degeneracy of the duality function. (D3) is the only substantial hypothesis, and it is the one that cannot be checked by bookkeeping; Proposition 7.7 gives the standard sufficient condition.

Lemma 7.3 (The semigroup identity; (T4) + shift invariant clock)**.** *In Setting 7.2, for $\varphi \in \{H, g\}$, $s,t \in \mathbb{T}$,*

$$(P_{s+t}\varphi)(x,y) \;=\; E^y\big[(P_t\varphi)(x,\pi_s)\,V_s\big]. \tag{74}$$

Proof. Step 1: the factor splits. By clock additivity (3),

$$V_{s+t} \;=\; V_s \cdot \exp\Big\{-\int_{\langle s,s+t\rangle_\iota} \beta(\pi_v)\,q(\,\mathrm{d}v)\Big\},$$

and substituting $v = s+u$, which is a bijection of $\langle 0,t\rangle_\iota$ onto $\langle s, s+t\rangle_\iota$ by Lemma 2.2, the second factor equals $\exp\{-\int_{\langle 0,t\rangle_\iota} \beta(\pi_{s+u})q(\,\mathrm{d}u)\} = V_t \circ \theta_s$. *This is the one place where shift invariance of q is used*: without it the substitution changes the measure and the shifted factor is not V_t.

Step 2: the Markov property. $\varphi(x,\pi_{s+t}) = (\varphi(x,\pi_t))\circ\theta_s$ and V_s is $\mathcal{F}_s^\circ$-measurable, so

$$(P_{s+t}\varphi)(x,y) = E^y\big[\varphi(x,\pi_t)\circ\theta_s \cdot (V_t\circ\theta_s)\cdot V_s\big] = E^y\Big[E^{\pi_s}\big[\varphi(x,\pi_t)V_t\big]V_s\Big],$$

by the Markov property of (P^y) and its time homogeneity. The inner expectation is $(P_t\varphi)(x,\pi_s)$, which is (74). All integrands are bounded by Setting 7.2, so every expectation is finite. ☐

Proposition 7.4 (The kernels form a transition semigroup; + (E1))**.** *In Setting 7.2, $\mu_0(x,\cdot) = \delta_x$ and*

$$\mu_{s+t}(x,\cdot) \;=\; \int_{E_1} \mu_s(x,\,\mathrm{d}x')\,\mu_t(x',\cdot), \qquad s,t\in\mathbb{T},\;\; x \in E_1. \tag{75}$$

Proof. Initial condition. $\langle 0,0\rangle_\iota = \emptyset$ gives $V_0 = 1$, and $P^y\{\pi_0 = y\} = 1$ gives $(P_0 H)(x,y) = H(x,y)$. So $\int \mu_0(x,\,\mathrm{d}x')H(x',y) = H(x,y)$ for every y, and (D2) forces $\mu_0(x,\cdot) = \delta_x$.

Chapman–Kolmogorov. Fix x,y,s,t and compute, using (D3) at x', then Fubini — legitimate because H is bounded and $\mu_s(x,\cdot)$ is a probability measure — then (D3) at x, then Lemma 7.3:

$$\begin{aligned}
\int \mu_s(x,\,\mathrm{d}x')\int \mu_t(x',\,\mathrm{d}x'')H(x'',y) &= \int \mu_s(x,\,\mathrm{d}x')\,(P_tH)(x',y)\\
&= \int \mu_s(x,\,\mathrm{d}x')\,E^y\big[H(x',\pi_t)V_t\big]\\
&= E^y\Big[\Big(\int \mu_s(x,\,\mathrm{d}x')H(x',\pi_t)\Big)V_t\Big]\\
&= E^y\big[(P_sH)(x,\pi_t)\,V_t\big]\\
&= (P_{t+s}H)(x,y) \;=\; \int \mu_{s+t}(x,\,\mathrm{d}x'')H(x'',y),
\end{aligned}$$

the commutativity of $\mathbb{T}$ being used once, in $t+s = s+t$. Since this holds for every y, (D2) gives (75). ☐

Theorem 7.5 (Existence by duality; [3], Theorem 2.1; (T4) + shift invariant clock + (E1))**.** *In Setting 7.2, let $\nu \in \mathcal{P}(E_1)$. Then there is a Markov process X on E_1 with initial distribution ν and transition kernels $(\mu_t)_{t\in\mathbb{T}}$, and it solves the martingale problem for*

$$\mathbb{X}^{(1)} \;=\; \Big\{ H(\pi_t, y) - \int_{\langle 0,t\rangle_\iota} g(\pi_u, y)\, q(\,\mathrm{d}u) \;:\; y \in E_2 \Big\}. \tag{76}$$

Moreover X and Y are in duality with potential β:

$$E\big[H(X_t, y) \mid X_0 = x\big] \;=\; E^y\big[H(x, \pi_t)V_t\big], \qquad x \in E_1,\ y \in E_2,\ t \in \mathbb{T}. \tag{77}$$

Proof. Step 1: the process. By Proposition 7.4 the family (μ_t) is a measurable transition semigroup on E_1, measurability being (D3). Since E_1 satisfies (E1) it is standard Borel, so the Kolmogorov extension theorem (Fact 2.45) applies to the projective family of finite-dimensional distributions built from ν and (μ_t); this is [4], Theorem 4.1.1. Let X be the resulting Markov process. Then (77) is (D3) for $\varphi = H$.

Step 2: the one-dimensional identity. Fix x and y and take P^y-expectations in (D1) between 0 and t; Fubini is legitimate because g and V are bounded on $\mathbb{T}_{\le t}$ and $q(\mathbb{T}_{\le t}) < \infty$. This gives

$$(P_t H)(x, y) \;=\; H(x, y) + \int_{\langle 0,t\rangle_\iota} (P_u g)(x, y)\, q(\,\mathrm{d}u).$$

Now apply (D3) to both sides — *here it is used for g as well as for H, which is why Setting 7.2 asks for both*:

$$\int \mu_t(x,\, \mathrm{d}x')H(x', y) \;=\; H(x, y) + \int_{\langle 0,t\rangle_\iota} \Big(\int \mu_u(x,\, \mathrm{d}x')g(x', y)\Big) q(\,\mathrm{d}u). \tag{78}$$

Step 3: the martingale property. Let $s \le t$ in $\mathbb{T}$. By the Markov property of X and (78) applied at X_s with $t - s$ in place of t,

$$\begin{aligned} E\big[H(X_t, y) \mid \mathcal{F}^X_s\big] &= \int \mu_{t-s}(X_s,\, \mathrm{d}x')H(x', y) \\ &= H(X_s, y) + \int_{\langle 0,t-s\rangle_\iota} E\big[g(X_{s+u}, y) \mid \mathcal{F}^X_s\big] q(\,\mathrm{d}u), \end{aligned}$$

using the Markov property once more inside the integral. Substituting $v = s+u$ and using shift invariance of q together with Lemma 2.2, the last integral is $E[\int_{\langle s,t\rangle_\iota} g(X_v, y)q(\,\mathrm{d}v) \mid \mathcal{F}^X_s]$, Fubini being available as in Step 2. Hence

$$E\Big[H(X_t, y) - \int_{\langle 0,t\rangle_\iota} g(X_v, y)q(\,\mathrm{d}v) \;\Big|\; \mathcal{F}^X_s\Big] \;=\; H(X_s, y) - \int_{\langle 0,s\rangle_\iota} g(X_v, y)q(\,\mathrm{d}v).$$

Testing against products $\prod_k h_k(X_{t_k})$ with $t_k \in \mathbb{T}_{\le s}$ and invoking Proposition 3.8 upgrades this from $\mathcal{F}^X$ to the filtration $({}^*\mathcal{F}^X_t)$ of Definition 3.5. So $X \in \mathcal{M}(\mathbb{X}^{(1)}, \nu)$. □

Corollary 7.6 (Well-posedness)**.** *Assume in addition that $\mathbb{X}^{(1)}$ admits a shift system with $\mathbb{Z}^\circ$ determining, and that Section 6 applies to the pair $(\mathbb{X}^{(1)}, Y)$ — for instance under the integrability hypothesis of Theorem 6.5. Then the martingale problem for $\mathbb{X}^{(1)}$ is well posed for every $\nu \in \mathcal{P}(E_1)$, and its solution is the Markov process of Theorem 7.5.*

Proof. Existence is Theorem 7.5. For uniqueness let $\tilde{X} \in \mathcal{M}(\mathbb{X}^{(1)}, \nu)$. The pair $(\tilde{X}, Y)$ satisfies the hypotheses of Section 6 with the same H and the balance condition (D1), so the duality relation (77) holds for $\tilde{X}$ as well; by (D2) this determines the one-dimensional distributions of $\tilde{X}$, and Theorem 5.16 concludes. This is the closing of the circle: *the same duality relation is used twice, once forwards to construct the process and once backwards to show it is the only one.* □

Proposition 7.7 (Riesz–Markov: a sufficient condition for (D3); (E3))**.** *Suppose E_1 is compact metrizable, $H \in C_b(E_1 \times E_2)$, and $\mathcal{H}_1 := \operatorname{span}\{H(\cdot, y) : y \in E_2\}$ is a convergence determining algebra containing the constants. If for each t the map $H(\cdot, y) \mapsto (P_t H)(\cdot, y)$ is positive — $\varphi \geq 0$ implies $P_t\varphi \geq 0$ — and $P_t\mathbf{1} = \mathbf{1}$, then the kernels of (D3) exist for $\varphi = H$.*

Proof. Fix t and x. By hypothesis $\mathcal{H}_1$ is a subalgebra of $C(E_1)$ containing the constants and separating points — being convergence determining — so it is dense in $C(E_1)$ by Stone–Weierstrass (Fact 2.34). The map $\varphi \mapsto (P_t\varphi)(x, \cdot)$ is linear, positive and unital on $\mathcal{H}_1$, hence bounded, hence extends to a positive unital linear functional on $C(E_1)$; the Riesz–Markov theorem provides $\mu_t(x, \cdot) \in \mathcal{P}(E_1)$ representing it. Measurability in (t, x) follows because $(t, x) \mapsto \mu_t(x, \cdot)$ is then continuous into $\mathcal{P}(E_1)$ with the weak topology, $\mathcal{H}_1$ being convergence determining; this is [3], Proposition 2.5. □

Remark 7.8 (What is new here, and what is [3]'s). The architecture is [3], Theorem 2.1. Three things are more general here, and in each case the source's hypothesis turns out to have been used for convenience rather than of necessity.

(i) *The clock.* [3] have $\mathbb{T} = \mathbb{R}_+$ with $q = \lambda$. The proofs above use the clock only through the additivity (3) and the local finiteness $q(\mathbb{T}_{\leq t}) < \infty$, so an arbitrary shift invariant clock is admissible — in particular $\mathbb{T} = \mathbb{N}_0$ with counting measure, where Theorem 7.5 becomes a statement about a Markov chain constructed from its dual, and $\iota = \mathrm{p}$ makes it the Doob decomposition of Remark 7.16.

(ii) *The index.* Nothing above uses (T2a): the semigroup identity is Lemma 7.3, which needs (T4), and Chapman–Kolmogorov needs commutativity of $+$. A two-parameter Markov process on $\mathbb{T} = \mathbb{R}_+^2$ with a dual is therefore admissible, which is not the case for Section 7.5.

(iii) *The state spaces.* Theorem 7.5 needs (E1) on E_1 — for the Kolmogorov extension in Step 1 — and only (E0) on E_2. The compactness of [3] enters solely in Proposition 7.7, i.e. in *checking* (D3), not in the theorem.

What does *not* generalize is the reach of the method. The relation (77) is a statement about one-dimensional distributions from a fixed initial state, so it presupposes exactly the Markov structure that Sections 5.1 and 5.4 took pains to avoid, and it has nothing to say about the path-dependent problems of Examples 5.40 and 5.41 or about Theorem 7.52. The natural repair is [3]'s own outlook, their Section 5: dualize the *historical* process, taking E_1 to be a path or genealogy space. That changes E_1 from compact to not even locally compact and puts the weight back on (D3); it is a project, not a remark.

The time-inhomogeneous version. Dropping shift invariance of q, or homogeneity of Y, replaces (P_t) by a two-parameter family $(P_{r,t})$ and Y by a shift system in the sense of Definition 5.10; the dual then runs on the reflected clock, which by Remark 6.6 is the q-reflection $s \mapsto Q^{-1}(Q(t) - Q(s))$ and not $t - s$. Lemma 7.3 becomes $P_{r,s}P_{s,t} = P_{r,t}$ and Proposition 7.4 the corresponding two-parameter Chapman–Kolmogorov relation, exactly as in Theorem 5.19. Nothing in the proofs resists; it is not carried out here only to keep the notation readable.

7.3 Jump processes

Throughout this subsection $\mathbb{T} = \mathbb{R}_+$ with q Lebesgue measure, so that the two conventions of Definition 2.3 agree and ι plays no rôle. The state space needs only (E0) for the construction and the martingale property; (E2) enters only when one wants to say that the paths lie in $D_E[0,\infty)$.

Setting 7.9 (Jump data; (T3) + (E0))**.** Let $\lambda : E \to [0,\infty)$ be measurable (the *rate*) and let μ be a transition kernel on E (the *jump distribution*). Define

$$Af(x) \;=\; \lambda(x) \int_E \big(f(y) - f(x)\big)\, \mu(x, \mathrm{d}y), \qquad f \in B(E), \tag{79}$$

and $A = \{(f, Af) : f \in B(E)\}$. If $\bar\lambda := \sup_x \lambda(x) < \infty$ then A maps $B(E)$ into itself with $\|Af\| \le 2\bar\lambda\|f\|$; in general Af need not be bounded.

Definition 7.10 (The construction; (T3) + (E0))**.** Let $\nu \in \mathcal{P}(E)$. On some probability space let $(Y_n)_{n\ge 0}$ be a Markov chain with initial distribution ν and transition kernel μ, and let $\varepsilon_1, \varepsilon_2, \dots$ be i.i.d. exponential random variables of parameter 1, independent of (Y_n). Put

$$\tau_0 = 0, \qquad \tau_{n+1} = \tau_n + \frac{\varepsilon_{n+1}}{\lambda(Y_n)}, \qquad \zeta = \lim_n \tau_n, \tag{80}$$

with the convention $1/0 = \infty$, and

$$X_t = Y_n \quad \text{for } \tau_n \le t < \tau_{n+1}, \qquad t < \zeta. \tag{81}$$

We call ζ the *explosion time* and (τ_n) the *jump times*.

Three observations before the theorem. The paths of X are piecewise constant and right continuous by construction, hence lie in $D_E[0,\infty)$ as soon as E carries a topology, i.e. under (E2). If $\lambda(Y_n) = 0$ then $\tau_{n+1} = \infty$ and X is absorbed at Y_n. And each τ_n is a functional of the path — the time of its n-th jump — which is what will make it a legitimate localizing sequence in the sense of (L1).

Lemma 7.11 (No explosion for bounded rates; (T3) + (E0))**.** *If $\bar\lambda < \infty$ then $\zeta = \infty$ almost surely.*

Proof. $\tau_{n+1} - \tau_n \ge \varepsilon_{n+1}/\bar\lambda$, and $\sum_n \varepsilon_{n+1}/\bar\lambda = \infty$ a.s. by the strong law of large numbers, the ε_n being i.i.d. with mean 1. □

Theorem 7.12 (Jump processes solve their martingale problem; (T3) + (E0))**.** *Let $\bar\lambda < \infty$ and let X be as in Definition 7.10. Then X solves the martingale problem for (A,ν) of (79).*

Proof. Fix $f \in B(E)$ and write $\mu f(x) = \int f(y)\mu(x,\mathrm{d}y)$, so that $Af = \lambda\cdot(\mu f - f)$. Let $\mathcal{G}_t = \sigma(X_u : u \le t)$ and

$$M_t \;=\; f(X_t) - f(X_0) - \int_0^t Af(X_r)\,\mathrm{d}r.$$

Step 1: the paths make both terms explicit. Since X is constant on $[\tau_n, \tau_{n+1})$,

$$\int_0^t Af(X_r)\,\mathrm{d}r = \sum_{n\ge 0} Af(Y_n)\big((\tau_{n+1}\wedge t) - (\tau_n \wedge t)\big),$$
$$f(X_t) - f(X_0) = \sum_{n\ge 1} \big(f(Y_n) - f(Y_{n-1})\big)\mathbf{1}_{\{\tau_n \le t\}},$$

both sums being finite a.s. by Lemma 7.11. Hence

$$M_t = \sum_{n\geq 0} D_n, \qquad D_n = \big(f(Y_{n+1})-f(Y_n)\big)\mathbf{1}_{\{\tau_{n+1}\leq t\}} - Af(Y_n)\big((\tau_{n+1}\wedge t)-(\tau_n\wedge t)\big). \quad (82)$$

Step 2: each bracket is a martingale difference. Let $\mathcal{H}_n = \sigma(Y_0, \dots, Y_n, \tau_1, \dots, \tau_n)$. Conditionally on $\mathcal{H}_n$, and on the event $\{\tau_n \leq t\}$, put $x = Y_n$, $\lambda = \lambda(x)$ and $a = t - \tau_n \geq 0$. By construction $\tau_{n+1} - \tau_n$ is exponential with parameter λ and independent of $Y_{n+1} \sim \mu(x, \cdot)$, so

$$E\big[(f(Y_{n+1}) - f(Y_n))\mathbf{1}_{\{\tau_{n+1}\leq t\}} \bigm| \mathcal{H}_n\big] = \big(\mu f(x) - f(x)\big)\big(1 - e^{-\lambda a}\big),$$
$$E\big[(\tau_{n+1} \wedge t) - \tau_n \bigm| \mathcal{H}_n\big] = E\big[\mathrm{Exp}(\lambda) \wedge a\big] = \frac{1 - e^{-\lambda a}}{\lambda}.$$

Multiplying the second identity by $Af(x) = \lambda(\mu f(x) - f(x))$ gives exactly the first. If $\lambda = 0$ both displays are read as 0 — then $\tau_{n+1} = \infty$, so the first term of D_n vanishes, and $Af(x) = 0$, so the second does too. Hence $E[D_n \mid \mathcal{H}_n] = 0$ on $\{\tau_n \leq t\}$, and $D_n = 0$ on $\{\tau_n > t\}$, an event in $\mathcal{H}_n$. *This cancellation — the factor λ in (79) against the mean $1/\lambda$ of the holding time — is the whole content of the theorem.*

Step 3: summation. The family (D_n) is dominated:

$$\sum_{n\geq 0} |D_n| \;\leq\; 2\|f\| N_t + \|Af\|\, t, \qquad N_t \coloneqq \#\{n \geq 1 : \tau_n \leq t\},$$

because the second terms telescope. Since $\tau_{n+1} - \tau_n \geq \varepsilon_{n+1}/\bar{\lambda}$, the variable N_t is stochastically dominated by a Poisson variable of parameter $\bar{\lambda} t$ and is in particular integrable. So dominated convergence applies to (82), and Step 2 gives

$$E[M_t] \;=\; \sum_{n\geq 0} E\big[E[D_n \mid \mathcal{H}_n]\big] \;=\; 0 \qquad \text{for every } t \geq 0 \quad (83)$$

and every initial law ν.

Step 4: the Markov property of the construction. Fix $s \geq 0$ and let N_s be as above. On $\{N_s = n\}$ the σ-field $\mathcal{G}_s = \sigma(X_u : u \leq s)$ is generated by $Y_0, \dots, Y_n, \tau_1, \dots, \tau_n$ and the event $\{\tau_{n+1} > s\}$. Conditionally on these, the residual holding time $\tau_{n+1} - s$ is again exponential with parameter $\lambda(Y_n)$ — memorylessness — and is independent of $(Y_{n+1}, Y_{n+2}, \dots)$ and of $\varepsilon_{n+2}, \varepsilon_{n+3}, \dots$. Hence, conditionally on $\mathcal{G}_s$, the process $(X_{s+u})_{u\geq 0}$ is again a process of Definition 7.10, with the same λ and μ and with initial state X_s.

Step 5: conclusion. For $s \leq t$,

$$M_t - M_s \;=\; \big(f(X_t) - f(X_s)\big) - \int_s^t Af(X_r)\, \mathrm{d}r$$

is the variable (82) at time $t - s$ for the restarted process of Step 4. By Step 4 and (83) applied with $\nu = \delta_{X_s}$,

$$E\big[M_t - M_s \bigm| \mathcal{G}_s\big] \;=\; 0.$$

As $|M_t| \leq 2\|f\| + 2\bar{\lambda}\|f\| t$ is bounded on bounded intervals, M is a $(\mathcal{G}_t)$-martingale, and since ${}^*\mathcal{F}_t^X = \mathcal{F}_t^X = \mathcal{G}_t$ for the right continuous X (Section 2.1), the same holds for the filtration of Definition 3.5. □

The construction gives existence. Uniqueness comes from Section 5, and the input it needs is a Gronwall estimate.

Proposition 7.13 (Well-posedness for bounded rates; (T3) + (E2))**.** *Let $\bar{\lambda} < \infty$. Then the martingale problem for (A, ν) is well posed for every $\nu \in \mathcal{P}(E)$, every solution is a Markov process, and if E satisfies (E2) it is strong Markov.*

Proof. Existence is Theorem 7.12. For uniqueness it suffices, by Theorem 5.16, to show that the one-dimensional distributions are determined by ν, and since $B(E)$ is separating it suffices to determine $E[f(X_t)]$ for every $f \in B(E)$. We show that they are given by the exponential of a *bounded* operator.

Step 1: A is bounded on $B(E)$. For $f \in B(E)$, $\|Af\| \le 2\bar{\lambda}\|f\|$ by Setting 7.9, so $A^k f \in B(E)$ is defined for every $k \ge 0$ with

$$\|A^k f\| \;\le\; (2\bar{\lambda})^k \|f\|. \tag{84}$$

This is the only place where $\bar{\lambda} < \infty$ is used, and it is what makes the iteration below converge.

Step 2: one step of the iteration. Let X be any solution of the martingale problem for (A, ν) and let $h \in B(E)$. Applying the martingale property to the pair $(h, Ah) \in A$ between 0 and t and taking expectations gives

$$E[h(X_t)] \;=\; E[h(X_0)] + \int_0^t E\big[Ah(X_r)\big]\,\mathrm{d}r, \tag{85}$$

Fubini being legitimate because Ah is bounded. Note that (85) is available for *every* $h \in B(E)$, which is what allows it to be iterated; a martingale problem posed on a smaller domain would not.

Step 3: the iteration. Insert (85) into itself, starting from $h = f$ and using it at each stage for $h = A^k f$:

$$E[f(X_t)] \;=\; \sum_{k=0}^{n} \frac{t^k}{k!} E\big[A^k f(X_0)\big] \;+\; R_n, \qquad R_n = \int_{\Delta_{n+1}(t)} E\big[A^{n+1} f(X_{r_{n+1}})\big]\,\mathrm{d}r,$$

where $\Delta_{n+1}(t) = \{0 \le r_{n+1} \le \cdots \le r_1 \le t\}$ is the simplex of volume $t^{n+1}/(n+1)!$. By (84),

$$|R_n| \;\le\; \frac{t^{n+1}}{(n+1)!}(2\bar{\lambda})^{n+1}\|f\| \;\xrightarrow[n\to\infty]{}\; 0.$$

Hence

$$E[f(X_t)] \;=\; \sum_{k\ge 0} \frac{t^k}{k!} \int_E A^k f\,\mathrm{d}\nu \;=\; \int_E \big(e^{tA} f\big)\,\mathrm{d}\nu, \tag{86}$$

the series converging absolutely in $B(E)$ by (84). The right hand side depends only on A and ν, so the one-dimensional distributions are determined and Theorem 5.16 applies.

The strong Markov property is Theorem 5.19, whose hypotheses hold because the paths are piecewise constant — so the shift system is measurable — and M has increments bounded on bounded intervals. □

Remark 7.14 (Which contraction, and where Gronwall really sits). The proof above is Picard iteration for the linear equation $\dot{v} = Av$ in the Banach space $B(E)$, and (84) is the estimate that makes the Picard operator a contraction on every bounded time interval; Gronwall's lemma would serve equally well in place of the explicit remainder R_n. It is worth contrasting it with the *first-jump equation*

$$v(t, x) \;=\; e^{-\lambda(x)t} f(x) + \int_0^t \lambda(x) e^{-\lambda(x)s} \big(\mu\, v(t-s, \cdot)\big)(x)\,\mathrm{d}s, \tag{87}$$

which one is tempted to use instead, and which yields the same conclusion by Gronwall. The temptation should be resisted at this point: (87) presupposes that the solution *is* a jump process with the prescribed first jump law, and that is a statement about the path structure which the martingale problem does not hand over — Proposition 3.8 characterizes finite-dimensional distributions, not jump times. Equation (87) is a valid identity for the constructed process of Definition 7.10, where it follows by conditioning on τ_1; but as an argument for *uniqueness* it would be circular. The iteration of Step 3 avoids the issue because it uses nothing but the martingale property.

Remark 7.15 (Explosion, and why localization is not academic). Drop the assumption $\bar{\lambda} < \infty$. Then ζ may be finite with positive probability — take $E = \mathbb{N}$, $\mu(n, \cdot) = \delta_{n+1}$ and $\lambda(n) = 2^n$, so that $E[\zeta] = \sum_n 2^{-n} < \infty$ — and (81) defines X only on $[0, \zeta)$. There is then no solution of the *global* martingale problem for A on $\mathbb{R}_+$, and the honest statement is the local one:

- Step 2 of the proof of Theorem 7.12 is a finite computation and is unaffected, so M^{τ_n} is a martingale for every n;
- τ_n is a functional of the path, the same for every solution, and $\tau_n \uparrow \zeta$;
- hence (τ_n) is a localizing system in the sense of Definition 5.22, and (L1) holds by construction rather than by assumption — exactly as Lemma 5.26 predicts, since the stopped test processes are bounded.

So X solves the local martingale problem for A on $[0, \zeta)$, and Section 5.3 applies verbatim. This is the example that Definition 5.22 was written for; up to this point it had none.

Remark 7.16 (Discrete time, and the two conventions). Deleting the holding times from Definition 7.10 leaves the chain (Y_n) itself, which is the case $\mathbb{T} = \mathbb{N}_0$ with q counting measure. There $A = \mu - I$ under the predictable convention $\iota = \mathrm{p}$, so that $M_n = f(Y_n) - f(Y_0) - \sum_{k<n}(\mu f - f)(Y_k)$ is the Doob decomposition; under $\iota = \mathrm{o}$ one would obtain the condition $\mu f - f = \mu(\mu f - f)$ instead, which is not what one wants. This is the concrete form of Remark 2.5, and it is also the case in which Corollary 6.14 lives.

Remark 7.17 (What this example is for). Besides being the only explicit solution in the manuscript, the construction exercises three of its abstractions and is the reason they were made. The state space needs only (E0): λ measurable and μ a kernel, no topology, no metric, no Polish structure — so jump processes on a standard Borel E, for instance on a space of measures or of distributions, are covered by Theorem 7.12 as it stands. The localizing system of Remark 7.15 is a genuine instance of (L1)–(L3). And the discrete case of Remark 7.16 is a genuine instance of a clock with atoms.

7.4 From stochastic differential equations

For $E = \mathbb{R}^d$ and continuous paths there is a route that is more important than all the others put together, and we state it without proofs. Let $\sigma : \mathbb{R}^d \to \mathbb{R}^{d\times m}$ and $b : \mathbb{R}^d \to \mathbb{R}^d$ be measurable, put $a = \sigma\sigma^\top$, and let

$$Af(x) \;=\; \tfrac{1}{2}a^{ij}(x)\partial_{ij}f(x) + b^i(x)\partial_i f(x), \qquad f \in C_c^\infty(\mathbb{R}^d). \tag{88}$$

A *weak solution* of the equation (σ, b) is a pair (X, B) on some filtered probability space with B an m-dimensional Brownian motion and $X_t = X_0 + \int_0^t \sigma(X_s)\,\mathrm{d}B_s + \int_0^t b(X_s)\,\mathrm{d}s$.

Fact 7.18 (Picard–Lindelöf for SDEs). If σ and b are Lipschitz, then for every x the equation (σ, b) has a strong solution with $X_0 = x$, and pathwise uniqueness holds. The proof is Picard iteration in L^2, with the Itô isometry in place of the triangle inequality.

Fact 7.19 (Yamada–Watanabe). Pathwise uniqueness together with weak existence implies uniqueness in law.

Fact 7.20 (Stroock–Varadhan; [7], Theorem 32.7). For progressive σ, b and $P \in \mathcal{P}(C_{\mathbb{R}_+,\mathbb{R}^d})$ the following are equivalent: P is the law of a weak solution of (σ, b); and P solves the local martingale problem for A of (88).

Corollary 7.21 (Well-posedness for Lipschitz coefficients; (T3), $E = \mathbb{R}^d$)**.** *If σ and b are Lipschitz, the local martingale problem for A of* (88) *is well posed for every initial distribution, and its solution is a strong Markov process.*

Proof. Facts 7.18 and 7.19 give weak existence and uniqueness in law; Fact 7.20 transfers both to the local martingale problem. The Markov properties are Theorems 5.30 and 5.19, applied with the localizing system of Remark 5.23 — which is where (L1)–(L3) were verified for exactly this operator — or directly [7], Theorem 32.11. □

Remark 7.22 (What a proof would cost). Section 2.5 contains nothing about stochastic integration, and this is the only place in the manuscript that reaches outside [4], Chapters 1–3. A self-contained treatment of Corollary 7.21 would need the Itô integral against a continuous local martingale, quadratic variation and the Kunita–Watanabe inequality, Itô's formula, and Lévy's characterization of Brownian motion — the last being what turns a solution of the martingale problem back into a solution of the equation in Fact 7.20. That is a chapter, not a section, and it is deliberately not attempted here.

For the formalization the situation is less bad than it looks: the `brownian-motion` development of Remark 8.1 is building stochastic integration in Lean, so the dependency may become available rather than having to be created. Until then, Section 7.3 is the route that can actually be carried out.

7.5 By convergence: the Ethier–Kurtz version

The remaining route obtains solutions as weak limits of solutions of approximating martingale problems. We state the classical result first, and then the abstract theorem of which it is a corollary; its proof below is a forward reference, and deliberately so — the point of the arrangement is that Lemma 7.23 has no proof of its own.

Lemma 7.23 ([4], Lemma 4.5.1; (T3))**.** *Assume (E3), let $A \subset C_b(E) \times C_b(E)$ and $A_n \subset B(E) \times B(E)$ for $n \in \mathbb{N}$. Suppose that for each $(f, g) \in A$ there exist $(f_n, g_n) \in A_n$ with*

$$\lim_{n\to\infty} \|f_n - f\| = 0, \qquad \lim_{n\to\infty} \|g_n - g\| = 0. \tag{89}$$

If for each n the process X_n is a solution of the martingale problem for A_n with sample paths in $D_E[0, \infty)$, and if $X_n \Rightarrow X$ in $D_E[0, \infty)$, then X is a solution of the martingale problem for A.

Proof. We verify (C1)–(C3) of Theorem 7.27, with $F = D_E[0, \infty)$ — Polish by (E3) and Fact 2.37 — and $\mathbb{X} = \mathbb{X}_A(X)$, whose canonical version is $Y_t^\circ(\omega) = f(\omega(t)) - \int_0^t g(\omega(u))\,du$ for $(f, g) \in A$. Fix $(f, g) \in A$ and an approximating sequence (f_n, g_n) as in (89), and put

$$D \;=\; \{u \ge 0 : P\{X(u) = X(u-)\} = 1\},$$

which has countable complement by Fact 2.39 and is therefore dense; $\mathbb{T} = \mathbb{R}_+$ has no greatest element, so the proviso in Theorem 7.27 on D is vacuous.

(C1) and (C2). The first is the assumed weak convergence $X_n \Rightarrow X$ on $D_E[0,\infty)$. For the second take $\mathbb{Z}^\circ$ to be the determining set of Example 3.4 built from D, i.e. the products $Z_s^\circ = \prod_{i\le k} h_i(\pi_{t_i})$ with $h_i \in C_b(E)$ and $t_i \in D \cap \mathbb{T}_{\le s}$.

(a). For $t \in D$ the evaluation $\omega \mapsto \omega(t)$ is J_1-continuous at every ω with $\omega(t) = \omega(t-)$, hence PX^{-1}-a.s. The integral term is continuous at *every* ω: if $\omega_m \to \omega$ in $D_E[0,\infty)$, then $\omega_m(u) \to \omega(u)$ for every u outside the at most countable set of discontinuities of ω, hence Lebesgue almost everywhere, so $g(\omega_m(u)) \to g(\omega(u))$ by continuity of g and bounded convergence gives $\int_0^t g(\omega_m(u))\,du \to \int_0^t g(\omega(u))\,du$. So Y_t° is P-continuous at X, and so is $Y_t^\circ Z_s^\circ$, the factor Z_s° being a product of evaluations at times in D.

(b). f and g are bounded, so $|Y_r^\circ| \le \|f\| + t\|g\|$ for $r \le t$, and a uniformly bounded family is uniformly integrable.

(c). Write $Y_t^{\circ,n}(\omega) = f_n(\omega(t)) - \int_0^t g_n(\omega(u))\,du$. Since X_n solves the martingale problem for A_n and $(f_n, g_n) \in A_n$, Proposition 3.8 gives

$$E^{P^n}\big[\big(Y_t^{\circ,n}(X_n) - Y_s^{\circ,n}(X_n)\big) Z_s^\circ(X_n)\big] \;=\; 0$$

for all $s \le t$ and $Z_s^\circ \in \mathbb{Z}_s^\circ$. Moreover

$$\big|Y_r^\circ(\omega) - Y_r^{\circ,n}(\omega)\big| \;\le\; \|f_n - f\| + r\|g_n - g\| \qquad \text{for every } \omega \in D_E[0,\infty),$$

a bound uniform in ω, so with $C = \sup_i \|h_i\|^k$,

$$\Big|E^{P^n}\big[(Y_t^\circ(X_n) - Y_s^\circ(X_n)) Z_s^\circ(X_n)\big]\Big| \;\le\; 2C\big(\|f_n - f\| + t\|g_n - g\|\big) \xrightarrow[n\to\infty]{} 0,$$

which is (c). *This is the only place where* (89) *is used, and it is used only as a uniform bound* — which is why Theorem 7.30 can replace it by the far weaker (91)–(92).

Conclusion. The paths of X lie in $D_E[0,\infty)$, so every $Y \in \mathbb{X}$ is right continuous, and $\mathbb{T} = \mathbb{R}_+$ satisfies (T2b) with D dense. Theorem 7.27 therefore gives $P \in \mathcal{M}(\mathbb{X})$, i.e. X solves the martingale problem for A. □

Remark 7.24 (Why this is a corollary and not a theorem). [4] prove Lemma 7.23 directly, and an earlier version of this manuscript reproduced that proof. It is worth being explicit that nothing is lost by not doing so, because the two arguments are the same argument: [4]'s "let $D(X)$ be the continuity times, take limits there, extend by right continuity" is precisely Steps 1 and 3 of Theorem 7.27, and their appeal to bounded convergence is its Steps 0 and 2 in the special case where (b) holds because f and g are bounded. Reproducing it would duplicate the abstract proof in a notation that hides which hypothesis does what.

What the derivation makes visible is the division of labour that the direct proof conceals. Of the four hypotheses of Lemma 7.23, weak convergence is (C1); $A \subset C_b \times C_b$ is what buys (a); boundedness of f and g is what buys (b); and the uniform approximation (89) is what buys (c) — one hypothesis per condition, with no overlap. Each can then be weakened separately, and Theorem 7.30 weakens the last two while Section 7.9 weakens (a).

Remark 7.25 ([4], Remark 4.5.2). Lemma 7.23 presupposes convergence. Suppose (E, r) is Polish and $\mathcal{D}(A)$ contains an algebra that separates points and vanishes nowhere; by Fact 2.34 such an algebra is separating, and it is dense in $C_b(E)$ in the topology of uniform convergence on compact sets. Then $\{X_n\}$ is relatively compact provided (89) holds for each $(f, g) \in A$ and the compact containment condition (15) holds. This is Facts 2.41, 2.42, 2.40 and 2.28. Combining with Lemma 7.23: every limit point of $\{X_n\}$ solves the martingale problem for A; in particular the martingale problem for A has a solution with sample paths in $D_E[0,\infty)$. If in addition uniqueness holds (e.g. by Theorem 5.43 or Corollary 6.18), then $X_n \Rightarrow X$ for the unique solution X.

7.6 By convergence: the abstract theorem

Theorem 7.30 below is an instance of one statement, which uses neither an operator, nor càdlàg paths, nor the Skorokhod topology, nor any structure on the time index beyond a dense subset — and none at all if that subset is all of $\mathbb{T}$. It is [2], Theorem 3.14 and Corollary 3.17, and it is the cheapest genuine theorem in this manuscript; we give it with its proof, since it is what the formalization should target (step **(F4)** of Section 8). Lemma 7.23 is a special case of it directly, as its proof shows, and a fortiori of Theorem 7.30; so all of Section 7 rests on Theorem 7.27.

Definition 7.26 (P-continuity at X; (T0) + (E3))**.** A Borel map $\psi : F \to \mathbb{R}$ is *P-continuous at X* if there is $C \in \mathcal{B}(F)$ with $P\{X \in C\} = 1$ such that $\psi(\alpha_m) \to \psi(\alpha)$ whenever $\alpha_m \to \alpha$ in F with $\alpha \in C$.

Theorem 7.27 (Abstract convergence; [2], Theorem 3.14 and Corollary 3.17; (T0) + (E3))**.** *In Setting 3.1, let $\mathbb{X}$ be canonical for X, let $D \subset \mathbb{T}$, and for each n let $B^n = (\Omega^n, \mathcal{F}^n, \mathbb{F}^n, P^n)$ be a filtered probability space carrying a process X^n with paths in F. Assume:*

(C1) $X^n \Rightarrow X$ weakly on F;

(C2) $\mathbb{X}$ admits a determining set $\mathbb{Z}^\circ$ (Definition 3.3);

(C3) for every $Y \in \mathbb{X}$ there is a canonical version (Y_t°) such that for all $t \in D$, $s \in D \cap \mathbb{T}_{\le t}$ and $Z_s^\circ \in \mathbb{Z}_s^\circ$:

(a) Y_t° and $Y_t^\circ Z_s^\circ$ are P-continuous at X;

(b) $\{Y_r^\circ(X^n) : r \in D \cap \mathbb{T}_{\le t},\ n \in \mathbb{N}\}$ is uniformly integrable;

(c) $\lim_{n\to\infty} E^{P^n}\big[\big(Y_t^\circ(X^n) - Y_s^\circ(X^n)\big) Z_s^\circ(X^n)\big] = 0$.

Then $E^P[Y_t \mid \mathcal{F}_s] = Y_s$ for all $s \le t$ in D and all $Y \in \mathbb{X}$. If moreover $D = \mathbb{T}$, then $P \in \mathcal{M}(\mathbb{X})$. The same conclusion holds for a proper $D \subsetneq \mathbb{T}$ provided $\mathbb{T}$ satisfies (T2b) with D its countable dense subset, D contains the greatest element of $\mathbb{T}$ if there is one, and every $Y \in \mathbb{X}$ is right continuous.

Proof. Fix $Y \in \mathbb{X}$ with canonical version Y°, and $t \in D$.

Step 0: integrability. For $r \in D \cap \mathbb{T}_{\le t}$ the map $|Y_r^\circ|$ is P-continuous at X by (a), so $|Y_r^\circ(X^n)| \Rightarrow |Y_r^\circ(X)|$ by (C1) and Fact 2.26; the family is uniformly integrable by (b), so Fact 2.43 gives $E^{P^n}|Y_r^\circ(X^n)| \to E^P|Y_r^\circ(X)| < \infty$. In particular $Y_r = Y_r^\circ(X) \in L^1(P)$.

Step 1: the martingale identity on D. Let $s \in D \cap \mathbb{T}_{\le t}$ and $Z_s^\circ \in \mathbb{Z}_s^\circ$, and put $\psi = (Y_t^\circ - Y_s^\circ) Z_s^\circ : F \to \mathbb{R}$. Both $Y_t^\circ Z_s^\circ$ and $Y_s^\circ Z_s^\circ$ are P-continuous at X by (a) — the second by applying (a) with t replaced by s, legitimate since $s \in D$ and $s \in D \cap \mathbb{T}_{\le s}$ — and Definition 7.26 is stable under differences, the exceptional sets being intersected; so ψ is P-continuous at X. Hence $\psi(X^n) \Rightarrow \psi(X)$ by (C1) and Fact 2.26. The family $\{\psi(X^n)\}_n$ is uniformly integrable, being dominated by $\|Z_s^\circ\|(|Y_t^\circ(X^n)| + |Y_s^\circ(X^n)|)$ and Z_s° bounded by Definition 3.3(i). So Fact 2.43 applies and, with (c),

$$E^P\big[(Y_t - Y_s) Z_s^\circ(X)\big] \;=\; E^P\big[\psi(X)\big] \;=\; \lim_{n\to\infty} E^{P^n}\big[\psi(X^n)\big] \;=\; 0,$$

the first equality because $\mathbb{X}$ is canonical. As $Z_s^\circ \in \mathbb{Z}_s^\circ$ was arbitrary and $Y_s, Y_t \in L^1(P)$ by Step 0, part (ii) of Definition 3.3 yields $E^P[Y_t \mid \mathcal{F}_s] = Y_s$. If $D = \mathbb{T}$ this is already the assertion.

Step 2: uniform integrability under P. Let $\varphi_N(x) = |x| - |x| \wedge N$. For $r \in D \cap \mathbb{T}_{\le t}$ the map $\varphi_N \circ Y_r^\circ$ is P-continuous at X, and $\{\varphi_N(Y_r^\circ(X^n))\}_n$ is uniformly integrable because it is dominated by $|Y_r^\circ(X^n)|$. So Fact 2.43 applies twice and gives

$$E^P\big[\varphi_N(Y_r)\big] \;=\; \lim_{n\to\infty} E^{P^n}\big[\varphi_N(Y_r^\circ(X^n))\big] \;\le\; \sup_{n\in\mathbb{N}} E^{P^n}\big[\varphi_N(Y_r^\circ(X^n))\big].$$

Taking the supremum over $r \in D \cap \mathbb{T}_{\le t}$ and letting $N \to \infty$, the right-hand side tends to 0 by (b) and the second characterization of uniform integrability in Fact 2.43. Hence $\{Y_r : r \in D \cap \mathbb{T}_{\le t}\}$ is uniformly integrable under P.

Step 3: from D to $\mathbb{T}$. Assume (T2b) and Y right continuous, and let $s < t$ in $\mathbb{T}$ be arbitrary; the case $s = t$ is trivial. Choose t_m as follows: if t is the greatest element of $\mathbb{T}$, put $t_m = t$, which lies in D by hypothesis; otherwise (T2b) provides $t_m \downarrow t$ in D with $t_m > t$. Since $s < t$, the same argument gives $s_m \downarrow s$ in D with $s \le s_m < t_m$ for all large m. Let $G \in \mathcal{F}_s$. Then $G \in \mathcal{F}_{s_m}$ because $s \le s_m$, so Step 1 gives $E^P[Y_{t_m}\mathbf{1}_G] = E^P[Y_{s_m}\mathbf{1}_G]$. By right continuity $Y_{t_m} \to Y_t$ and $Y_{s_m} \to Y_s$ P-a.s. All the indices involved lie in $D \cap \mathbb{T}_{\le t_1}$, so both families are uniformly integrable by Step 2 applied at t_1; Fact 2.43 applied to $Y_{t_m}\mathbf{1}_G$ and $Y_{s_m}\mathbf{1}_G$ yields $E^P[Y_t\mathbf{1}_G] = E^P[Y_s\mathbf{1}_G]$. As $G \in \mathcal{F}_s$ was arbitrary, $E^P[Y_t \mid \mathcal{F}_s] = Y_s$, and $P \in \mathcal{M}(\mathbb{X})$. □

Remark 7.28 (Where the index enters). Steps 0–2 use no structure on $\mathbb{T}$ whatsoever beyond the preorder: D is any subset, and $\mathbb{T}_{\le t}$ is a down-set. The only place where more is needed is Step 3, and it is needed only to pass from D to its complement. With $D = \mathbb{T}$ — in particular for $\mathbb{T} = \mathbb{N}_0$, where every point is isolated from the right and no approximation is available or necessary — Step 3 is empty and the theorem holds under (T0). This is why step **(F4)** of Section 8 can be done immediately after **(F3)**, before any of the Skorokhod theory.

Remark 7.29 (What has to be checked, and what does not). Of the three hypotheses, (C1) is the input, (C2) is a property of the path space and is supplied once and for all by Example 3.4 or by Proposition 3.8, and (C3) is the work. Within (C3), (a) is where the topology of F enters — and it is the only place, which is what Lemma 7.40 makes precise and Theorem 7.41 exploits; for $F = D_E[0,\infty)$ it is the reason for the "countable complement" hypothesis of Theorem 7.30, since $\alpha \mapsto \alpha(t)$ is J_1-continuous exactly at paths not jumping at t. Condition (c) is where the approximating martingales are used, and (b) is what makes the two limits interchangeable.

7.7 By convergence: the general version

The following theorem is [2], Theorem 4.1; it is the specialization of the abstract convergence theorem [2], Corollary 3.17, to the Markovian martingale problem, and it generalizes Lemma 7.23 — which Theorem 7.27 has already delivered — in three independent directions:

(1) the approximating objects are *not* required to solve a martingale problem for an operator; they are arbitrary pairs (ξ^n, φ^n) of progressively measurable processes such that $\xi^n - \int_0^\cdot \varphi_s^n\,\mathrm{d}s$ is a martingale;

(2) the approximation (89) in the uniform norm is replaced by the much weaker *asymptotic* conditions (91) and (92), which are conditions on the laws only;

(3) uniform boundedness is replaced by uniform integrability (90).

Theorem 7.30 ([2], Theorem 4.1; (T3)). *Assume (E3). Let $B = (\Omega, \mathcal{F}, \mathbb{F}, P)$ and $B^n = (\Omega^n, \mathcal{F}^n, \mathbb{F}^n, P^n)$ be filtered probability spaces carrying E-valued càdlàg adapted processes X and X^n, and assume that $\mathbb{F}$ is generated by X. Let $A \subset C_b(E) \times C_b(E)$ and let $\mathbb{X} = \mathbb{X}_A(X)$ be as in (20). For each n, let $\mathbb{X}^n$ be a set of pairs (ξ, φ) of real-valued progressively measurable processes on B^n such that $\sup_{s \le T} E^{P^n}[|\xi_s| + |\varphi_s|] < \infty$ for all $T > 0$ and such that $\xi - \int_0^\cdot \varphi_s \, \mathrm{d}s$ is a martingale on B^n.*

Suppose that $X^n \Rightarrow X$ weakly on $D_E[0, \infty)$ with the Skorokhod J_1 topology, and that there is a set $\Gamma \subset \mathbb{R}_+$ with countable complement such that for each $(f, g) \in A$ and $T > 0$ there is a sequence $(\xi^n, \varphi^n) \in \mathbb{X}^n$ with

$$\sup_{n \in \mathbb{N}} \sup_{s \le T} E^{P^n}\big[|\xi^n_s| + |\varphi^n_s|\big] < \infty, \tag{90}$$

$$\lim_{n \to \infty} E^{P^n}\Big[\big(\xi^n_t - f(X^n_t)\big) \prod_{i=1}^k h_i(X^n_{t_i})\Big] = 0, \tag{91}$$

$$\lim_{n \to \infty} E^{P^n}\Big[\int_s^t \big(\varphi^n_u - g(X^n_u)\big) \, \mathrm{d}u \prod_{i=1}^k h_i(X^n_{t_i})\Big] = 0, \tag{92}$$

for all $k \in \mathbb{N}$, $t_1, \dots, t_k \in \Gamma \cap [0, t]$, $t \in \Gamma \cap [0, T]$, $h_1, \dots, h_k \in C_b(E)$. Then $P \in \mathcal{M}(\mathbb{X})$, i.e. X solves the martingale problem for A.

Proof. We verify (C1)–(C3) of Theorem 7.27.

The set D, and (C2). Set $D = \{t \in \Gamma : P\{X_t \ne X_{t-}\} = 0\}$. By Fact 2.39 and the hypothesis on Γ, D^c is countable, so D is dense. Let $\mathbb{Z}^\circ$ be the determining set of Example 3.4 built from D; this is (C2). Hypothesis (C1) is the assumed weak convergence $X^n \Rightarrow X$ on $D_E[0, \infty)$.

(b). Let $Y = f(X) - \int_0^\cdot g(X_s) \, \mathrm{d}s \in \mathbb{X}$ with canonical version $Y^\circ_t(\omega) = f(\omega(t)) - \int_0^t g(\omega(s)) \, \mathrm{d}s$. Since f, g are bounded, Y°_r is bounded on $r \in [0, t]$, so $\{Y^\circ_r(X^n) : r \in D \cap [0, t],\ n \in \mathbb{N}\}$ is uniformly integrable.

(a). The evaluation $\omega \mapsto \omega(t)$ is J_1-continuous at every ω with $\omega(t) = \omega(t-)$, so the definition of D makes Y°_t and $Y^\circ_t Z^\circ_s$ P-continuous at X in the sense of Definition 7.26, for $s, t \in D$. This is the only point at which the Skorokhod topology is used, and it is what the hypothesis on Γ is for.

(c). It remains to verify, for $s, t \in D$ with $s \le t$ and $Z^\circ_s = \prod_i h_i(\cdot(t_i))$,

$$\lim_{n \to \infty} E^{P^n}\big[\big(Y^\circ_t(X^n) - Y^\circ_s(X^n)\big) Z^\circ_s(X^n)\big] \;=\; 0. \tag{93}$$

Put $Y^n = \xi^n - \int_0^\cdot \varphi^n_u \, \mathrm{d}u$, a martingale on B^n. Because $Z^\circ_s(X^n)$ is $\mathcal{F}^n_s$-measurable, $E^{P^n}[(Y^n_t - Y^n_s) Z^\circ_s(X^n)] = 0$, so

$$E^{P^n}\big[(Y^\circ_t(X^n) - Y^\circ_s(X^n)) Z^\circ_s(X^n)\big] = E^{P^n}\big[(Y^\circ_t(X^n) - Y^n_t + Y^n_s - Y^\circ_s(X^n)) Z^\circ_s(X^n)\big],$$

and the right-hand side tends to 0 by (91) and (92), since $Y^\circ_t(X^n) - Y^n_t = (f(X^n_t) - \xi^n_t) + \int_0^t (\varphi^n_u - g(X^n_u)) \, \mathrm{d}u$. Thus (93) holds, which is (c).

Theorem 7.27 now gives $E^P[Y_t \mid \mathcal{F}_s] = Y_s$ for $s \le t$ in D, and its Step 3 — which uses the uniform integrability established in its Step 2 from (b), together with the right continuity of Y — extends this to arbitrary $s < t$. Hypothesis (90) is what makes all the expectations in (91) and (92) finite, uniformly in n; it is not what supplies (b), which here comes for free from the boundedness of f and g. $\square$

Remark 7.31. Theorem 7.30 concludes that P solves the *global* martingale problem. To obtain a solution of a *local* martingale problem in the limit, apply the theorem to a localized family of test processes, i.e. to $\mathbb{X}^{(m)} = \{Y^{\tau_m} : Y \in \mathbb{X}\}$ for a suitable exhausting sequence of stopping times τ_m, and let $m \to \infty$. This is how [2] obtain their semimartingale results (their Section 4.3), and it is the same reduction as in Remark 5.45.

Remark 7.32 (Path spaces other than $D_E[0,\infty)$). Theorem 7.27 is stated for an arbitrary Polish path space $F \subset E^{\mathbb{T}}$, and Theorem 7.30 is the instance $F = D_E[0,\infty)$. Nothing in the abstract theorem refers to càdlàg paths, to a Markov structure, or to the Skorokhod topology; F may for instance be $L^p_{\mathrm{loc}}(\mathbb{R}_+,\mathbb{R}^d)$, which is the state space [2] use for solutions of Volterra equations (their Example 3.13), or a product of such spaces. What changes from one F to another is only which maps are P-continuous at X, i.e. hypothesis (a) — see Remark 7.29. [2] also give a version in which the uniform integrability of (b) is dropped (their Corollary 3.21) and one with control variables and weak-strong convergence (their Theorem 3.14). The part of the latter that concerns different clocks is Section 7.8, the part that concerns weak-strong convergence is Section 7.9. Theorem 7.41 is a third instance of the same freedom in F — there the enlargement is by finitely or countably many coordinates rather than by a change of topology — and it is the one that does the work here.

7.8 Approximation on different clocks

Nothing so far requires the approximating processes to run on the same clock as the limit, and this subsection makes that precise. The observation to begin with is negative and structural: *Theorem 7.27 does not mention the approximating martingale problems at all.* Its hypotheses (C1)–(C3) speak only of the canonical version Y° of the *limit* test process, evaluated along X^n. Where the approximating martingales come from, on which clock they are compensated, and with which convention, is not part of the theorem — it is part of the verification of (c). So different clocks are not a difficulty for the theorem; they are a difficulty for one of its hypotheses, and the following isolates it.

Setting 7.33 (Approximants with their own clocks; (T3) + (E3))**.** In the situation of Theorem 7.27, let the limit test process be

$$Y^\circ_t \;=\; f(\pi_t) - \int_{\langle 0,t\rangle_\iota} g(u,\pi_u)\, q(\,\mathrm{d}u),$$

and for each n let q^n be a clock on $\mathbb{T}$, ι_n a convention, and ξ^n, ψ^n progressively measurable processes on B^n such that

$$Y^n_t \;=\; \xi^n_t - \int_{\langle 0,t\rangle_{\iota_n}} \psi^n_u\, q^n(\,\mathrm{d}u) \tag{94}$$

is a P^n-martingale. Write, for $s \le t$,

$$\Delta^n_{s,t} \;\coloneqq\; \int_{\langle s,t\rangle_{\iota_n}} \psi^n_u\, q^n(\,\mathrm{d}u) - \int_{\langle s,t\rangle_\iota} \psi^n_u\, q(\,\mathrm{d}u) \tag{95}$$

for the *clock discrepancy* along the integrands.

Theorem 7.34 (Clock change; (T3) + (E3))**.** *Assume (C1), (C2), (a) and (b) of Theorem 7.27, and suppose that for all $s \le t$ in D and all $Z^\circ_s \in \mathbb{Z}^\circ_s$:*

(K1) $\sup_n \sup_{r\le T} E^{P^n}\big[|\xi^n_r| + |\psi^n_r|\big] < \infty$ *for every T;*

(K2) $\lim_n E^{P^n}\left[(\xi^n_t - f(X^n_t))\, Z^\circ_s(X^n)\right] = 0$;

(K3) $\lim_n E^{P^n}\left[\int_{\langle s,t\rangle_\iota} (\psi^n_u - g(u, X^n_u))q(\,\mathrm{d}u)\; Z^\circ_s(X^n)\right] = 0$;

(K4) $\lim_n E^{P^n}\left[\Delta^n_{s,t}\, Z^\circ_s(X^n)\right] = 0$.

Then (c) holds, and consequently $P \in \mathcal{M}(\mathbb{X})$.

Proof. Fix $s \le t$ in D and $Z^\circ_s \in \mathbb{Z}^\circ_s$. Since Y^n is a P^n-martingale and $Z^\circ_s(X^n)$ is $\mathcal{F}^n_s$-measurable, $E^{P^n}[(Y^n_t - Y^n_s)Z^\circ_s(X^n)] = 0$, so

$$E^{P^n}\left[(Y^\circ_t(X^n) - Y^\circ_s(X^n))Z^\circ_s(X^n)\right] = E^{P^n}\left[\left(R^n_t - R^n_s\right)Z^\circ_s(X^n)\right], \qquad R^n_r := Y^\circ_r(X^n) - Y^n_r. \tag{96}$$

By (94) and the definition of Y°,

$$R^n_r = \left(f(X^n_r) - \xi^n_r\right) + \int_{\langle 0,r\rangle_{\iota_n}} \psi^n_u\, q^n(\,\mathrm{d}u) - \int_{\langle 0,r\rangle_\iota} g(u, X^n_u)\, q(\,\mathrm{d}u),$$

Inserting $\pm\int_{\langle 0,r\rangle_\iota} \psi^n_u\, q(\,\mathrm{d}u)$ splits the last two terms into $\Delta^n_{0,r}$ and

$$\int_{\langle 0,r\rangle_\iota} (\psi^n_u - g(u, X^n_u))q(\,\mathrm{d}u).$$

Taking the difference at $r = t$ and $r = s$, and using the additivity (3) of both interval families,

$$R^n_t - R^n_s = \left(f(X^n_t) - \xi^n_t\right) - \left(f(X^n_s) - \xi^n_s\right) + \Delta^n_{s,t} + \int_{\langle s,t\rangle_\iota} \left(\psi^n_u - g(u, X^n_u)\right)q(\,\mathrm{d}u).$$

Each of the four terms has vanishing limit against $Z^\circ_s(X^n)$, by (K2) twice, (K4) and (K3); (K1) makes all the expectations finite. So the left-hand side of (96) tends to 0, which is (c), and Theorem 7.27 applies. $\square$

Remark 7.35. Condition (K4) is *not* weak convergence $q^n \Rightarrow q$. It is convergence of the clocks *paired with the integrands*, and it does not follow from $q^n \Rightarrow q$ unless the family $\{\psi^n\}$ is equicontinuous in a suitable sense — which for the piecewise constant ψ^n of Example 7.36 it is not. This is why [2], Assumption 5.3 states it as a hypothesis, their condition (5.10), rather than deriving it; and Theorem 7.34 is their Theorem 5.4, stripped of the control variables. Note also that (K4) absorbs the difference between ι_n and ι: the two conventions for one clock differ by a boundary term, and (95) does not care which of the two discrepancies it is measuring.

Example 7.36 (Rescaled Markov chains; the invariance principle)**.** Let $E = \mathbb{R}^d$, $F = D_E[0,\infty)$, $q = \lambda$ for the limit, and for the n-th approximant

$$q^n = \frac{1}{n}\sum_{k\ge 1} \delta_{k/n}, \qquad \iota_n = \mathrm{o}.$$

Let $(\Xi^n_k)_{k\ge 0}$ be a Markov chain with one-step kernel P_n and put $X^n_t = \Xi^n_{\lfloor nt\rfloor}$, a piecewise constant element of $D_E[0,\infty)$. With $\xi^n = f(X^n)$ and the *predictable* integrand $\psi^n_u = n\,(P_n f - f)(X^n_{u-})$ we get, since $X^n_{k/n-} = \Xi^n_{k-1}$,

$$\int_{(0,t]} \psi^n_u\, q^n(\,\mathrm{d}u) = \frac{1}{n}\sum_{1\le k\le nt} \psi^n_{k/n} = \sum_{j<\lfloor nt\rfloor} (P_n f - f)(\Xi^n_j),$$

so that (94) reads $Y_t^n = f(\Xi^n_{\lfloor nt\rfloor}) - \sum_{j<\lfloor nt\rfloor}(P_n f - f)(\Xi^n_j)$. This is the Doob decomposition of $f(\Xi^n)$ read along the embedded grid, it is constant on each $[k/n,(k+1)/n)$, and it is a genuine $(\mathcal{F}^{X^n}_t)_{t\geq 0}$-martingale.

Then (K2) holds trivially, $\xi^n - f(X^n)$ being identically zero; (K3) is the classical requirement that the rescaled generators $n(P_n f - f)$ converge to Af — the discrepancy between X^n_{u-} and X^n_u is immaterial there, the two differing only on the Lebesgue null set $\frac{1}{n}\mathbb{N}$; and (K4) holds because ψ^n is constant on each interval $(\frac{k-1}{n},\frac{k}{n}]$, on which the atomic and the Lebesgue integral give the same value $(P_n f - f)(\Xi^n_{k-1})$, so that the two integrals agree except on the two boundary intervals and $|\Delta^n_{s,t}| \leq \frac{2}{n}\sup_u|\psi^n_u| = 2\|P_n f - f\|$, which tends to 0 as soon as $n(P_n f - f)$ is bounded.

So "a rescaled Markov chain converges to a diffusion" is, in this formalism, a statement about clocks: $q^n \to q$ along the integrands, with the chain's clock atomic and the limit's atomless.

Remark 7.37 (Why the convention flips on embedding). Remark 7.16 says that on $\mathbb{T} = \mathbb{N}_0$ with counting measure the Doob decomposition is the *predictable* convention $\iota = \mathrm{p}$, and Example 7.36 uses $\iota_n = \mathrm{o}$. There is no contradiction, and the point is worth isolating because it is easy to get wrong — an earlier version of this manuscript did.

On $\mathbb{T} = \mathbb{N}_0$ there is nothing between two time points, and $f(\Xi_m) - \sum_{k<m}(P_n f - f)(\Xi_k)$ is a martingale for the only filtration there is. Embed the grid into $\mathbb{R}_+$, however, and the same formula with $\iota_n = \mathrm{p}$ and $\psi^n_u = n(P_n f - f)(X^n_u)$ gives

$$\int_{[0,t)} \psi^n_u\, q^n(\mathrm{d}u) \;=\; \sum_{k<nt}(P_n f - f)(\Xi^n_k),$$

whose value for $t \in (\frac{m}{n},\frac{m+1}{n})$ is $\sum_{k\leq m}(P_n f - f)(\Xi^n_k)$, one summand *more* than at $t = m/n$. So Y^n drops by $(P_n f - f)(\Xi^n_m)$ immediately after m/n while $\mathcal{F}^{X^n}$ does not change, and Y^n is a martingale along the grid but *not* in continuous time. The mismatch is exactly the one Section 4 warns about: with $\iota = \mathrm{p}$ an atom of the clock at u is compensated *after* u, whereas $f(X^n)$ jumps *at* u.

The repair is the one the semimartingale literature makes by reflex: take $\iota_n = \mathrm{o}$ and evaluate the integrand at $u-$. Then atom and jump occur at the same instant and cancel. This is the sharpest argument in the manuscript for carrying both conventions: the same object needs p intrinsically and o after embedding, and a manuscript committed to one of them could not state Example 7.36 at all.

Remark 7.38 (What this example costs, and what paid for it). Three earlier decisions are needed to write Example 7.36 down at all, and none of them was made with it in view.

(i) *Atoms* (D7 of the progress log, Theorem 4.3). Without a clock that may have atoms, the chain's martingale problem is not a martingale problem; one would have to interpolate the chain into continuous time, which is precisely the manoeuvre the clock makes unnecessary.

(ii) *Clocks as measures on* $\mathbb{T}$ (Definition 2.3). Different clocks on one $\mathbb{T}$ subsume different time indices, as long as the latter embed: $\frac{1}{n}\mathbb{N}_0 \subset \mathbb{R}_+$ carries q^n, and no separate index set is needed.

(iii) *Both conventions* (Remark 2.5). The chain needs $\iota = \mathrm{p}$ intrinsically and $\iota = \mathrm{o}$ once its grid is embedded in $\mathbb{R}_+$ (Remark 7.37); the limit does not care. Had one convention been fixed, Example 7.36 could not be stated as it is. That the two differ by $\psi^n_t\, q^n(\{t\}) = (P_n f - f)(X^n_{t-}) = O(1/n)$, and so *wash out in the limit*, is what makes the flip harmless; it is a retrospective argument for carrying both rather than choosing.

What is *not* supplied here is tightness: Theorem 7.34 assumes (C1), and establishing $X^n \Rightarrow X$ on $D_E[0,\infty)$ is the Skorokhod half of the theory, cited in Remark 7.25 and not proved.

7.9 Relaxing the continuity hypothesis

Theorem 7.27 has two hypotheses that can fail in practice, and this manuscript now has a section for each. Section 7.8 relaxes (c), the approximation condition. This one relaxes (a), the continuity condition — and it has to, because (a) fails outright in precisely the situation that the clock of Definition 2.3 was widened to admit.

There are two repairs, and they are not of the same weight. The first, given here in full, enlarges the path space by the values at the atoms of the clock and needs nothing beyond the continuous mapping theorem; it settles every case that arises in this manuscript. The second is the weak-strong convergence of [2], which is what the literature states and what is needed when the times at which continuity fails are *random*. Remark 7.44 sets the two side by side.

Example 7.39 (Why (a) is not a technicality; [2], Section 4.3)**.** Take $\mathbb{T} = \mathbb{R}_+$, $E = \mathbb{R}$, $F = D_{\mathbb{R}}[0,\infty)$, $g(x) = x$ and the atomic clock $q = \delta_1$. The compensator of Definition 3.5 is then the functional

$$F_t(\omega) \;=\; \int_{\langle 0,t\rangle_\iota} \omega(s)\, q(\,\mathrm{d}s) \;=\; \omega(1)\,\mathbf{1}_{\{t>1\}} \quad (\iota = \mathrm{p}), \qquad \omega(1)\,\mathbf{1}_{\{t\geq 1\}} \quad (\iota = \mathrm{o}),$$

since $[0,t)$ contains the atom 1 exactly when $t > 1$, and $(0,t]$ exactly when $t \geq 1$. Either way F_t is, for every t beyond the atom, the evaluation $\omega \mapsto \omega(1)$, and *that* map is J_1-discontinuous at every path jumping at 1: with

$$\omega_n = \mathbf{1}_{[1+1/n,\infty)} \;\longrightarrow\; \omega = \mathbf{1}_{[1,\infty)} \quad \text{in } J_1,$$

one has $F_t(\omega_n) = \omega_n(1) = 0 \not\to 1 = \omega(1) = F_t(\omega)$. For t before the atom $F_t \equiv 0$ is continuous, so the times at which F_t is continuous at ω form the bounded set $\mathbb{T}_{\leq 1}$.

Consequently, if the limit law P charges paths jumping at the atom — $P\{X_1 \neq X_{1-}\} > 0$, which is the generic situation when the clock has an atom at 1 — then hypothesis (a) of Theorem 7.27 fails for *every* $t > 1$, a set of times of full measure. This is a different failure from the one Theorem 7.30 handles: there the bad times form a countable set, namely the fixed discontinuities of X itself, and a dense Γ avoids them; here the bad *index* t ranges over a half-line because the discontinuity of the functional sits at the atom of the clock and not at t. No choice of Γ helps. The cause is exactly the atom of q, and hence exactly the generality that Remark 2.5 and Section 7.3 insist on. Processes with fixed times of discontinuity are not a curiosity here; they are the case this manuscript admits and [4] do not.

An elementary repair: enlarge the path space by the atoms

Before importing machinery it is worth asking how much of it the failure in Example 7.39 actually calls for. The answer is: none. The discontinuity there sits at a *deterministic* place — an atom of the clock — and a functional that is discontinuous only because it evaluates the path at finitely or countably many prescribed times becomes continuous as soon as those values are carried along as separate coordinates. That is the whole idea, and it costs one lemma.

Lemma 7.40 (Where continuity is used in Theorem 7.27)**.** *In the proof of Theorem 7.27, hypotheses (C1) and (a) are used only in the following way: to conclude, from a Borel $\psi : F \to \mathbb{R}$ which is P-continuous at X, that $\psi(X^n) \Rightarrow \psi(X)$. They are used for $\psi \in \{|Y_r^\circ|,\ \varphi_N \circ Y_r^\circ,\ (Y_t^\circ - Y_s^\circ)Z_s^\circ\}$ and for nothing else. In particular neither hypothesis enters the filtration, the determining set, or the conclusion, all of which are statements about F and P alone.*

Proof. Inspection. Step 0 applies Fact 2.26 to $|Y_r^\circ|$ and then Fact 2.43; Step 1 does the same for $(Y_t^\circ - Y_s^\circ)Z_s^\circ$; Step 2 for $\varphi_N \circ Y_r^\circ$. Step 3 uses right continuity and Step 2, and no continuity on F at all. The conclusion $E^P[Y_t \mid \mathcal{F}_s] = Y_s$ and the hypothesis (C2) refer only to F. ☐

Lemma 7.40 says that the continuity may be demanded on *any* space through which the relevant functionals factor, provided the convergence (C1) is available there. So we enlarge.

Theorem 7.41 (Abstract convergence, augmented form; (T0) + (E3))**.** *In the situation of Theorem 7.27, let G be Polish, let $\gamma : F \to G$ be Borel, and put*

$$\hat{F} = F \times G, \qquad \hat{\gamma} : F \to \hat{F}, \quad \omega \mapsto (\omega, \gamma(\omega)), \qquad \hat{X}^n = \hat{\gamma}(X^n), \quad \hat{X} = \hat{\gamma}(X).$$

Replace (C1) and (a) by

(C1′) $\hat{X}^n \Rightarrow \hat{X}$ weakly on $\hat{F}$;

(C3a′) *there are Borel $\hat{Y}_t^\circ : \hat{F} \to \mathbb{R}$ with $\hat{Y}_t^\circ \circ \hat{\gamma} = Y_t^\circ$ such that $\hat{Y}_t^\circ$ and $\hat{Y}_t^\circ\,(Z_s^\circ \circ \mathrm{pr}_F)$ are $\hat{P}$-continuous at $\hat{X}$, where $\hat{P} = P \circ \hat{X}^{-1}$.*

Keep (C2), (b) and (c) unchanged. Then the conclusion of Theorem 7.27 holds verbatim.

Proof. By Lemma 7.40 it suffices to produce $\psi(X^n) \Rightarrow \psi(X)$ for the three functionals listed there. Each factors through $\hat{\gamma}$: $\psi = \hat{\psi} \circ \hat{\gamma}$ with $\hat{\psi} \in \{|\hat{Y}_r^\circ|,\ \varphi_N \circ \hat{Y}_r^\circ,\ (\hat{Y}_t^\circ - \hat{Y}_s^\circ)(Z_s^\circ \circ \mathrm{pr}_F)\}$, by the defining property $\hat{Y}_t^\circ \circ \hat{\gamma} = Y_t^\circ$ and because $\mathrm{pr}_F \circ \hat{\gamma} = \mathrm{id}_F$. Each $\hat{\psi}$ is $\hat{P}$-continuous at $\hat{X}$ — for the first and second because $x \mapsto |x|$ and φ_N are continuous, for the third by (C3a′) and stability of Definition 7.26 under differences. So $\hat{\psi}(\hat{X}^n) \Rightarrow \hat{\psi}(\hat{X})$ by (C1′) and Fact 2.26, and $\hat{\psi}(\hat{X}^n) = \psi(X^n)$, $\hat{\psi}(\hat{X}) = \psi(X)$. Every other step of the proof of Theorem 7.27 is untouched. ☐

Note that (C1′) implies (C1), by projection; the augmented statement is a genuine strengthening of the input and a genuine weakening of the work. The point is what G has to be for an atomic clock.

Proposition 7.42 (Atomic clocks need only the values at the atoms; (T3) + (E2))**.** *Let $F = D_E[0,\infty)$, let $q = q_{\mathrm{a}} + q_{\mathrm{c}}$ with $q_{\mathrm{a}} = \sum_{a \in A} m_a \delta_a$ for a countable $A \subset \mathbb{T}$ and q_{c} atomless, and let*

$$Y_t^\circ(\omega) \;=\; f(\omega(t)) - \int_{\langle 0,t\rangle_\iota} g(u, \omega(u))\, q(\mathrm{d}u)$$

with $f \in C_b(E)$, g bounded, $g(u,\cdot) \in C_b(E)$ for each u and $u \mapsto g(u,x)$ measurable. Put $G = E^A$ with the product topology, $\gamma(\omega) = (\omega(a))_{a \in A}$, and

$$\hat{Y}_t^\circ(\omega, \xi) \;=\; f(\omega(t)) - \int_{\langle 0,t\rangle_\iota} g(u, \omega(u))\, q_{\mathrm{c}}(\mathrm{d}u) - \sum_{a \in A \cap \langle 0,t\rangle_\iota} m_a\, g(a, \xi_a). \tag{97}$$

Then $\hat{Y}_t^\circ \circ \hat{\gamma} = Y_t^\circ$, and (C3a′) *holds for every t outside the countable set $\{t : P\{X_t \neq X_{t-}\} > 0\}$ — the same exceptional set as in the atomless case, and in particular independent of A.*

Proof. The identity $\hat{Y}_t^\circ \circ \hat{\gamma} = Y_t^\circ$ is the splitting $q = q_{\mathrm{a}} + q_{\mathrm{c}}$, the atomic integral being $\sum_{a \in A \cap \langle 0,t\rangle_\iota} m_a g(a, \omega(a))$; note that the index set $A \cap \langle 0, t\rangle_\iota$ is *deterministic*, which is the whole point. We check continuity of the three summands of (97) on $\hat{F} = D_E[0,\infty) \times E^A$.

The atomic sum. Each $(\omega, \xi) \mapsto g(a, \xi_a)$ is continuous, the coordinate projections of a product being continuous and $g(a, \cdot)$ being continuous. The series converges uniformly on $\hat{F}$, since $\sum_{a \in A \cap \langle 0,t\rangle_\iota} m_a \le q(\mathbb{T}_{\le t}) < \infty$ by Definition 2.3 and $|g| \le \|g\|$. A uniform limit of continuous functions is continuous, so this summand is continuous *everywhere* on $\hat{F}$ — no exceptional set at all. *This is where the augmentation does its work*: on $D_E[0,\infty)$ alone the same sum is discontinuous at every path jumping at an atom, which is Example 7.39.

The diffuse integral. Let $\omega_m \to \omega$ in J_1. Then $\omega_m(u) \to \omega(u)$ for every u outside the at most countable set of discontinuities of ω, hence q_{c}-almost everywhere, q_{c} being atomless; by continuity of $g(u, \cdot)$ and bounded convergence the integral converges. So this summand too is continuous everywhere, for *every* ω, and it does not use the second coordinate.

The evaluation $f(\omega(t))$. This is the classical term: it is J_1-continuous at every ω with $\omega(t) = \omega(t-)$, so it is $\hat{P}$-continuous at $\hat{X}$ as soon as $P\{X_t = X_{t-}\} = 1$, and by Fact 2.39 the set of t failing this is countable.

Finally $Z_s^\circ \circ \mathrm{pr}_F$ is bounded and, for s in the same exceptional-free set, continuous at $\hat{X}$ $\hat{P}$-a.s. by Example 3.4; products of $\hat{P}$-continuous functions are $\hat{P}$-continuous. ☐

Remark 7.43 (What (C1′) asks, concretely). For $G = E^A$ the hypothesis (C1′) reads: *$X^n \Rightarrow X$ in J_1, jointly with convergence of the values at the atoms of the clock.* This is ordinary weak convergence on a product of Polish spaces, so Prohorov (Fact 2.28) applies and tightness is the tightness of $\{X^n\}$ in $D_E[0,\infty)$ together with that of the finitely or countably many E-valued families $\{X_a^n\}$, which for E Polish is automatic once $\{X^n\}$ is tight in $D_E[0,\infty)$. What is *not* automatic is the identification of the limit: one must know that $X_a^n \Rightarrow X_a$ and not, say, $X_a^n \Rightarrow X_{a-}$. That is exactly the content of Example 7.39, and it is a condition on the approximating scheme — it holds whenever the X^n jump *at* the atoms rather than near them, which is the case for schemes whose grid contains A, and in particular in Example 7.36.

It is worth being explicit that (C1′) is strictly stronger than (C1): the sequence $\omega_n = \mathbf{1}_{[1+1/n,\infty)}$ of Example 7.39 converges in J_1 but not in $D_E[0,\infty) \times E^{\{1\}}$. Something has to be strengthened, and the question is only what.

Remark 7.44 (Augmentation versus weak-strong convergence). Proposition 7.42 settles Example 7.39 with the continuous mapping theorem and Vitali's theorem, both of which the manuscript already has (Facts 2.26 and 2.43). Set against the alternative below, the ledger reads:

Augmentation (Theorem 7.41)	**Weak-strong** (Theorem 7.48)
one Borel map $\gamma : F \to G$	a control space U, Definitions 7.45 and 7.46
ordinary weak convergence on $F \times G$	a new mode of convergence
Facts 2.26, 2.43	Fact 7.47 (Jacod–Mémin), cited unproved
discontinuity at *deterministic* times	discontinuity at *random* times

The last line is the real division, and it is what decides which to use. The atoms of a clock are fixed in advance, so the bad set of paths is $\{\omega : \omega(a) \neq \omega(a-)$ for some $a \in A\}$ with A deterministic, and carrying the finitely or countably many numbers $(\omega(a))_{a \in A}$ removes it. [2] need more because in their Section 5.3 the times at which continuity fails are the jump times of the approximating processes, which are random and different for each n; no fixed countable family of coordinates captures them, and the exceptional set must be allowed to depend on a control variable.

For everything in this manuscript the deterministic case is the case that occurs. Theorem 7.41 is therefore the tool to reach for, and Theorem 7.48 is recorded for completeness and because it is what the literature states.

The general repair of [2]

The repair of [2] is to weaken not the continuity but the *space on which it is demanded*: a continuity that holds along random sections is enough, provided the sections are indexed by a variable that converges in a stronger sense than weakly. This is the setting we have so far silently avoided by taking the control space to be a singleton.

Definition 7.45 (Weak-strong convergence; [2], Definition 2.1)**.** Let $(U, \mathcal{U})$ be a measurable space, let F be Polish, and put $S = U \times F$ with $\mathcal{S} = \mathcal{U} \otimes \mathcal{B}(F)$. Let C_S be the set of bounded $\mathcal{S}$-measurable $f : S \to \mathbb{R}$ such that $\omega \mapsto f(\alpha, \omega)$ is continuous for *every* $\alpha \in U$. Probability measures P^n on S converge to P in the *weak-strong* sense, $P^n \to_{\text{ws}} P$, if $E^{P^n}[f] \to E^P[f]$ for every $f \in C_S$.

Continuity is thus required in the second variable only; in the first, measurability suffices. If U is a singleton this is ordinary weak convergence ([2], Remark 2.2(i)), which is why nothing of it has appeared until now.

Definition 7.46 ((P^n, P)-continuity; [2], Definition 2.7)**.** An $\mathcal{S}$-measurable $g : S \to \mathbb{R}$ is (P^n, P)-*continuous* if there is $A \in \mathcal{S}$ with $P^n(A) \to 1$ and $P(A) = 1$ such that

$$\big\{(\alpha, \omega) \in A \;:\; A_\alpha \ni \zeta \mapsto g(\alpha, \zeta) \text{ is discontinuous at } \omega\big\}$$

is P-null, where $A_\alpha = \{\zeta : (\alpha, \zeta) \in A\}$.

Two relaxations at once: continuity is asked for only along the sections A_α, and only on a set of asymptotically full measure. For a singleton U and $A = U \times C$ this is Definition 7.26.

Fact 7.47 (Continuous mapping, Jacod–Mémin; [2], Theorem 2.9). If $P^n \to_{\text{ws}} P$ and $g : S \to \mathbb{R}$ is (P^n, P)-continuous with $\sup_n E^{P^n}[|g|\mathbf{1}_{\{|g|>a\}}] \to 0$ as $a \to \infty$, then $E^{P^n}[g] \to E^P[g]$.

Theorem 7.48 (Abstract convergence with control variables; [2], Theorem 3.14)**.** *Theorem 7.27 holds with X^n, X replaced by pairs $(L^n, X^n), (L, X)$ taking values in $S = U \times F$, with (C1) replaced by $(L^n, X^n) \to_{\text{ws}} (L, X)$ and the P-continuity of (a) replaced by (Q^n, Q)-continuity, where Q^n and Q are the laws of (L^n, X^n) and (L, X).*

Proof, or rather: what has to be changed. Nothing, beyond two citations. Steps 0, 1 and 2 of the proof of Theorem 7.27 each consist of applying the continuous mapping theorem (Fact 2.26) to obtain weak convergence and then Fact 2.43 to upgrade it to convergence of expectations. Fact 7.47 performs both steps at once under the weaker hypotheses, and is the only place where the difference between weak and weak-strong convergence is felt. Step 3 uses no continuity at all. All the work is therefore in Fact 7.47, which we do not prove. □

Remark 7.49 (What the control variable does). The point of U is that the exceptional set may depend on α. In [2], Section 5.3 the control variables L^n are càdlàg processes chosen so that, with probability tending to one, *the jump times of X^n are dominated by those* *of* L^n (their Assumption 5.1). On the resulting sections the Skorokhod J_1 and the local uniform topology coincide (their Proposition 2.12), and the functional of Example 7.39 — discontinuous in J_1 at every $t \geq 1$ — becomes continuous. One may then "treat the control variables as deterministic", which is the practical content of Definition 7.45.

So the answer to why weak-strong convergence has been absent is that Theorem 7.27 is [2], Corollary 3.17, the case of a singleton control space, and that in that case (a) is available whenever the limit has no fixed times of discontinuity. Atoms in the clock destroy that, but — and this is the conclusion of Remark 7.44 — they destroy it at deterministic times, which Theorem 7.41 handles with the continuous mapping theorem. So the price atoms exact is *not* a new mode of convergence but a joint convergence hypothesis, (C1′). Weak-strong convergence is needed for a different reason: random times of discontinuity, which no fixed family of coordinates captures. Section 6, where one might also have expected atoms to cost something, survives them intact (Remark 6.7).

7.10 Hawkes processes and their Volterra limit

This last subsection is a single example, and it is here because it is the first place where every strand of the manuscript is used at once: the explicit construction of Section 7.3, the localization of Section 5.3, the *non-Markovian* uniqueness of Section 5.1, and the convergence theorem of Section 7.6. Throughout, $\mathbb{T} = \mathbb{R}_+$ and q is Lebesgue measure.

Jump processes with a path-dependent rate

Section 7.3 constructed jump processes from a rate $\lambda(x)$ and a kernel $\mu(x, \cdot)$ depending on the current state. Nothing in that construction needs the dependence to be Markovian, and one gain is worth the change of notation: the holding time ceases to be exponential, and the computation that carried Theorem 7.12 survives intact.

Setting 7.50 (Path-dependent jump data; (T3) + (E0))**.** Let F be the set of piecewise constant right continuous paths $\mathbb{T} \to E$ with finitely many jumps on bounded intervals, and let

$$\Lambda : \mathbb{T} \times F \to [0, \infty), \qquad \mu : \mathbb{T} \times F \to \mathcal{P}(E)$$

be *predictable*: $\Lambda(t, \omega)$ and $\mu(t, \omega, \cdot)$ depend on ω only through $\omega|_{[0,t)}$, and $t \mapsto \Lambda(t, \omega)$ is locally integrable for every ω. Put

$$\mathcal{A}_t f(\omega) \;=\; \Lambda(t, \omega) \int_E \big(f(y) - f(\omega(t-))\big)\, \mu(t, \omega, \mathrm{d}y), \qquad f \in B(E), \tag{98}$$

and let $\mathbb{X}^\circ$ consist of the processes $f(\pi_t) - \int_0^t \mathcal{A}_s f \, \mathrm{d}s$, $f \in B(E)$.

Definition 7.51 (The construction; (T3) + (E0))**.** Let $\varepsilon_1, \varepsilon_2, \ldots$ be i.i.d. exponential of parameter 1. Set $\tau_0 = 0$, let $Y_0 \sim \nu$, and given (τ_n, Y_n) let $\omega^{(n)}$ be the path built so far, continued constantly at Y_n; put

$$\tau_{n+1} \;=\; \inf\Big\{ t > \tau_n \;:\; \int_{\tau_n}^t \Lambda(u, \omega^{(n)})\, \mathrm{d}u > \varepsilon_{n+1} \Big\}, \tag{99}$$

and draw $Y_{n+1} \sim \mu(\tau_{n+1}, \omega^{(n)}, \cdot)$. Define X by (81) and $\zeta = \lim_n \tau_n$ as before.

Writing $A_n(u) = \int_{\tau_n}^u \Lambda(v, \omega^{(n)})\, \mathrm{d}v$, the construction gives, conditionally on $\mathcal{H}_n = \sigma(Y_0, \ldots, Y_n, \tau_1, \ldots, \tau_n)$, the survival function $P(\tau_{n+1} > u \mid \mathcal{H}_n) = e^{-A_n(u)}$ and the jump-time density $\Lambda(u, \omega^{(n)}) e^{-A_n(u)}$. For Λ constant this is the exponential law of Definition 7.10.

Theorem 7.52 (Path-dependent jump processes; (T3) + (E0)). *Let X be as in Definition 7.51 and let $N_t = \#\{n \geq 1 : \tau_n \leq t\}$ be its jump counter.*

(a) X solves the local *martingale problem for $\mathbb{X}^\circ$ of Setting 7.50 on $[0, \zeta)$, with initial law ν and with (τ_n) a localizing system in the sense of Definition 5.22. No hypothesis beyond Setting 7.50 is needed.*

(b) If $E[N_t] < \infty$ for every t — equivalently, by (a), $E\big[\int_0^t \Lambda(u, X)\,\mathrm{d}u\big] < \infty$ — then $\zeta = \infty$ a.s. and X solves the global *martingale problem for $\mathbb{X}^\circ$.*

Remark 7.53 (Why the local statement is the primary one). In Setting 7.9 the bound $\bar\lambda < \infty$ gave $E[N_t] \leq \bar\lambda t$ for free, and the global statement was the natural one. With a path-dependent rate this is no longer so, and the failure is not about explosion: even when $\zeta = \infty$ a.s., the compensator $\int_0^t \Lambda(u, X)\,\mathrm{d}u$ need not be integrable, so $f(X_t) - \int_0^t \mathcal{A}_s f\,\mathrm{d}s$ need not lie in L^1 and the phrase "is a martingale" has no content. Stopping at τ_n repairs this *unconditionally*, because

$$\int_{\tau_k}^{\tau_{k+1}} \Lambda(u, \omega^{(k)})\,\mathrm{d}u \;=\; \varepsilon_{k+1} \qquad \text{on } \{\tau_{k+1} < \infty\} \tag{100}$$

by the very definition (99) of τ_{k+1}, so that $\int_0^{t\wedge\tau_n} \Lambda\,\mathrm{d}u \leq \varepsilon_1 + \cdots + \varepsilon_n$ is integrable. This is a clean instance of the phenomenon Definition 5.22 exists for, and unlike Remark 7.15 it is not about paths leaving every compact set.

Proof. The decomposition of Step 1 in the proof of Theorem 7.12 is unchanged, so the first task is to show $E[D_n \mid \mathcal{H}_n] = 0$ on $\{\tau_n \leq t\}$ for

$$D_n = \big(f(Y_{n+1}) - f(Y_n)\big)\mathbf{1}_{\{\tau_{n+1} \leq t\}} - \int_{\tau_n \wedge t}^{\tau_{n+1}\wedge t} \mathcal{A}_s f\,\mathrm{d}s.$$

Write $\Lambda_u = \Lambda(u, \omega^{(n)})$ and $\mu_u f = \int f\,\mathrm{d}\mu(u, \omega^{(n)}, \cdot)$, and note that on $\{\tau_{n+1} > s\}$ the path agrees with $\omega^{(n)}$, so that $\mathcal{A}_s f = \Lambda_s(\mu_s f - f(Y_n))$ there. Then

$$\begin{aligned} E\big[(f(Y_{n+1}) - f(Y_n))\mathbf{1}_{\{\tau_{n+1} \leq t\}} \,\big|\, \mathcal{H}_n\big] &= \int_{\tau_n}^t \Lambda_u e^{-A_n(u)}\big(\mu_u f - f(Y_n)\big)\,\mathrm{d}u, \\ E\Big[\int_{\tau_n}^{\tau_{n+1}\wedge t} \mathcal{A}_s f\,\mathrm{d}s \,\Big|\, \mathcal{H}_n\Big] &= \int_{\tau_n}^t \Lambda_s\big(\mu_s f - f(Y_n)\big)\,P(\tau_{n+1} > s \mid \mathcal{H}_n)\,\mathrm{d}s, \end{aligned}$$

the first because the jump time has density $\Lambda_u e^{-A_n(u)}$ and the jump distribution at time u is $\mu(u, \omega^{(n)}, \cdot)$, the second by Fubini. Since $P(\tau_{n+1} > s \mid \mathcal{H}_n) = e^{-A_n(s)}$, the two expressions *coincide*, and $D_n = 0$ on $\{\tau_n > t\}$. Both displays are finite because $|\mu_u f - f(Y_n)| \leq 2\|f\|$ and $\int_{\tau_n}^t \Lambda_u e^{-A_n(u)}\,\mathrm{d}u \leq 1$.

(a). Stop at τ_m. Then (82) has only the m summands $D_0, \ldots, D_{m-1}$, each integrable by the previous paragraph, so no domination argument is needed and $E[M_{t\wedge\tau_m}] = 0$ for every t. For the increment from s to t, observe that conditionally on $\mathcal{F}_s^\circ$ and on $\{N_s = n\}$ the residual jump time satisfies

$$P\big(\tau_{n+1} > u \,\big|\, \mathcal{F}_s^\circ\big) \;=\; \exp\Big\{-\int_s^u \Lambda(v, \omega^{(n)})\,\mathrm{d}v\Big\}, \qquad u \geq s,$$

by (99) and the lack of memory of ε_{n+1} — which here concerns the *compensator* A_n, not the time; and that the subsequent construction is (99) again, with τ_n replaced by s and with the past $\omega|_{[0,s]}$ carried along. The computation above then applies with τ_n replaced by s and $\mathcal{H}_n$ by $\mathcal{F}_s^\circ$, and gives $E[M_{t\wedge\tau_m} - M_{s\wedge\tau_m} \mid \mathcal{F}_s^\circ] = 0$. Note that the restart carries the whole past, not merely X_s: this is Lemma 5.5, and it is the reason Theorem 7.52 needs no shift system. Since $\tau_m \uparrow \zeta$ and each τ_m is a functional of the path — indeed a *strict* stopping time, being the m-th jump time of a right continuous piecewise constant path — (τ_m) is a localizing system, with (L1) holding by construction, (L2) because the m-th jump time after r is a jump time, and (L3) by (100).

(b). Assume $E[N_t] < \infty$. Then $\tau_m \uparrow \infty$ a.s., so $\zeta = \infty$. Moreover

$$\sum_{n\geq 0} |D_n| \;\leq\; 2\|f\|N_t + 2\|f\| \int_0^t \Lambda(u, X)\,\mathrm{d}u,$$

and by (100) the last integral equals $\sum_{k<N_t} \varepsilon_{k+1} + \int_{\tau_{N_t}}^t \Lambda\,\mathrm{d}u \leq \sum_{k\leq N_t} \varepsilon_{k+1}$, whose expectation is $E[N_t] + 1 < \infty$ by Wald's identity, the ε_k being i.i.d. of mean 1 and $N_t + 1$ a stopping time for their natural filtration. So dominated convergence applies to (82), and letting $m \to \infty$ in (a) turns the local statement into the global one. □

Remark 7.54. The cancellation is the same one as in Theorem 7.12: the rate in the generator against the survival function of the holding time. What has gone is only the Markov structure — Λ and μ may depend on the whole past — and with it the shift system. So Theorem 7.52 produces solutions to which Proposition 5.9 applies and Theorem 5.16 does not, which is the situation Remark 5.42 describes and which has so far been illustrated only by the cited Examples 5.40 and 5.41.

Example 7.55 (Hawkes processes). Take $E = \mathbb{N}_0$, $\mu(t, \omega, \cdot) = \delta_{\omega(t-)+1}$ and

$$\Lambda(t, \omega) \;=\; \mu_0 + \int_{[0,t)} \phi(t-s)\,\mathrm{d}\omega_s, \qquad \mu_0 > 0,\; \phi : \mathbb{R}_+ \to [0, \infty), \tag{101}$$

with ϕ locally integrable. Then $X = N$ is a *Hawkes process*, and Theorem 7.52 says that $f(N_t) - \int_0^t \Lambda_s(f(N_s + 1) - f(N_s))\,\mathrm{d}s$ is a martingale for every bounded f; taking $f = \mathrm{id}$ on bounded sets recovers the defining property that $N - \int_0^\cdot \Lambda_s\,\mathrm{d}s$ is a martingale.

The stability threshold is $\|\phi\|_1 = 1$: for $\|\phi\|_1 < 1$ the renewal equation $E[\Lambda_t] = \mu_0 + \int_0^t \phi(t-s)E[\Lambda_s]\,\mathrm{d}s$ gives $\sup_t E[\Lambda_t] \leq \mu_0/(1 - \|\phi\|_1) < \infty$, hence $E[N_t] < \infty$ and Theorem 7.52(b) applies, while for $\|\phi\|_1 \geq 1$ explosion is possible and Theorem 7.52 delivers the *local* problem, localized by the jump times. That is not a defect of the example: the scaling limit below lives at $\|\phi\|_1 \uparrow 1$, so the interesting regime is exactly the one in which Section 5.3 is needed.

The Volterra limit

The observation that makes the limit theorem an application rather than an analogy is that (101) *is* a Volterra equation.

Remark 7.56 (A Hawkes process is a Volterra SDE). Put $Z = N$ and $X = \Lambda$. Then (101) reads

$$X_t \;=\; g_0(t) + \int_{[0,t)} K_{t-s}\,\mathrm{d}Z_s, \qquad g_0 \equiv \mu_0, \quad K = \phi,$$

which is verbatim the constraint of Example 5.40, and Z is a semimartingale whose characteristics are driven by X: $a = 0$ and $\nu(x, \mathrm{d}y) = x\,\delta_1(\mathrm{d}y)$. A Hawkes process is therefore a solution of a Volterra stochastic differential equation with pure-jump noise, and the pair (Λ, N) lies in the path space $L^p_{\mathrm{loc}}(\mathbb{R}_+, \mathbb{R}) \times D_{\mathbb{R}}[0, \infty)$ of Example 5.40. Approximants and limit below are thus elements of one and the same solution set, for two triplets of the same shape — one purely discontinuous, one continuous.

Theorem 7.57 (Hawkes to Volterra, conditionally on tightness; (T3) + (E3))**.** *For each n let (Λ^n, N^n) be a Hawkes process with kernel ϕ^n and baseline μ_0^n, rescaled in time, and let (X^n, Z^n) be the corresponding pair of Remark 7.56. Let $\mathbb{X}^\circ$ be the Volterra family of Example 5.40 for a kernel K, a triplet (b, a, ν) and g_0. Suppose*

(a) $(X^n, Z^n) \Rightarrow (X, Z)$ weakly on $L^p_{\mathrm{loc}}(\mathbb{R}_+, \mathbb{R}) \times D_{\mathbb{R}}[0, \infty)$;

(b) the rescaled generators converge in the sense of (c), i.e. for $f \in C_b^2$ and $s \le t$ in a dense D, $\lim_n E\big[\big(\int_s^t (\mathcal{A}_u^n f - Lf(X_u^n, Z_u^n))\,\mathrm{d}u\big) Z_s^\circ(X^n, Z^n)\big] = 0$;

(c) $\{f(Z_r^n) - \int_0^r Lf(X_u^n, Z_u^n)\,\mathrm{d}u : r \in D \cap [0, t],\ n \in \mathbb{N}\}$ is uniformly integrable.

Then (X, Z) solves the Volterra martingale problem $\mathbb{X}^\circ$. If moreover that problem propagates agreement in the sense of Definition 5.7 — uniqueness in law from any prescribed past — then (X, Z) is determined, and the whole sequence converges to it.

Proof. Hypotheses (a)–(c) are (C1), (c) and (b) of Theorem 7.27; (C2) is Example 3.4 on the second coordinate together with the integrated form of the Volterra constraint, and (a) holds because the first coordinate enters only through Lebesgue integrals — which is why Example 5.40 states the constraint in integrated form — and the second through evaluations at times where the limit does not jump. The approximating martingales required by (c) are those of Theorem 7.52. Theorem 7.27 then gives $(X, Z) \in \mathcal{M}(\mathbb{X}^\circ)$. The final assertion is Proposition 5.9, whose hypothesis (34) is uniqueness in law of the Volterra equation restarted from a prescribed past; it upgrades to uniqueness of the whole law with no Markov structure — which is essential, since a Volterra process has none. See Example 5.40 and Remark 5.8. □

Remark 7.58 (What is proved here and what is cited). Hypothesis (a) is tightness, and it is not supplied — here as everywhere in this section (Remark 7.38). For the Hawkes scaling it is the substance of the theorem: the interesting regime is $\|\phi^n\|_1 \uparrow 1$ with a heavy-tailed ϕ, the kernel rescales to $K(t) = t^{\alpha-1}/\Gamma(\alpha)$, and the limit is a rough stochastic Volterra equation. That is the content of [6], and it is cited, not proved; the present theorem is the identification step, and it is the part the abstract machinery does supply.

Three remarks on scope. First, *neither weak-strong convergence nor the augmentation of Theorem 7.41 is needed*: the clock is Lebesgue, so there are no atoms, the jumps of N^n have height one and the limit is continuous; the limit therefore has no fixed times of discontinuity and (a) holds on a dense set with countable complement. Section 7.9 stays out of it entirely. Second, *the clock does not change* if one rescales the continuous-time Hawkes process, so Theorem 7.27 suffices and Section 7.8 is not used; it would be, if one approximated instead by the discrete branching representation, whose clock is atomic. Third, and this is the point of the example: *no result of Section 5.2 is used anywhere.* Neither the approximants nor the limit has a shift system, and none is wanted.

8 Notes for the formalization

8.1 What is available and what has to be built

Section 2 lists the prerequisites; the following is a stocktaking of which of them Mathlib already supplies.

Available. Filtrations, adapted and progressively measurable processes, stopping times and $\mathcal{F}_\tau$ (Definition 2.15, Fact 2.16); martingales and the optional stopping theorem (Definition 2.17, Fact 2.20); Doob's inequalities (Fact 2.21); Polish spaces; weak convergence of measures with the portmanteau theorem, the Lévy–Prokhorov metric and Prohorov's theorem (Facts 2.25, 2.27, 2.28); uniform integrability (Fact 2.43); monotone class theorems (Fact 2.44); the Riesz–Markov–Kakutani representation of a positive linear functional on compactly supported continuous functions, in the form `integral_rieszMeasure`, which is what Proposition 7.7 needs; and part of what Fact 2.45 needs — see the list there.

To be built.

- the Skorokhod space $D_E[0, \infty)$ with the J_1 metric, and the facts that it is Polish, that its Borel field is generated by the coordinates, and the characterization of its compact sets (Definition 2.36, Facts 2.37, 2.38, 2.35);
- regularization of submartingales along a dense set (Facts 2.18, 2.19) — but see Remark 8.1: this exists already, outside Mathlib;
- local martingales in the [4] sense (Definition 2.22, Fact 2.23);
- separating and convergence determining classes and their Stone–Weierstrass theory (Definition 2.32, Facts 2.33, 2.34, 2.35, 2.46);
- bounded pointwise convergence and bp-closures (Definition 2.29, Fact 2.30) — but only for Corollary 3.13, which is itself optional: the working statement is Lemma 3.11, and that is dominated convergence;
- the continuous mapping theorem in the form of Fact 2.26 and the Skorokhod representation (Fact 2.27);
- convergence of finite-dimensional distributions in $D_E[0, \infty)$ and the relative compactness criteria (Facts 2.40, 2.41, 2.42);
- dissipative operators and the full generator of a measurable contraction semigroup (Definitions 2.11, 2.12, Fact 2.13); these are needed only for Proposition 3.18 and Remark 3.19, which are themselves optional context;
- the Kolmogorov extension theorem for an uncountable index (Fact 2.45). Mathlib has the scaffolding and the sequential case but not the theorem; this is the one external prerequisite of Section 7.2, and for $\mathbb{T} = \mathbb{N}_0$ it is already there.

Remark 8.1 (Prior art: the `brownian-motion` project). The càdlàg half of **(F7)** is already formalized, not in Mathlib but in the Lean development accompanying [1], in the repository `RemyDegenne/brownian-motion` on GitHub. Its file `Cadlag.lean` defines `IsRightContinuous` for a preorder and `IsCadlag` as a structure, and develops jump sets, the discreteness of large jumps, local boundedness and the usual closure properties, with no remaining gaps. Its `CadlagModification.lean` then constructs the càdlàg modification; the blueprint chapter proves that a submartingale has a càdlàg modification if and only if $t \mapsto E[X_t]$ is right continuous, and that a martingale always has one. These are precisely Facts 2.18 and 2.19, and they are proved for *quasimartingales*, which is more general than what Theorem 4.3 needs.

Three consequences for the present plan. First, **(F7)** loses its most expensive prerequisite. Second, the index there is a preorder with an order topology, which agrees with the bundle (T2b) of Definition 2.1 — the two developments make the same choice independently. Third, what is *not* there is the Skorokhod space: no J_1 metric, no $\mathcal{B}(D_E[0, \infty)) = \sigma(\pi_t)$, no compactness criterion. So **(F8)** is untouched, and Fact 2.37 and Fact 2.38 remain to be built.

This dictates a natural order of work. Each item names the bundle of Definition 2.1 it targets; the point of doing so is that **(F1)**, **(F4)** and half of **(F3)** live under `[Preorder ι]`, which is how Mathlib already indexes `Filtration` and `Martingale`, and can therefore be stated without any bespoke index theory.

(F1) *Targets (T0) + (E0) + clock.* Definitions only: Definition 3.2, Definition 3.5, the filtration (4), Definition 3.20 (which needs (T2b)), and Proposition 3.8. The clock is a `MeasureTheory.Measure` on $\mathbb{T}$ together with the hypothesis that every down-set is measurable and of finite mass; the interval algebra (2) is then definitional. These need nothing beyond what Mathlib has and should be done first; note that Proposition 3.8 is what turns every subsequent statement into a statement about finite-dimensional distributions.

(F2) *Targets (T3) + (E0).* The jump processes of Section 7.3. Theorem 7.12 needs a Markov chain, i.i.d. exponential variables and the computation of Step 2, and nothing else; Proposition 7.13 adds the Picard iteration (86) for the bounded operator A, which in Lean is the exponential series of a bounded linear map and needs no analysis beyond `NormedSpace`. It is by a wide margin the cheapest theorem with real content in this manuscript, it is the only one that exhibits a solution, and it supplies the missing example for Definition 5.22. It is placed here, before any of the theory, as a way of testing the definitions of **(F1)** against something concrete.

(F3) *Targets (T0) + (T4) + (E0); (T2a) for the uniqueness half, (E1) for Lemma 5.3.* Lemma 5.2, Lemma 5.13 and Theorem 5.16, then Theorem 5.43 and Corollary 5.44 as corollaries. Modulo (19) this is pure measure theory, and Theorem 5.16 needs no path regularity at all, so it is the best entry point after **(F1)**. Almost all of the work sits in Lemma 5.13, which is four lines; the two theorems that follow are bookkeeping. Theorem 5.19 additionally needs Fact 2.20 and càdlàg paths and should be deferred until after **(F7)**, and so should the local theory of Section 5.3 — Definition 5.22 is about stopping times. Note that Lemma 5.24(a) and Lemma 5.26 are the only genuinely new proofs there; Lemma 5.29 is Lemma 5.13 plus the one identity (44), and Theorem 5.30 is bookkeeping.

(F4) *Targets (T0) + (E3), with $D = \mathbb{T}$.* Theorem 7.27, the abstract convergence theorem. It needs an abstract Polish path space, a determining set, P-continuity at X, uniform integrability and Vitali's theorem — and *no* Skorokhod topology, no càdlàg paths and no Markov structure. It is placed here, immediately after **(F3)**, and not with the Skorokhod theory of **(F8)** where the ordering of the literature would put it: it is one of the cheapest genuine theorems in this list, not part of the most expensive one. Take Lemma 7.40 and Theorem 7.41 with it; they are two lines each once Theorem 7.27 is in place.

(F5) *Targets (T0) + (E0) for Lemma 6.1, (T2a) + $\iota = \mathrm{p}$ for Lemma 6.4 and Theorem 6.5 (which needs (T2b)), (T3) + (E2) for the rest.* Lemma 6.1 — an exact telescoping identity, and the cheapest item in this list — then Proposition 6.3, Theorem 6.5, Corollary 6.14, and finally Lemma 6.9, Theorem 6.10 and their corollaries. Self-contained: Fubini, absolute continuity, dominated convergence. No Skorokhod space is needed. Note that Corollary 6.14 needs none of the analysis and can be formalized directly after Lemma 6.1. Together with **(F3)** this already yields Corollary 6.18, a complete uniqueness theory.

(F6) *Targets (T0) + (T4) + (E1) + a shift invariant clock.* Section 7.2: Lemma 7.3, Proposition 7.4 and Theorem 7.5. Everything in the proofs is the Markov property, Fubini and (3). The one external input is Fact 2.45, and it is a real one: Mathlib has the scaffolding and the sequential Ionescu–Tulcea case but not the Kolmogorov extension theorem for an uncountable index. So this item splits: for $\mathbb{T} = \mathbb{N}_0$ it can be done at once, for $\mathbb{T} = \mathbb{R}_+$ it waits on that theorem. It uses **(F3)** for Theorem 5.16 and **(F5)** for Corollary 7.6, but Theorem 7.5 itself needs neither. Proposition 7.7 is separate; Mathlib's `integral_rieszMeasure` should carry it, but the theorem is stated with (D3) as a hypothesis so that it can be skipped.

(F7) *Targets (T2b) + (E2), $\iota = \mathrm{o}$.* Fact 2.18 and Fact 2.19, then Theorem 4.3 and, as a one-line corollary, Theorem 4.5. This requires the notion of a càdlàg path but *not* the Skorokhod topology. Formalize the abstract theorem, not the Markovian one: Definition 4.2 is three hypotheses on a pair of processes, whereas $\mathcal{D}(A)$ drags in the operator, its domain and the filtration (4), none of which the proof uses.

(F8) *Targets (T3) + (E3).* The Skorokhod space and its topology, then Lemma 7.23, Remark 7.25 and Theorem 7.30 as instances of **(F4)**. This is by far the largest item. Theorem 7.41 belongs with **(F4)**, not here: it is Lemma 7.40 plus a change of the ambient space, costs nothing once **(F4)** is done, and needs no Skorokhod theory. Only Proposition 7.42, which instantiates it for $F = D_E[0, \infty)$, does. Section 7.9 beyond that — Definitions 7.45 and 7.46 and Fact 7.47 — should be skipped: nothing in this manuscript needs it, by Remark 7.44.

8.2 Design decisions

(a) *Operators as relations.* A is a subset of $M(E) \times M(E)$, not a function. In Lean this is `A : Set ((E → R) × (E → R))` together with measurability side conditions, or a structure bundling the pair with its measurability. Keeping A multivalued costs nothing and is needed for Remark 3.10(b).

(b) *Two notions of solution.* Definition 3.5 defines "solution with respect to $(\mathcal{G}_t)$" as the primitive and "solution" as the special case $\mathcal{G}_t = {}^*\mathcal{F}_t^X$. Doing it the other way round duplicates every statement.

(c) *Compact containment as a predicate.* Definition 3.20 occurs in Theorem 4.5, in Remark 7.25 and implicitly in Remarks 5.45 and 7.31. It should be one definition with the two variants (29) (over D, single process) and (15) (over $[0, T]$, family) related by a lemma for right continuous processes.

(d) *Local martingale problems: localize the test processes, not the operator.* Define the local martingale problem at the level of Definition 3.2 and localize by stopping the elements of $\mathbb{X}^\circ$, as Definition 5.22 does. The alternative — replacing A by $A^{(m)} = \{(f\mathbf{1}_{K_m}, g\mathbf{1}_{K_m})\}$ and passing to the limit, which is [4], Section 4.6 and Remark 5.45 — should *not* be the primitive: in the abstract setting $\mathbb{X}^\circ$ is primitive and A is not, so the operator route has nothing to modify. See Remark 5.32. Concretely, the reusable object in Lean is the structure (L1)–(L3) of Definition 5.22, and the reusable lemma is Lemma 5.26, which *constructs* (L1) rather than assuming it. Note that the times in Σ must be *strict* stopping times; Remark 5.27 is the trap, and in Lean it is the difference between `IsStoppingTime` for the raw and for the right continuous filtration.

(e) *Separating classes.* Facts 2.46 and 2.35 should be developed independently of martingale problems; they are of general interest, and within Sections 4 and 5 they are the only place where Polishness of E is really used. (In Section 7 it is used throughout, and irreducibly: (E3) is what makes Prohorov's theorem available.)

(f) *No semigroup theory.* The manuscript rests on Chapters 2 and 3 of [4] alone, as envisaged in the project plan; of Chapter 1 only the notions of a dissipative operator and of the full generator are borrowed, and only for the optional Proposition 3.18 and Remark 3.19. See Remark 2.14. Section 7.2 does not disturb this: the family (P_t) there is a semigroup of *kernels*, and the balance condition (D1) is assumed in integrated form, so no strong continuity, no generator and no domain theory are needed — which is one reason for stating it that way rather than as [3] do.

(g) *Both conventions, carried and not chosen.* $\iota \in \{\mathrm{o}, \mathrm{p}\}$ should be a parameter of the compensator, not a fixed choice. The evidence is in Remark 7.37: the same Markov chain needs p on $\mathbb{T} = \mathbb{N}_0$ and o once its grid is embedded in $\mathbb{R}_+$. Everything except Section 4 is convention-agnostic, so the parameter costs a hypothesis in one section and nothing anywhere else.

(h) *Augmentation before weak-strong convergence.* Where a functional fails to be continuous at deterministic times — the atoms of the clock — enlarge the path space (Theorem 7.41) rather than weaken the mode of convergence. This keeps Section 7 inside the continuous mapping theorem and Vitali's theorem, and avoids formalizing Fact 7.47. See Remark 7.44.

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